Module 2: (6 Lectures)

- Geometric Design- Cross section elements, Sight distances
- Horizontal alignment Vertical alignment, Intersections
- Construction of Pavements
- Construction and Maintenance of Drainage
- Road Arboriculture

Geometric Design of Highway :

- The physical features of road are known **as road geometrics**.
- Geometric design of highway **deals with the dimension, layout & visible features or elements of the highway** such as horizontal, vertical alignment, sight distance, etc.
- Geometric design of highway deals with following elements:

1) **Cross Section Elements:** carriage way, road way, road margin, shoulder, side slope, kerb, formation level & width.
2) **Sight Distance:** SSD, OSD, Intermediate & Lateral sight dist.
3) **Horizontal Alignment**
4) **Vertical Alignment**
5) **Super-elevation**
6) **Intersection Elements**

- It is possible to design & construct the pavement of road in stages, but it is **very expensive & difficult to improve the geometric elements of a road** in stages **at later date**. Therefore design of geometric features at earlier stage is very important & crucial.

Design Control & Criteria :

- The important factors which control the geometric elements are:

1) **Design Speed:** Different speed for different road NH, SH, MDR, ODR & VR.
2) **Topography or Terrain:** Plain, steep, mountainous, rolling, etc.
3) **Traffic Factors:** Vehicular & human characteristics
4) **Design Hourly Volume & Capacity:** traffic flow & volume
5) **Environmental & Others Factors:** aesthetics, landscaping, air pollution, noise pollution, etc.

Objective of Geometric Design:

To promote & achieve the more speed, safety & comfort to road user.

Factors affecting geometric design

A number of factors affect the geometric design and they are discussed in detail in the following sections.

a. Design speed

- Design speed is defined as the highest continuous speed at which individual vehicles can travel with safety on the highway when weather conditions are conducive.
- Design speed is the single most important factor that affects the geometric design.
- It directly affects the sight distance, horizontal curve, and the length of vertical curves.
- Since the speed of vehicles vary with driver, terrain etc, a design speed is adopted for all the geometric design.

b. Topography

- The next important factor that affects the geometric design is the topography.
- It is easier to construct roads with required standards for a plain terrain.
- However, for a given design speed, the construction cost increases multiform with the gradient and the terrain.
- Therefore, geometric design standards are different for different terrain to keep the cost of construction and time of construction under control.

c. Other factors

In addition to design speed and topography, there are various other factors that affect the geometric design and they are briefly discussed below:

1. Vehicle: :The dimensions, weight of the axle and operating characteristics of a vehicle influence the design aspects such as width of the pavement, radii of the curve, clearances, parking geometrics etc. A *design vehicle* which has standard weight, dimensions and operating characteristics are used to establish highway design controls to accommodate vehicles of a designated type.

2. **Human:** The important human factors that influence geometric design are the physical, mental and psychological characteristics of the driver and pedestrians like the reaction time.

3. **Traffic:** A reasonable value of traffic volume is selected as the design hourly volume which is determined from the various traffic data collected. The geometric design is thus based on this design volume, capacity etc.

4. **Environmental:** Factors like air pollution, noise pollution etc. should be given due consideration in the geometric design of roads.

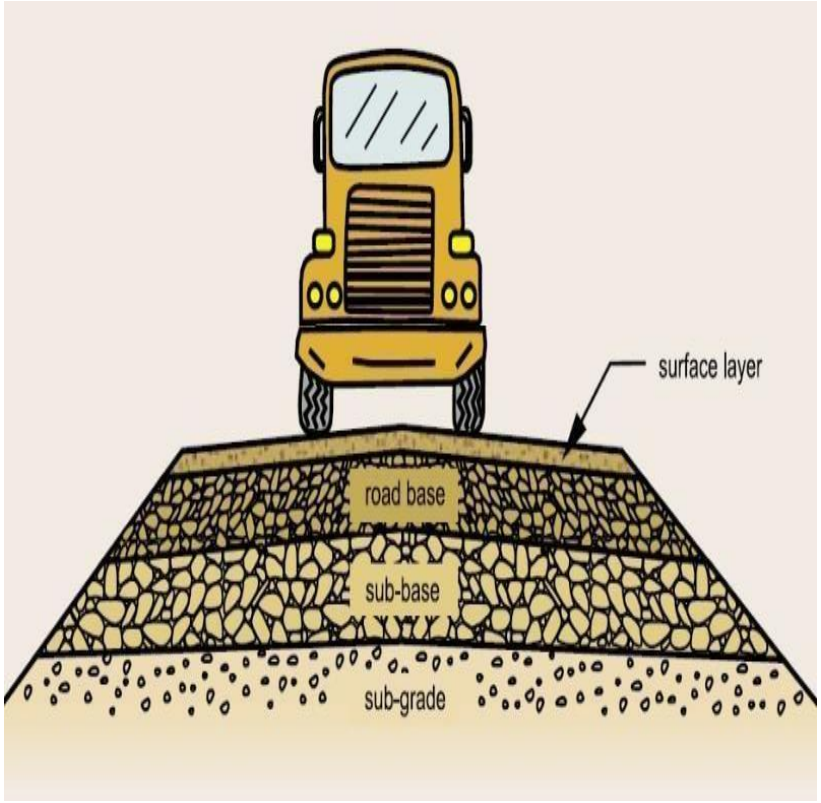
5. **Economy:** It should match with the funds alloted for capital cost and maintenance cost.

6. **Others:** Geometric design should be such that the aesthetics of the region is not affected.

Highway Cross Section Elements:

1) **Pavement Surface Characteristics:** Friction, unevenness, light reflecting characteristics, drainage of surface water.
2) Cross Slope or Camber:
3) Width of Pavement or Carriageway:
4) Width of Formation or Roadway:
5) Traffic Separators or Medians:
6) Kerbs:
7) Road Margins or Boundary:
8) Shoulder:
9) Side Slopes:
10) Gradient:
11) **Right of Way or Land Width or Permanent Land:** The area of **land acquired & reserved for construction & development of a road** along its alignment is known as Right of Way.

- The features of the cross-section of the **pavement influences the life of the pavement as well as the riding comfort and safety.**
- These, pavement surface characteristics affect both of these.
- Camber,kerbs, and geometry of various cross-sectional elements are important aspects to be considered in this regard.



1.Pavement surface characteristics

For safe and comfortable driving four aspects of the pavement surface are important;

- the **friction**between the wheels and the pavement surface,
- **smoothness** of the road surface,
- the **light reflection** characteristics of the top of the pavement surface, and **drainage** to water.

Cross Sectional Elements

a. Friction

Friction between the wheel and the pavement surface is a crucial factor in the design of horizontal curves and thus the safe operating speed. Further, it also **affect the acceleration and deceleration** ability of vehicles. Lack of adequate friction can cause skidding or slipping of vehicles.

b. Unevenness

It is always desirable to have an even surface, but it is seldom possible to have such a one. Unevenness **affects the vehicle operating cost, speed, riding comfort, safety, fuel consumption and wear and tear of tyres**.

c. Light reflection

1.White roads have good visibility at night, but caused glare during day time.

2.Black roads has no glare during day, but has poor visibility at night

3.Concrete roads has better visibility and less glare

d. Drainage

The pavement surface should be absolutely impermeable to prevent **seepage of water into the pavement** layers. Further, both the geometry and texture of the **pavement surface should help in draining out** the water from the surface in less time.

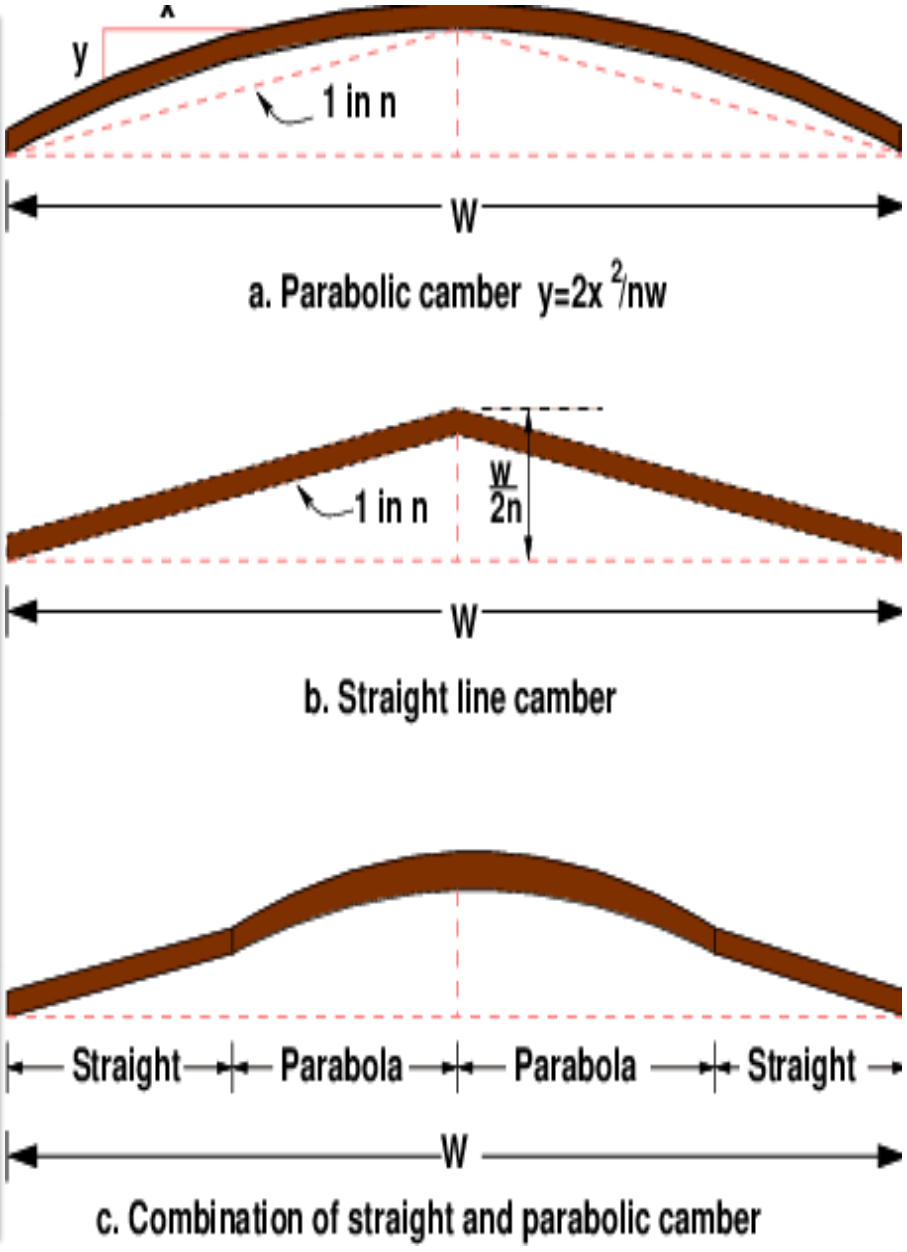
Camber or cant is the **cross slope provided to raise middle of the road surface** in the transverse direction to drain off rain water from road surface.

The objectives of providing camber are:

1. Surface protection especially for gravel and bituminous roads
2. Sub-grade protection by proper drainage
3. Quick drying of pavement which in turn increases safety

- Too steep slope is undesirable for it will erode the surface. Camber is measured in ***1 in n*** or *n%* (Eg. 1 in 50 or 2%) and the value depends on the type of pavement surface.

Type of Camber	Description	Uses & Advantages
Straight Camber	Two straight slopes meeting at the center of the carriageway.	Simple to construct, used in low to moderate rainfall areas.
Parabolic Camber	A continuous curved slope from the center to the edges.	Ensures smooth vehicle movement, used for high-speed roads.
Composite Camber	A combination of straight and parabolic camber.	Used to balance drainage and vehicle comfort, common in highways.
Sloped Camber (One-Sided Camber)	Entire surface is inclined in one direction.	Used on hill roads or where proper drainage is required in a single direction.



Sr.No	Type Of Road Surface	Range of camber in areas of	
		Heavy rainfall	Low rainfall
1	Cement concrete and thick bituminous surface	1 in 50 or 2 %	1 in 60 or 1.7 %
2	Thin bituminous surface	1 in 40 or 2.5%	1 in 50 or 2 %
3	Water bound macadam and gravel pavement	1 in 33 or 3%	1 in 40 or 2.5 %
4	Earth road	1 in 25 or 4%	1 in 33 or 3 %

S.No.	Type of Terrain	Gradient		
		Ruling	Limiting or Max	Exceptional
1.	Plain or Rolling	1 in 30 (3.3%)	1 in 20 (5%)	1 in 15 (6.7%)
2.	Mountainous and steep Terrain with	1 in 20 (5%)	1 in 16.7 (6%)	1 in 14.3 (7%)
3.	Elevation more than 3000 m. Steep terrain up to 3000 m height	1 in 16.7 (6%)	1 in 14.3 (7%)	1 in 12.7 (8%)

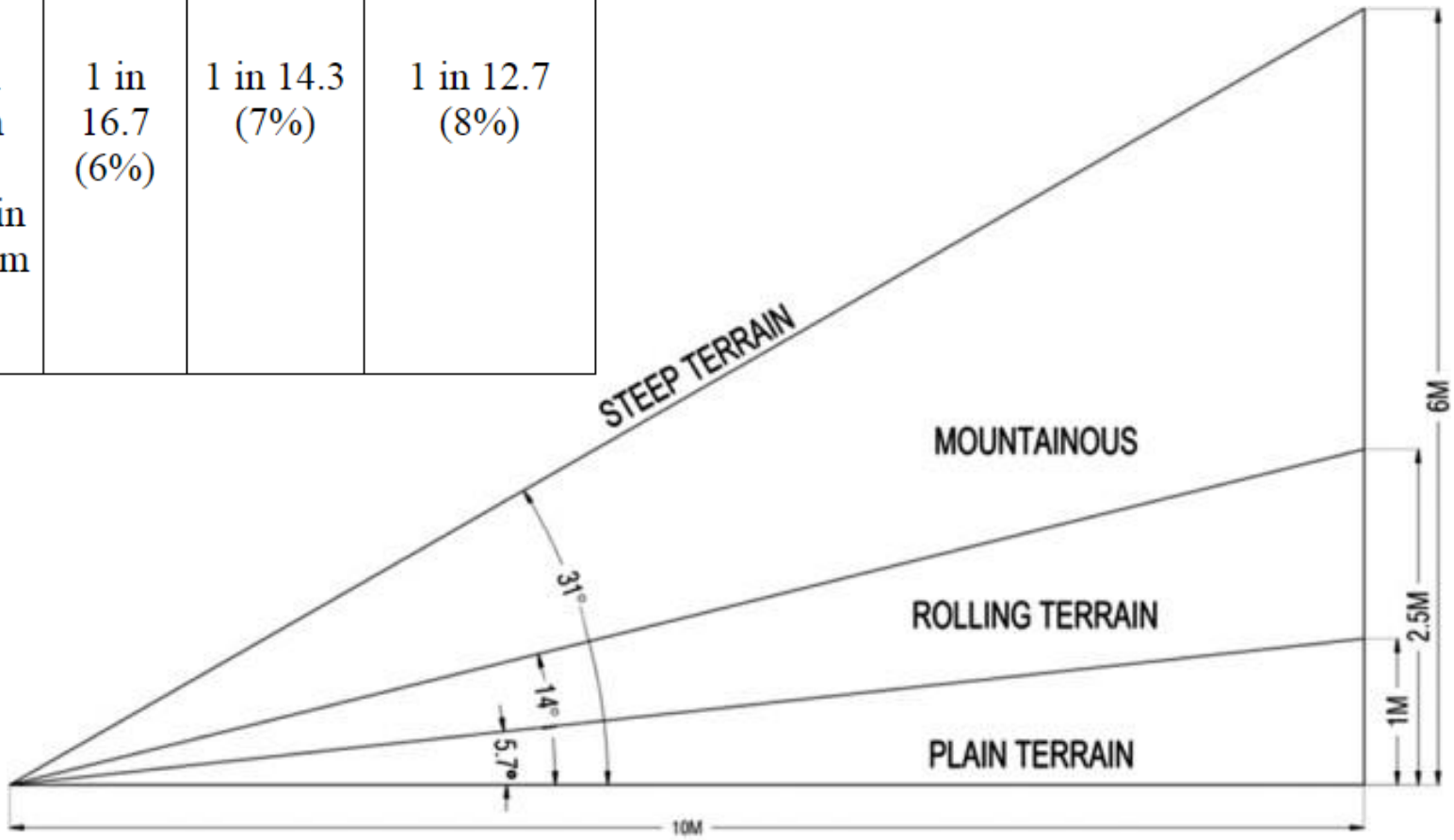
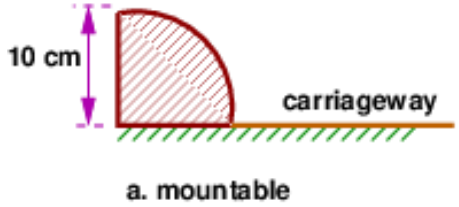


Fig. 6.2 Classification of Terrain

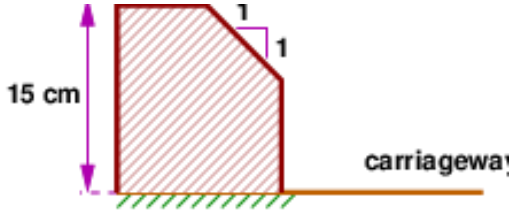
Kerbs

Kerbs indicate the boundary between the carriage way and the shoulder or islands or footpaths.

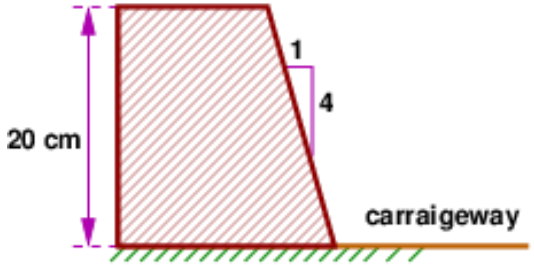
- **Low or mountable kerbs** : This type of kerbs encourage the traffic to remain in the through traffic lanes and also allow the driver to enter the shoulder area with little difficulty. The height of this kerb is about 10 cm above the pavement edge with a slope which allows the vehicle to climb easily.
- **Semi-barrier type kerbs** : When the pedestrian traffic is high, these kerbs are provided. Their height is 15 cm above the pavement edge.
- **Barrier type kerbs** : They are designed to discourage vehicles from leaving the pavement. They are placed at a height of 20 cm above the pavement edge with a steep batter.



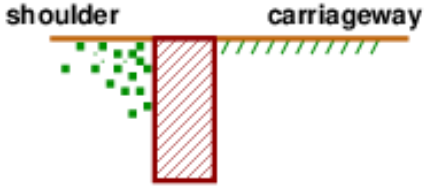
a. mountable



b. semi barrier type



c. barrier type

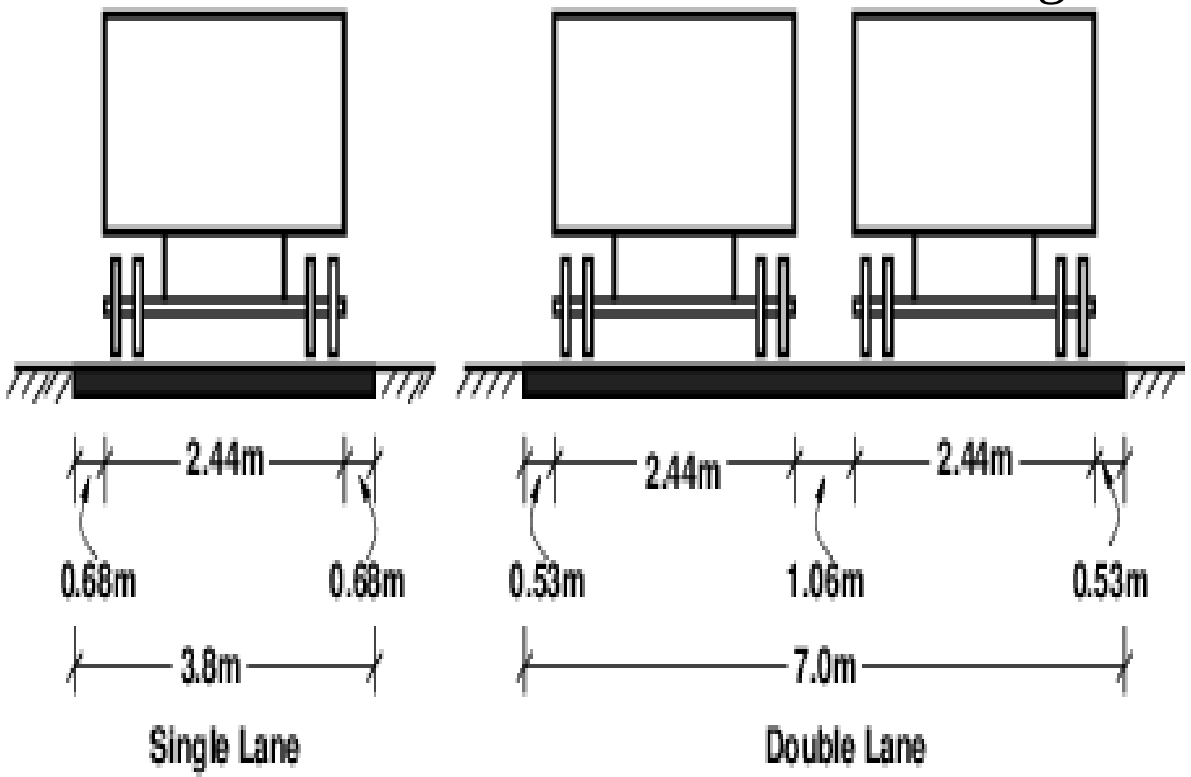


d. submerged

Width of carriage way

Width of the carriage way or the width of the pavement depends on the width of the traffic lane and number of lanes.

- Width of a traffic lane depends on the width of the vehicle and the clearance.
- Side clearance improves operating speed and safety.
- The maximum permissible width of a vehicle is 2.44 and the desirable side clearance for single lane traffic is 0.68 m.
- This require minimum of lane width of 3.75 m for a single lane road.



Type of Road	Number of Lanes	Width of Carriageway (meters)
Single Lane Road	1 Lane	3.75 m
Intermediate Lane Road	1.5 Lanes	5.50 m
Two-Lane Road (Undivided)	2 Lanes	7.00 m
Four-Lane Road (Undivided)	4 Lanes	14.00 m
Four-Lane Road (Divided)	4 Lanes (2 in each direction)	7.00 m (each direction)
Six-Lane Road (Divided)	6 Lanes (3 in each direction)	10.50 m (each direction)
Expressways	4+ Lanes (divided)	3.5 m per lane

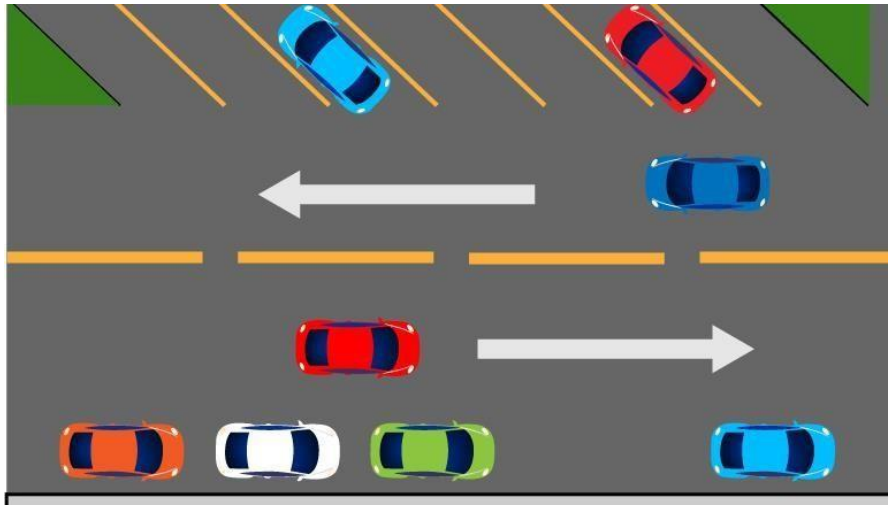
Road margins

The portion of the road **beyond the carriageway** and on the roadway can be generally called road margin.

A. Shoulders



- Shoulders are provided along the road edge and is intended for accommodation of stopped vehicles, serve as an emergency lane for vehicles and provide lateral support for base and surface courses.
- It is desirable to have a width of 4.6 m for the shoulders.
- A minimum width of 2.5 m is recommended for 2-lane rural highways in India.

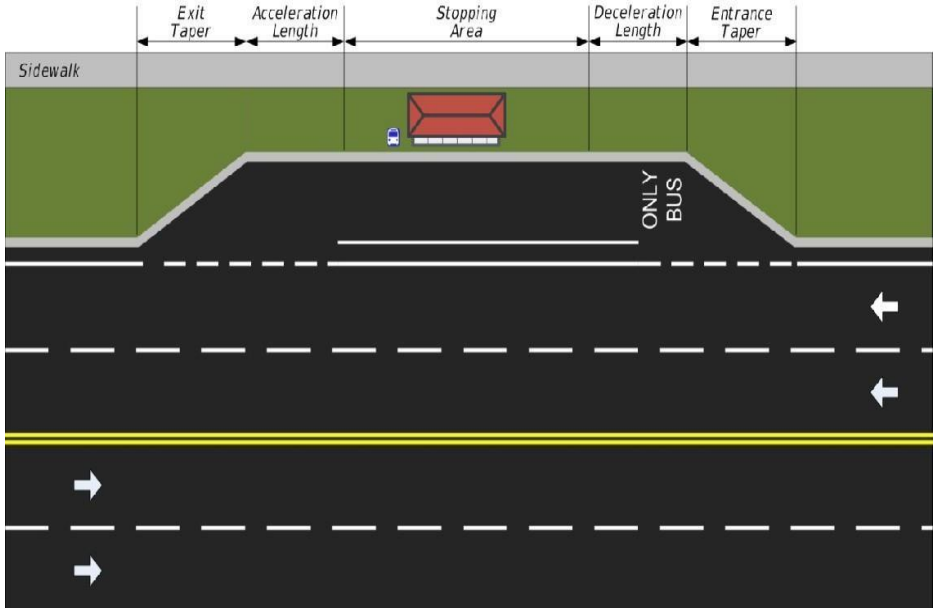


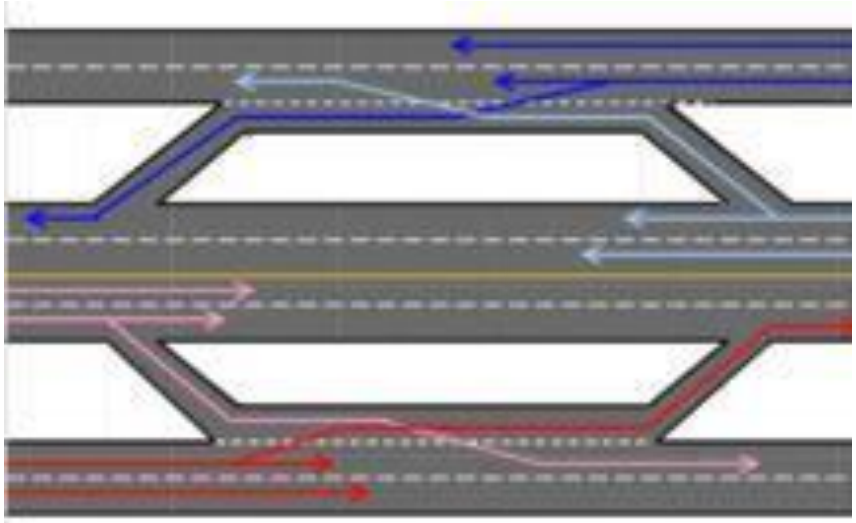
B. Parking lanes

- Parking lanes are provided in urban lanes for side parking.
- Parallel parking is preferred because it is safe for the vehicles moving on the road.
- The parking lane should have a minimum of 3.0 m width in the case of parallel parking.

C. Bus-bays

- Bus bays are provided by recessing the kerbs for bus stops.
  They are provided so that they do not obstruct the movement of vehicles in the carriage way.
- **They should be at least 75 meters away from the intersection** so that the traffic near the intersections is not affected by the bus-bay.





D. Service roads

- Service roads or frontage roads give access to access controlled highways like freeways and expressways.
- They run parallel to the highway and will be usually isolated by a separator and access to the highway will be provided only at selected points.
- These roads are provided to avoid congestion in the expressways and also the speed of the traffic in those lanes is not reduced.

E. Cycle track

- Cycle tracks are provided in urban areas when the volume of cycle traffic is high Minimum width of 2 meter is required, which may be increased by 1 meter for every additional track.





F. Footpath

- Footpaths are exclusive right of way to pedestrians, especially in urban areas.
- They are provided for the safety of the pedestrians when both the pedestrian traffic and vehicular traffic is high.
- Minimum width is 1.5 meter and may be increased based on the traffic.
- The footpath should be either as smooth as the pavement or more smoother than that to induce the pedestrian to use the footpath.

G. Guard rails

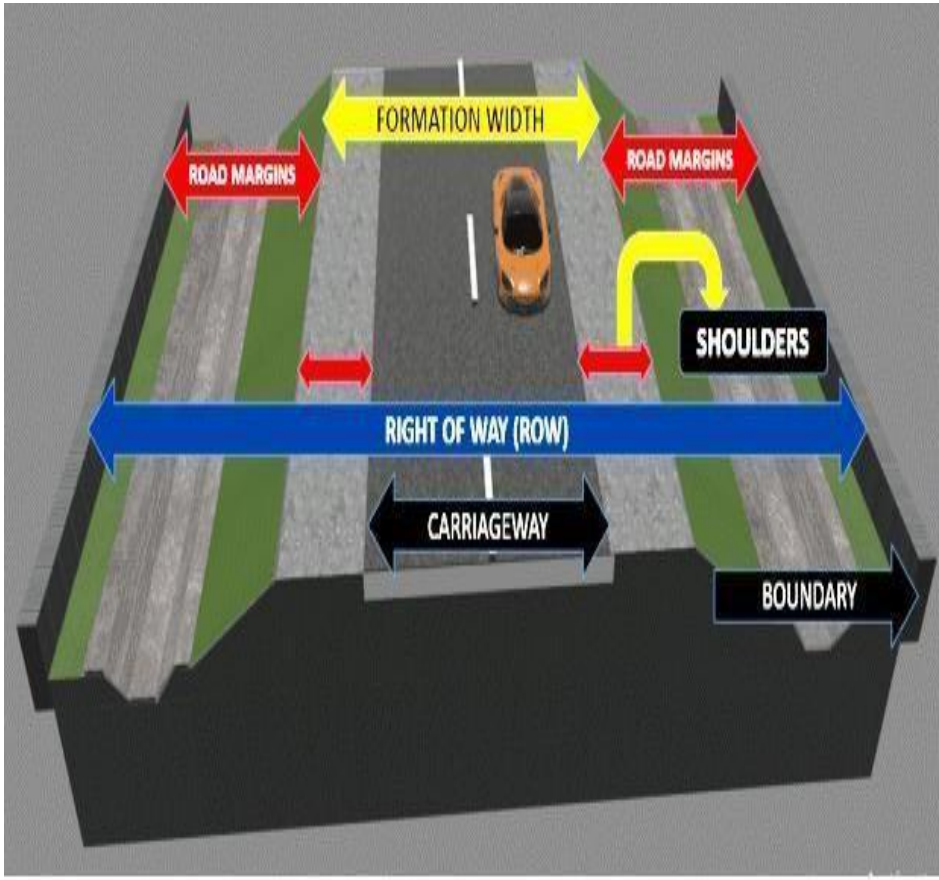
- They are provided at the edge of the shoulder usually when the road is on an embankment.
- They serve to prevent the vehicles from running off the embankment, especially when the height of the fill exceeds 3 m.
- Guard stones painted in alternate black and white are usually used.
- They also give better visibility of curves at night under headlights of vehicles.

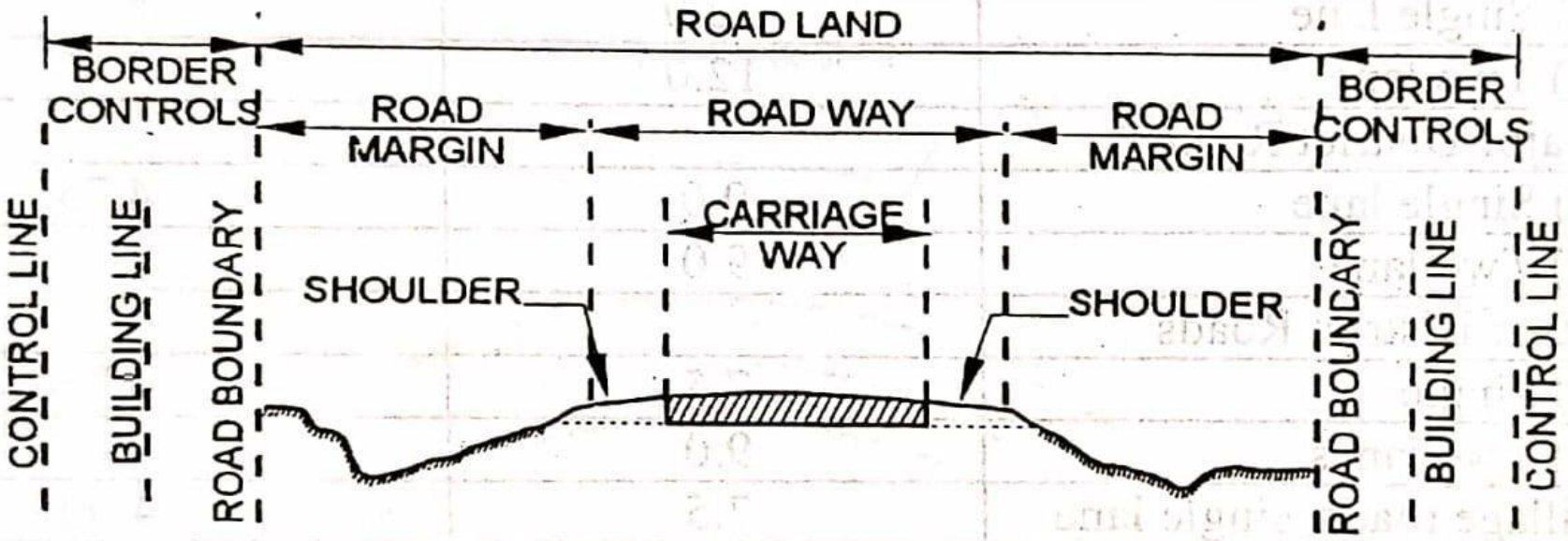


H. Width of formation

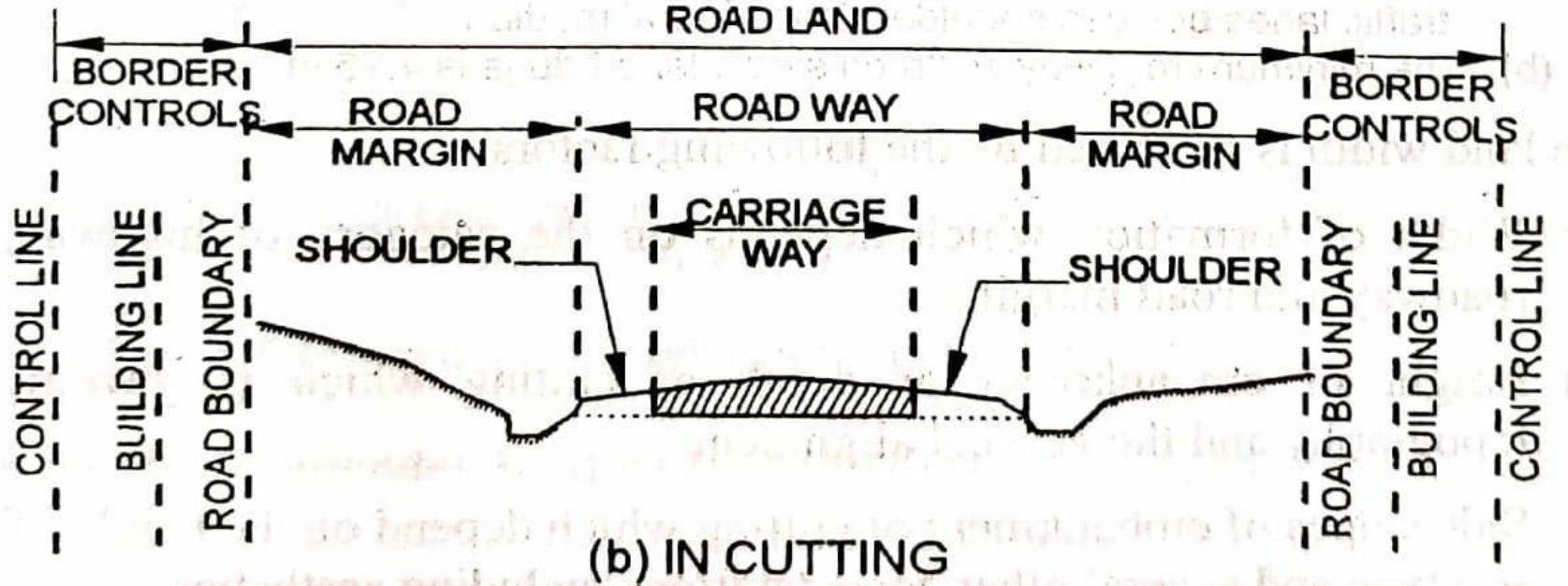
- Width of formation or roadway width is the sum of the widths of pavements or carriage way including separators and shoulders.
- This does not include the extra land information/cutting.
- The values suggested by IRC are given in Table.

Type of road	Plain and rolling terrain	Mountain or steep terrain
• NH and SH		
– Single lane	12 m	6.25 m
– Double lane	12 m	8.80 m
• Major District Road (MDR)		
– Single & double lane	9m	4.75 m
• Other District Road (ODR)		
– Single lane	7.5 m	4.75 m
– Double lane	9m	
• Village road/Single lane	7.5 m	4 m
• Single lane bridge	4.25 m	–





(a) IN EMBANKMENT



(b) IN CUTTING

Fig. 4.4 Cross section details

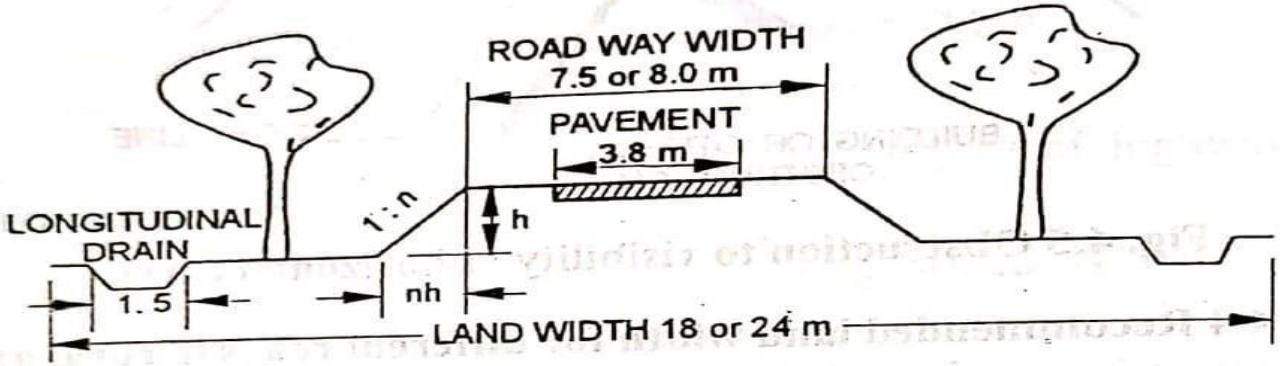


Fig. 4.6 Cross section of VR or ODR in embankment in rural area

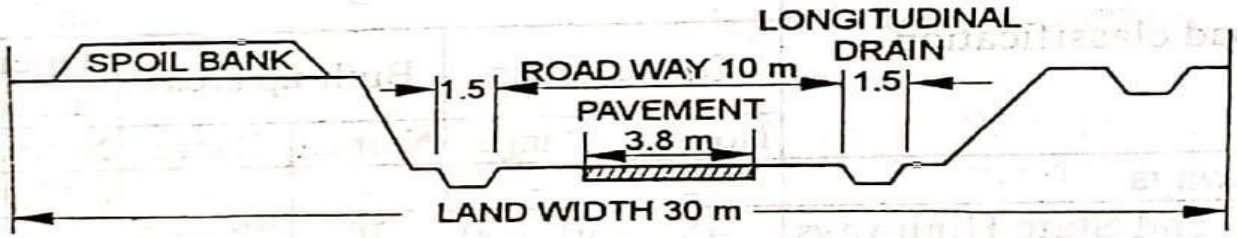


Fig. 4.7 Cross section of MDR in cutting in rural area

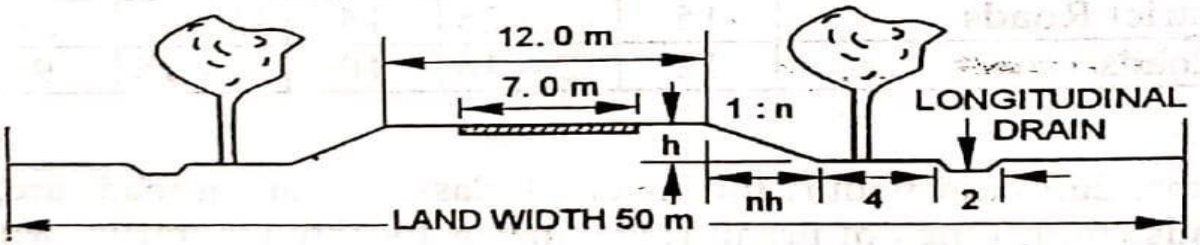


Fig. 4.8 Cross section of NH or SH in rural area



Fig. 4.9 Cross section of two-lane city road in built-up area

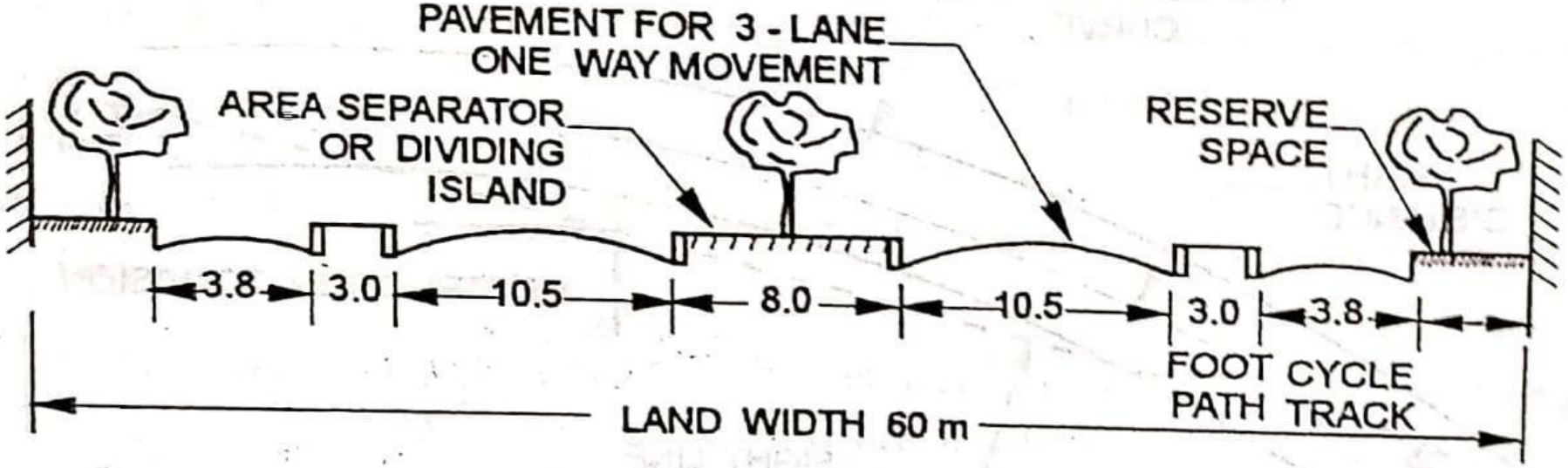


Fig. 4.10 Cross section of divided highway in urban area

Traffic Characteristics:

Road Users Characteristics:

The physical, mental & emotional characteristics of human affect their ability to operate motor vehicle safely or as a pedestrian.

1) **Physical Characteristics:** Permanent & Temporary Characteristics. Permanent characteristics are vision, hearing, strength & general reaction. Temporary Characteristics are fatigue, alcohol or drugs & illness.

2) **Mental Characteristics:** Knowledge, skill, experience, intelligence & literacy. Knowledge of traffic, vehicle, rules, regulation, road condition, etc.

3) **Psychological Characteristics:** Emotional factors such as attentiveness, fear, anger, superstition, impatience, general attitude, etc.

4) **Environmental Characteristics:** Traffic stream characteristics, roadside features, atmospheric condition, locality, etc.

Traffic Characteristics:

Vehicle Characteristics:

Standards for dimensions, weights, turning angle, height of driver seats of vehicle should be consistent with road & facility available. Power, speed, acceleration, deceleration & braking characteristics.

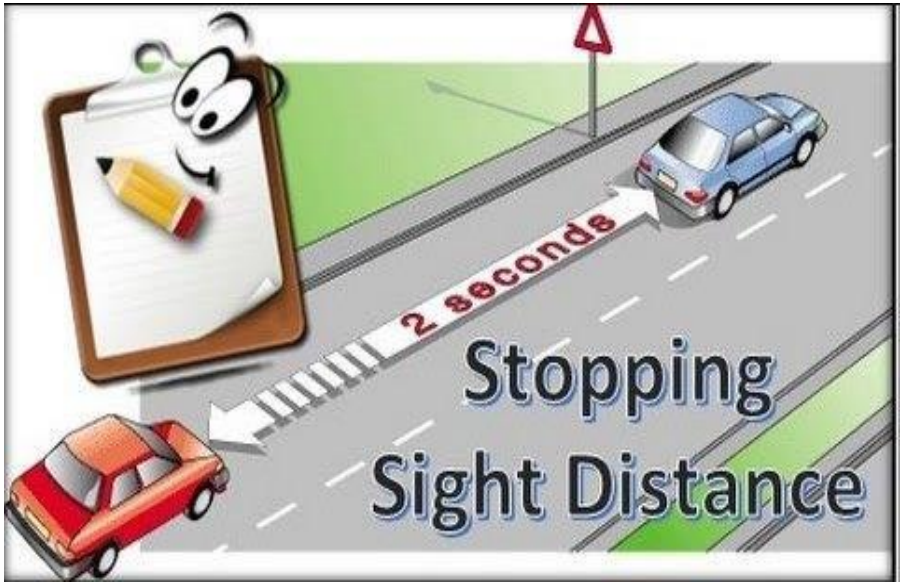
1) **Vehicle Dimension:** Width, length & height of vehicle.
2) **Weight of Loaded Vehicle:** Max. weight of loaded vehicle & the wheels & axles affect the design of pavements thickness & gradients.
3) **Power of Vehicle:** The power of heavies vehicle & their loaded weights govern the permissible & limiting values of gradient on roads.
4) **Speed of Vehicle:** Speed affects sight distance, super-elevation, length of transition curve, radius of curve, design gradient, capacity of traffic lane, design & control measure at intersection.
5) **Braking Characteristics:** types of braking system (Mechanical, air brake, fluid Brake, etc.)

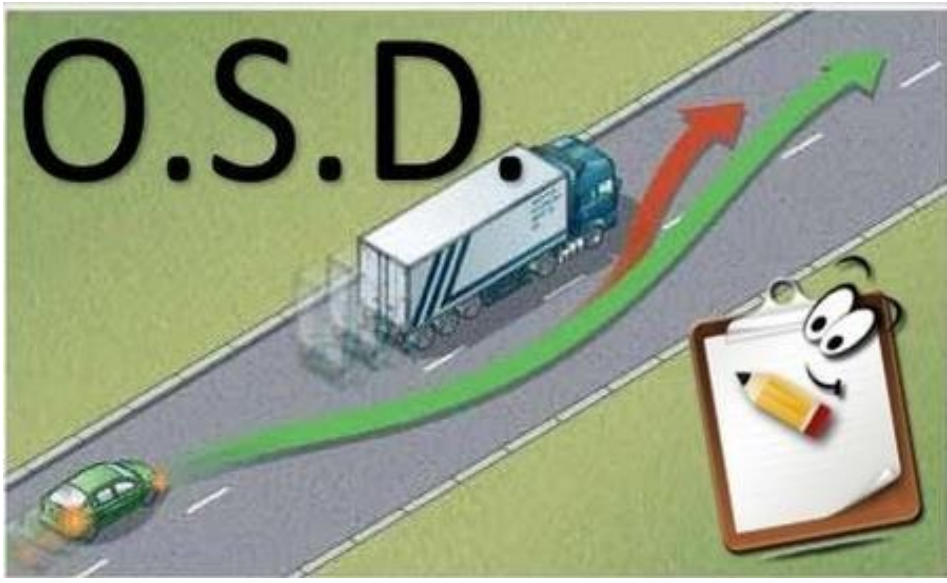
Sight Distance:

- Sight distance is the length of a road visible ahead to the driver at any instances.
- Sight distance required by drivers applies to both geometric design of highway & for traffic control.

Types of sight distance:

1) Stopping Sight Distance
2) Overtaking Sight Distance
3) Sight Distance for Entering into Uncontrolled Intersections.





Computation of Sight Distance

Driver's Reaction Time

- The time from the moment the driver observes the obstacle in front of him to the moment he applies brake is called as the reaction time of the driver.
- Based on the PIEV theory, the reaction time can be divided into 4 components.
- All these times will be combined to form a total perception reaction time while undergoing actual practice and design of highway.
- In India, as per IRC, the reaction time is set to be 2.5 seconds.

Speed of the Vehicle

- The speed of the vehicle clearly affects the sight distance.
- If the speed employed is high, the time required to stop the vehicle will be high.
- This means that with the increase in the speed there is increase in the sight distance.

Gradient of Road

- The sight distance is greatly influenced by the gradient of the road.
- There are chances for the vehicles to stop suddenly when the vehicle is climbing suddenly.
- This situation will ask for a small sight distance.
- When the vehicle is moving down, the movement is supported by the gravity action.
- This will hence ask for more time to stop the vehicle.

Friction between the Road and the Tire

- The stopping of a vehicle is also dependent on the frictional resistance between the tire and the road.
- Having a higher value of frictional resistance will result in efficient stopping of the vehicle when applied.
- Here, the sight distance required will be less.
- During the computation of sight distance, no special provision or consideration is given for the brake efficiency.
- In India, the value of longitudinal friction is between 0.35 and 0.4. This is as per the Indian Road Congress.

Brake Efficiency

- Many factors like the age of the vehicle, the characteristics of the vehicle will affect the brake efficiency of the vehicle.
- An efficiency of 100% implies that the vehicle will stop at the moment the brake is applied.
- Obtaining 100% is not practicable.
- This is an ideal condition of the vehicle.
- This means, for a lower value of brake efficiency, it is necessary to obtain a higher value of sight distance.
- In the process of determining a safe geometric design, it is required to assume 50% brake efficiency.

Stopping Sight Distance (SSD) :

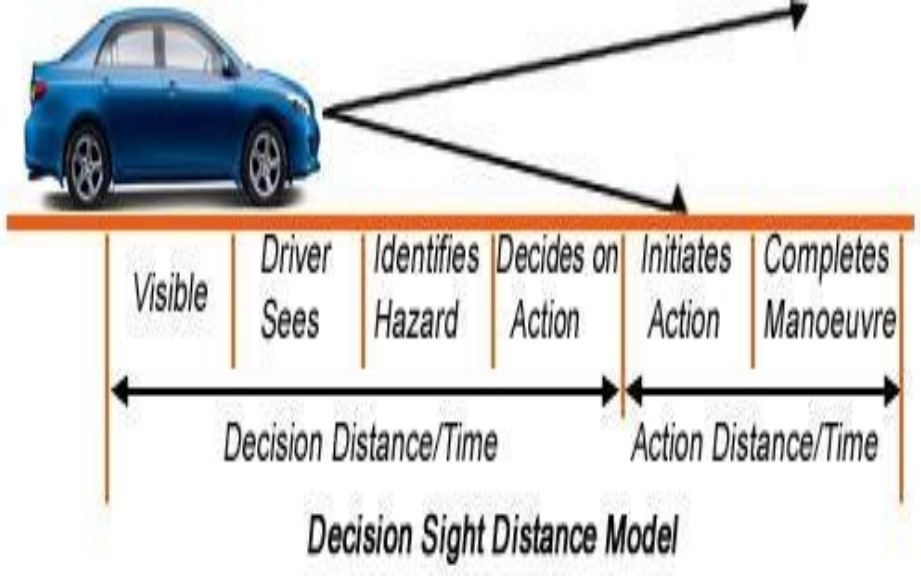
- The **minimum clear distance ahead** required by the driver of vehicle, travelling at a given speed, to bring his vehicle to a stop, after an object on the carriage way become visible is called stopping sight distance. (without collision with any other obstruction)

- Sight distance available to a driver travelling on a road at any instance depends on the following factors:

    1) Feature of road ahead
    2) Height of the driver's eye above the road surface
    3) Height of the object above the road surface.

- IRC suggested the height of eye level of driver as 1.2m & the object as 0.15m above the road surface.

Stopping Sight Distance (SSD) :



Stopping Sight Distance depends on:

1) Total reaction time of the driver
2) Speed of vehicle
3) Efficiency of brakes
4) Gradient of road.
5) Frictional resistance between road & tyres.

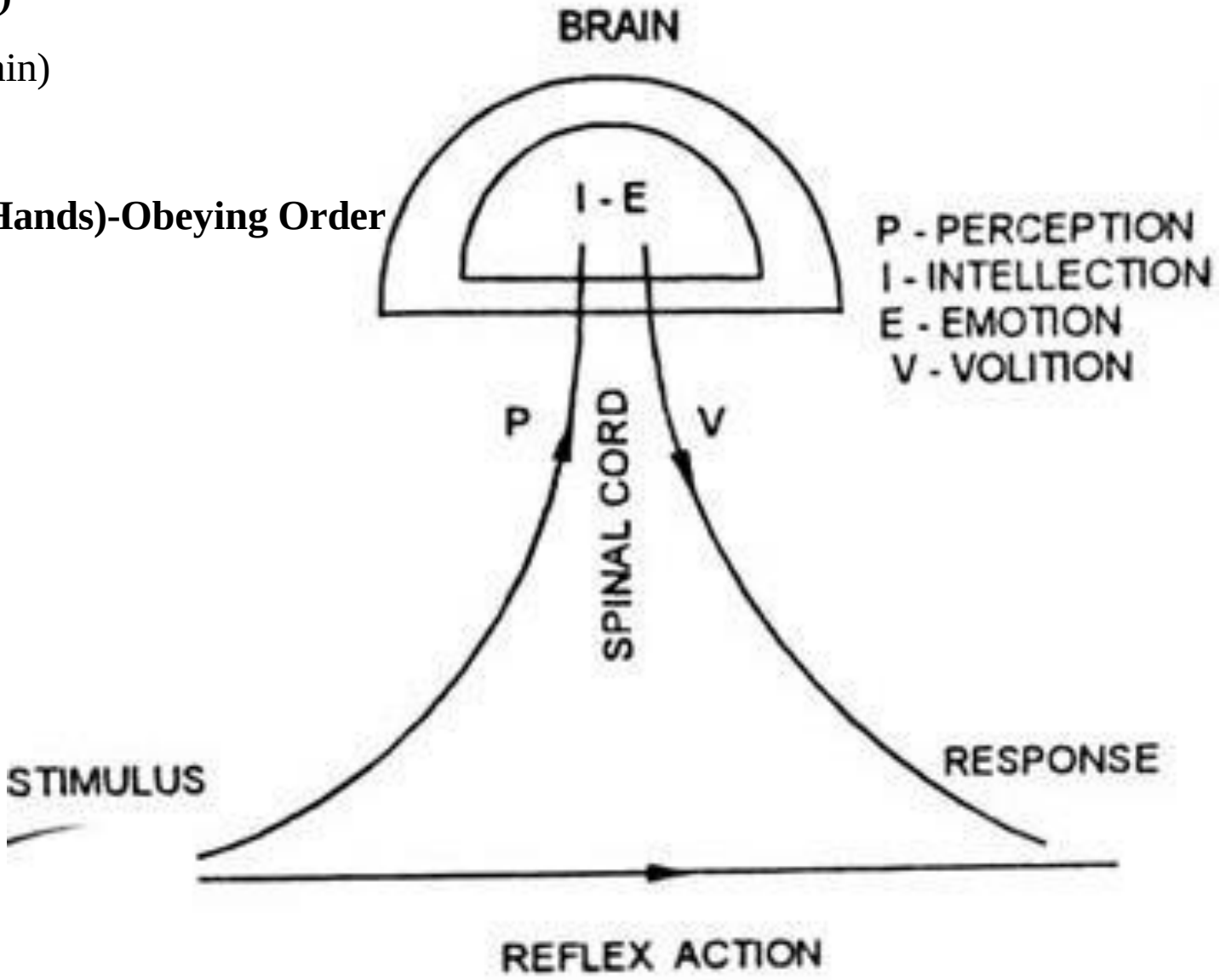
Total Reaction Time of the driver:

- Reaction time of the driver is the time taken from the instant the object is visible to the driver to the instant the brakes are effectively applied.
- Stopping distances increases with increase in reaction time of the driver.
- **Total Reaction Time** may be split into 1) Perception Time & 2) Brake Reaction Time
- **Perception Time** is the time required for driver to realize that brakes must be applied.
- **Brake Reaction Time** depends on skill of driver, type of problem & other environmental conditions.

PIEV Theory :

According to PIEV theory Reaction time of driver splits into 4 types:

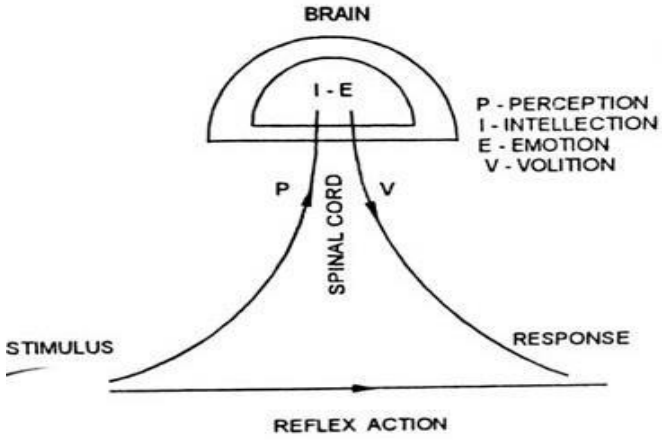
1) **Perception: (eyes)**
2) **Intellection:** (Brain)
3) **Emotion:** (Brain)
4) **Volition: (Legs/Hands)-Obeying Order**



PIEV Theory :

According to PIEV theory Reaction time of driver splits into 4 types:

1) **Perception:** Time required for sensations received by eyes or ears of the driver to be transmitted to the brain through the nervous system & spiral cord. Time required to perceive an situation.
2) **Intellection:** Time require to understand the situation.
3) **Emotion:** Time elapsed during emotional sensations & other mental disturbance such as fear, anger, superstitions, etc. Emotion time varies for different drivers but even for a particular driver the emotion time is likely to vary considerably depending upon the situation.
4) **Volition:** Time taken by the driver for the final action such as brake application.



Speed of Vehicle:

Higher the speed, higher will be the stopping distance.

Efficiency of Brakes:

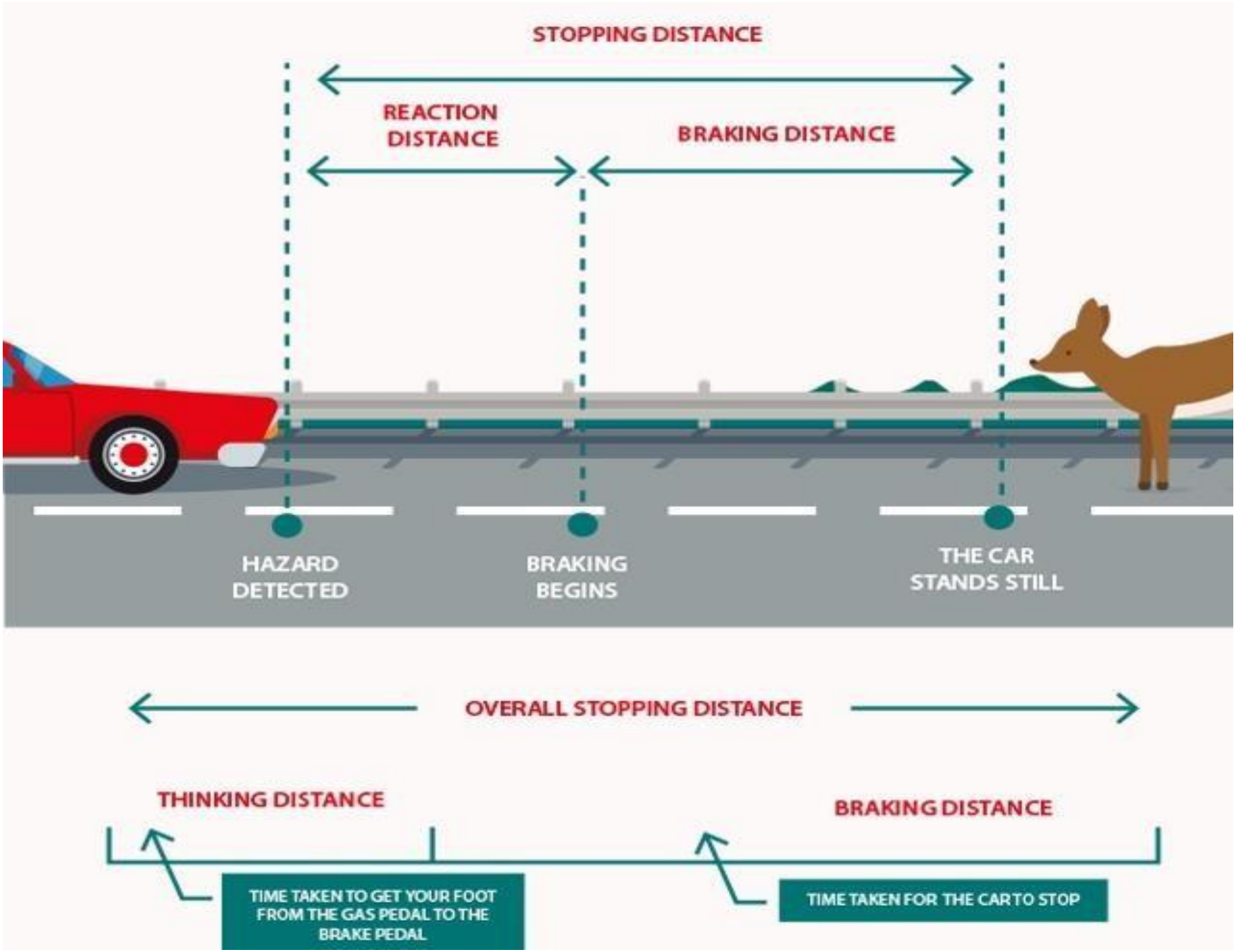
100 % braking efficiency will result in 100% skidding which is undesirable & dangerous. To avoid skid, the braking forces should not exceed the frictional forces.

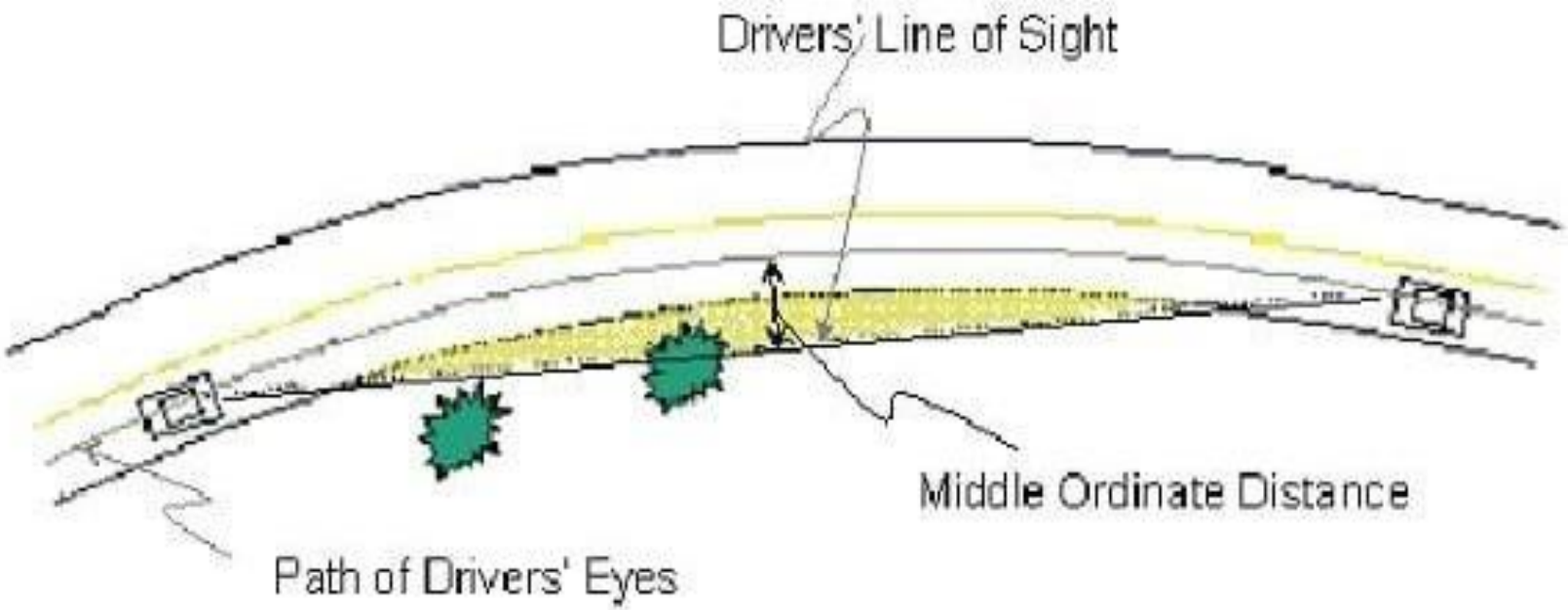
Frictional Resistance between Road & Tyres:

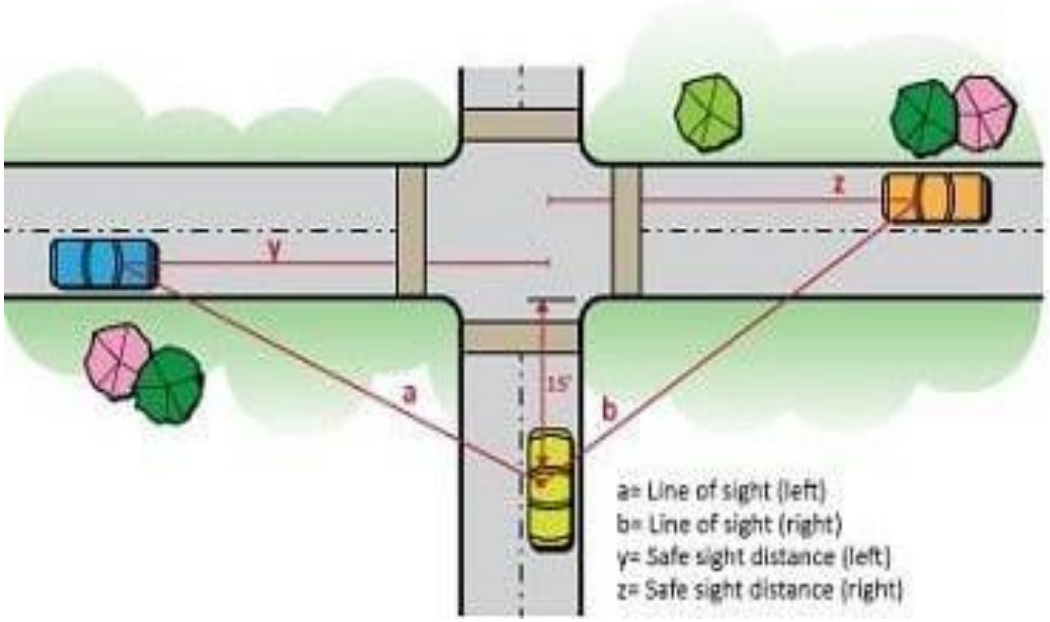
The IRC has specified design friction coefficient values in the range **0.35 to 0.40.**

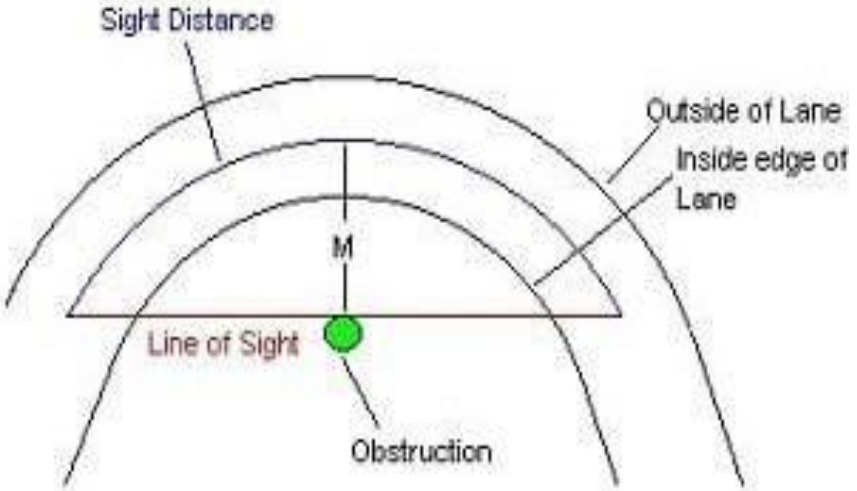
Gradient of Road:

More gradient downward or downgrade more stopping distance & more gradient upwards or upgrade less stopping distance.


STOPPING DISTANCE
REACTION DISTANCE
BRAKING DISTANCE
HAZARD DETECTED
BRAKING BEGINS
THE CAR STANDS STILL
OVERALL STOPPING DISTANCE
THINKING DISTANCE
BRAKING DISTANCE
TIME TAKEN TO GET YOUR FOOT FROM THE GAS PEDAL TO THE BRAKE PEDAL
TIME TAKEN FOR THE CAR TO STOP


Drivers' Line of Sight
Middle Ordinate Distance
Path of Drivers' Eyes


z
y
a
15'
b
a= Line of sight (left)
b= Line of sight (right)
y= Safe sight distance (left)
z= Safe sight distance (right)


Sight Distance
Outside of Lane
Inside edge of Lane
M
Line of Sight
Obstruction

Stopping Sight Distance (SSD): (Analysis of Stopping Sight Distance)

Stopping distance is the sum of lag distance & braking distance.

Lag Distance:

Distance travelled by the vehicle at uniform speed during the total reaction time, t

Lag distance = 0.278 * V * t

v is the design speed, kmph

t is the total reaction time in seconds = 2.5 seconds

Lag distance = 0.7 * V

Braking Distance:

Distance travelled after the brakes till the vehicle comes to a halt.

Braking distance = (V^2 / (254 * f))

f is design coefficient of friction = 0.4 to 0.35 (as per speed)

Braking distance = (V^2 / (254 * (f $\pm$ 0.01 * G)))

G is gradient. + ve sign for upgrades & -ve signs for downgrades

(1)

Que. Calculate the safe stopping sight distance on a level road stretch for design speed of 50 kmph for (a) two way traffic on a two lane road (b) two way traffic on a single lane road.

Assume coeff. of friction = 0.37

reaction time of drivers = 2.5 seconds.

Soln :- $V = 50$ kmph $\quad f = 0.37 \quad t = 2.5$ sec.

$g = 9.8 \text{ m/s}^2$

SSD on level road stretch $= (0.278 \times V \times t) + \frac{V^2}{254 \times f}$

$$SSD = (0.278 \times 50 \times 2.5) + \frac{50^2}{254 \times 0.37}$$

$$\boxed{SSD = 61.4 \text{ m}}$$

(a) SSD, for two way, two lane road = 61.4 m

(b) SSD, for two way, one lane road = $2 \times SSD$

$= 2 \times 61.4$

$= 122.8$ m.

(2)

Que (2) Calculate the SSD required to avoid a head on collision of two cars approaching from the opposite directions at 90 & 60 kmph. Take $t = 2.5$ seconds, coeff. of friction $= 0.7$ & brake efficiency $= 50\%$.

Sol^n :- Effective friction coeff. $= 0.5 \times 0.7 = 0.35$

$$SSD_1 \text{ of car-1} = (0.278 \times V \times t) + \frac{V^2}{254 \times f}$$

$$= (0.278 \times 90 \times 2.5) + \frac{90^2}{254 \times 0.35}$$

$$\boxed{SSD_1 = 153.6 \text{ m}}$$

$$SSD_2 \text{ of car-2} = (0.278 \times 60 \times 2.5) + \frac{60^2}{254 \times 0.35}$$

$$= 82.2 \text{ m}$$

∴ Total sight distance required to avoid head on collision of the two approaching cars :-

$$= SSD_1 + SSD_2$$

$$= 153.6 + 82.2$$

$$\boxed{SSD = 235.8 \text{ m}}$$

Que. ③ Calculate the SSD on highway at a descending gradient of 2% for a design speed of 80 kmph. Assume other data as per IRC recommendations.

Soln :- $t = 2.5$ seconds $\quad f = 0.35$

$V = 80$ kmph $\quad g = 9.8\ m/s^2$

Gradient = 2% (-ve) descending.

$$S.S.D. = (0.278 V \times t) + \frac{V^2}{254(f \pm G)}$$

$$= (0.278 \times 80 \times 2.5) + \frac{80^2}{254(0.35 - 0.02)}$$

$$\boxed{SSD = 132\ m}$$

Que ④ The design speed of a road is 65 kmph, the friction coeff. is 0.36 and reaction time of driver is 2.5 seconds. Calculate the values of @ Head Light sight Distance.
ⓑ Intermediate sight distance required for the road.

Sol^n :- $V = 65$ kmph $\qquad f = 0.36$

Reaction time, $t = 2.5$ seconds.

$\therefore$ Stopping Sight Distance, $(SSD) = (0.278 \times V \times t) + \frac{V^2}{254 \times f}$

$= (0.7 \times V) + \frac{V^2}{254 \times f}$

$= (0.7 \times 65) + \frac{65^2}{254 \times 0.36}$

$\boxed{SSD = 91.4 \text{ m}}$

@ Head Light Sight Distance = SSD = 91.4 m

ⓑ Intermediate Sight Distance = $(2 \times SSD)$

$= 2 \times 91.4$

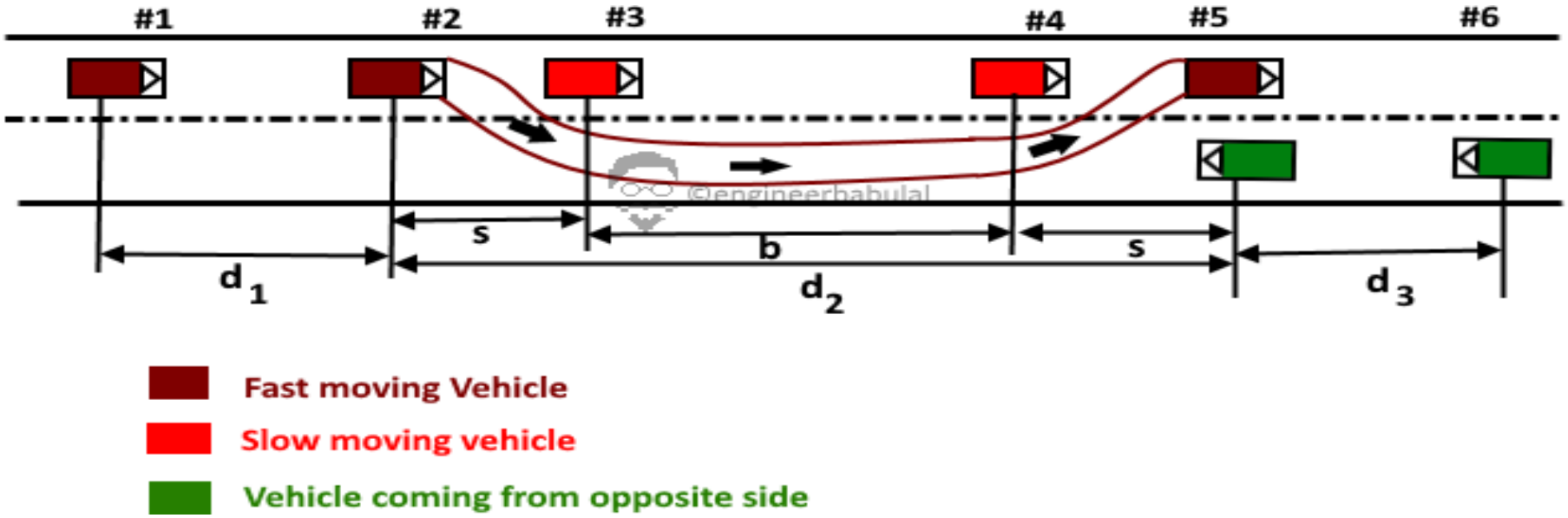
$= 182.8$ m

Overtaking Sight Distance (OSD) : Safe Passing Sight Distance

1) The minimum distance required for overtaking another vehicle safely & comfortably without interfering the speed of oncoming vehicle, travelling at the design speed is called overtaking sight distance.

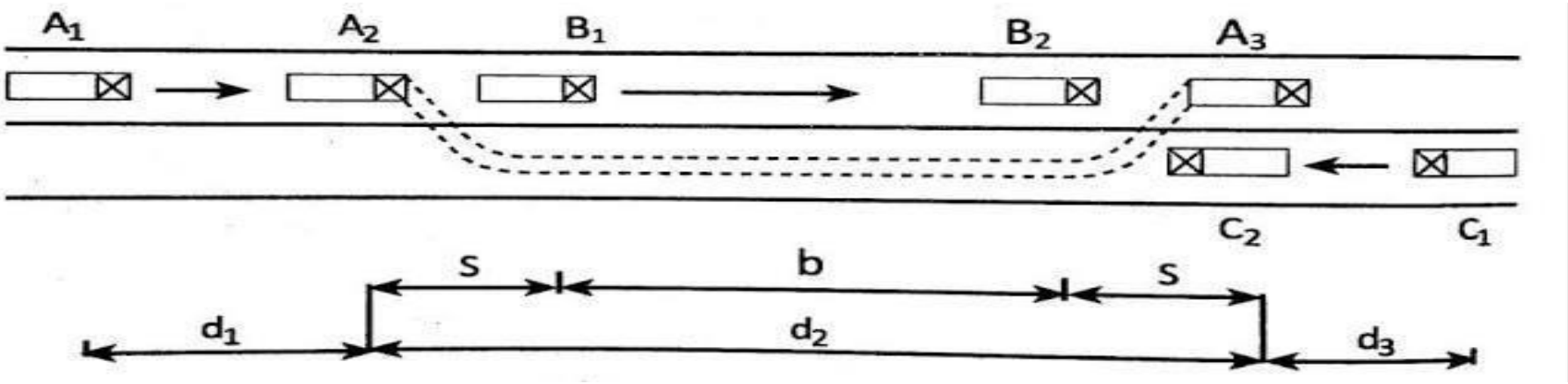
2) If all the vehicle travel along a road at the design speed, then there should be no need for any overtaking.

3) Speed, gradient of road, rate of acceleration of overtaking vehicle & skill of the driver affect the OSD.



Overtaking Sight Distance (OSD) : Safe Passing Sight Distance

1) Fast moving vehicle ‘A’ desires to overtake another slower vehicle ‘B’ . Vehicle ‘A’ has to accelerate, shift to adjacent right side lane, complete the overtaking manoeuvre & return to the left lane, before the on-coming vehicle ‘C’ approaches. As shown in figure.



2) d_1 is the distance (m) travelled by overtaking vehicle during the reaction time ‘t’ (sec) of the driver, from position A_1 to A_2 before starting to overtake the slow vehicle ‘B’

3) d_2 is the distance (m) travelled by overtaking vehicle during the actual overtaking operation during T (sec) from position A_2 to A_3.

4) d_3 is the distance (m) travelled by on-coming vehicle C during the actual overtaking operation of ‘A’ during T (sec) from position C_1 to C_2.

5) **OSD = $d_1 + d_2 + d_3$ (m)**

Overtaking Sight Distance (OSD) : Safe Passing Sight Distance

1) d_1 is the distance (m) travelled by overtaking vehicle during the reaction time 't' (sec) of the driver, from position A_1 to A_2 before starting to overtake the slow vehicle 'B'
2) d_2 is the distance (m) travelled by overtaking vehicle during the actual overtaking operation during T (sec) from position A_2 to A_3.
3) d_3 is the distance (m) travelled by on-coming vehicle C during the actual overtaking operation of 'A' during T (sec) from position C_1 to C_2.

$$\mathbf{OSD = d_1 + d_2 + d_3 \quad (m)}$$

$$\mathbf{OSD = (0.28 * V_b * t) + (0.28 * V_b * T) + (2 * s) + (0.28 * V * T)}$$

V_b = initial speed of overtaking vehicle, kmph

t = reaction time of driver = 2 sec.

V = speed of overtaking vehicle or design speed, kmph.

s = spacing of vehicles = (0.7 * Vb + 6)

A = average acceleration during overtaking, kmph/sec

$T = \sqrt{((14.4 * s) / A)}$

* Overtaking Sight Distance *

$$OSD = d_1 + d_2 + d_3$$

$$d_1 = 0.28 \times V_b \times t$$

$$d_2 = (0.28 \times V_b \times T) + (2 \times 3)$$

$$d_3 = 0.28 \times V \times T$$

$$\therefore \boxed{OSD = (0.28 \times V_b \times t) + (0.28 \times V_b \times T) + (2 \times 3) + (0.28 \times V \times T)}$$

Where, V_b = Initial speed of overtaking vehicle, kmph.

t = Reaction time of driver = 2 seconds.

V = speed of overtaking vehicle, kmph.

S = spacing of vehicle = $\boxed{(0.2 \times V_b) + 6 = S}$ in meter.

A = Average acceleration during overtaking kmph/second (Table in Book)

$$\boxed{T = \sqrt{\frac{14.4 \times S}{A}}}$$ in seconds.

Note :- If V_b is not given then assume it 16 kmph less than design speed of road i.e. $\boxed{V_b = V - 16}$

→ OSD for one way traffic = $d_1 + d_2$

→ OSD for two way traffic = $d_1 + d_2 + d_3$

Que. ⑤ The speeds of overtaking & overtaken vehicles are 70 & 40 kmph, respectively on a two way traffic road. The average acceleration during overtaking may be assumed as 0.99 m/sec². Calculate

@ Calculate safe overtaking sight distance?

ⓑ What is the minimum length of overtaking zone?

ⓒ Draw a neat sketch of overtaking zone.

Sol^n :- Overtaking vehicle speed, $V = 70$ kmph.

Overtaken vehicle speed, $V_b = 40$ kmph.

Avg. Acceleration, $A = 0.99 \frac{m}{s^2} \Rightarrow 0.99 \times 10^{-3} \times 60 \times 60$

$\therefore A = 3.564$ kmph/seconds.

$$O.S.D. = (0.28 \times V_b \times t) + (0.28 \times V_b \times T) + (2 \times 9) + (0.28 \times V \times T)$$

Take $t = 2$ seconds.

$$s = (0.2 \times V_b) + 6 \Rightarrow (0.2 \times 40) + 6 \Rightarrow 14$$

$$T = \sqrt{\frac{14.4 \times 9}{A}} \Rightarrow \sqrt{\frac{14.4 \times 14}{3.564}} \Rightarrow 7.52$$

$$\therefore OSD = (0.28 \times 40 \times 2) + (0.28 \times 40 \times 7.52) + (2 \times 14) + (0.28 \times 70 \times 7.52)$$

$$\boxed{OSD = 282.02 \text{ m}}$$

@ Safe overtaking sight distance $= 282.02$ m

(b) Minimum length of overtaking zone

$= 3 \times OSD$

$= 3 \times 282.02$

$= 846.06$ m

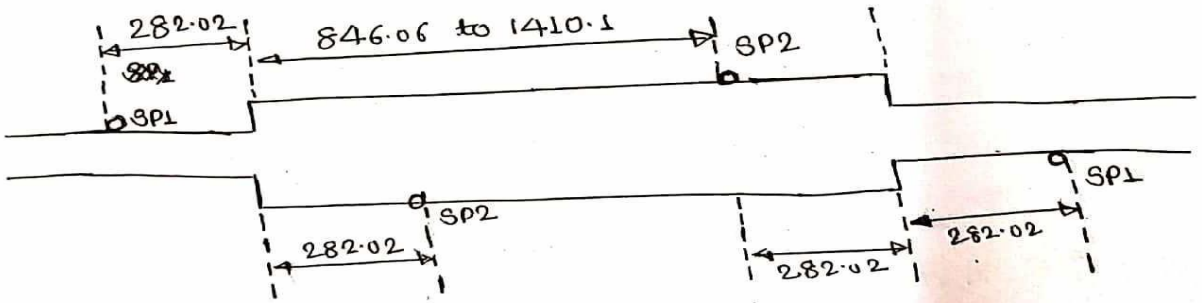
Desirable length of overtaking zone

$= 5 \times OSD$

$= 5 \times 282.02$

$= 1410.1$ m

(c) Overtaking zone



SP1 → Sign Post overtaking sign ahead

SP2 → Sign Post End of overtaking zone

Que. ⑥ Calculate the safe overtaking sight distance for a design speed of 96 kmph. Assume all other data suitably.

Sol^n :—

$V = 96$ kmph.

$V_b = V - 16 \Rightarrow 96 - 16 = 80$ kmph.

Reaction Time, $t = 2$ seconds.

Acceleration, $A = 2.5$ kmph/sec. ← from Table

$$S = (0.2 \times V_b) + 6 \Rightarrow (0.2 \times 80) + 6 \Rightarrow 22 \text{ m}$$

$$T = \sqrt{\frac{14.4 \times S}{A}} \Rightarrow \sqrt{\frac{14.4 \times 22}{2.5}} \Rightarrow 11.3 \text{ sec.}$$

$$d_1 = 0.28 \times V_b \times t \Rightarrow 0.28 \times 80 \times 2 \Rightarrow 44.8 \text{ m}$$

$$d_2 = (0.28 \times V_b \times T) + (2 \times S) \Rightarrow (0.28 \times 80 \times 11.3) + (2 \times 22)$$

$$d_2 = 297 \text{ m}$$

$$d_3 = 0.28 \times V \times T \Rightarrow 0.28 \times 96 \times 11.3 \Rightarrow 303.7 \text{ m}$$

OSD on one way traffic road $= d_1 + d_2 \Rightarrow 44.8 + 297$

$= 341.8 \approx 342$ m

OSD on two way traffic road $= d_1 + d_2 + d_3$

$= 645.5 \approx 646$ m

Road Alignment: Highway Alignment

1) The position occupied by the center line of a road in plan is called road alignment.

2) The position or layout of the center line of highway on the ground is called the alignment.

3) Highway alignment includes both horizontal & vertical alignment of a roadway.

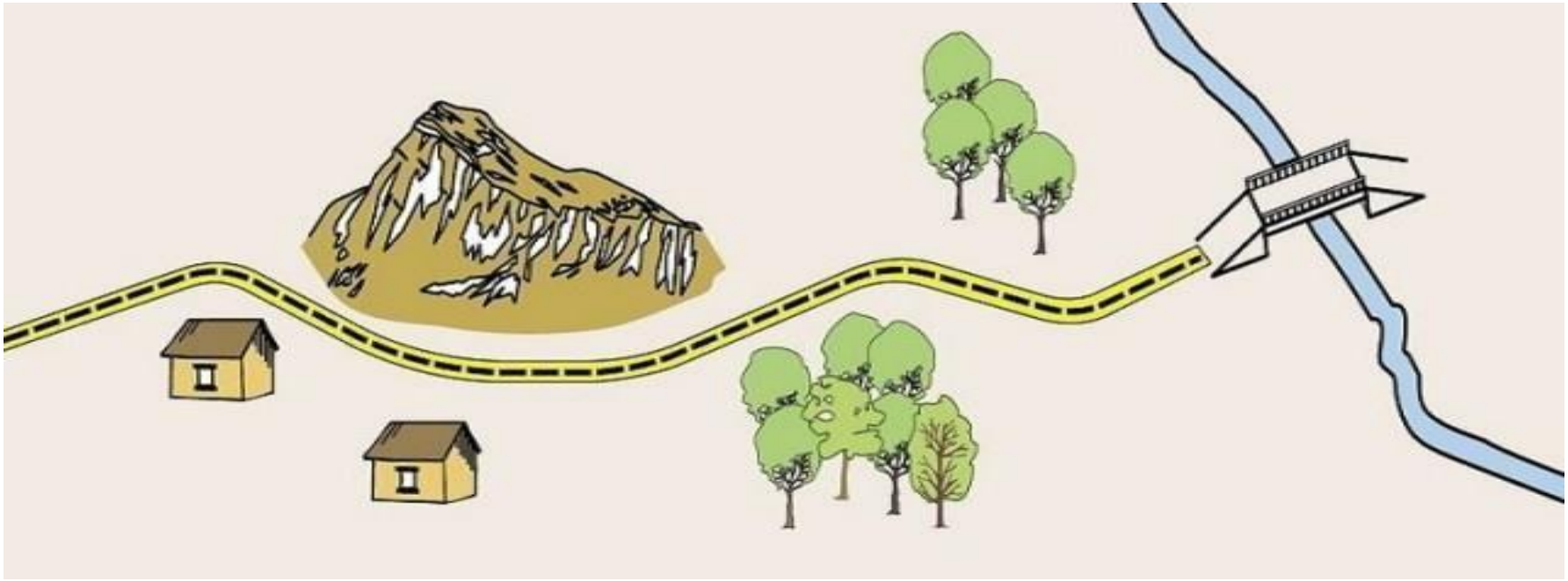
4) The **horizontal alignment** includes the straight path, the deviations and horizontal curves.

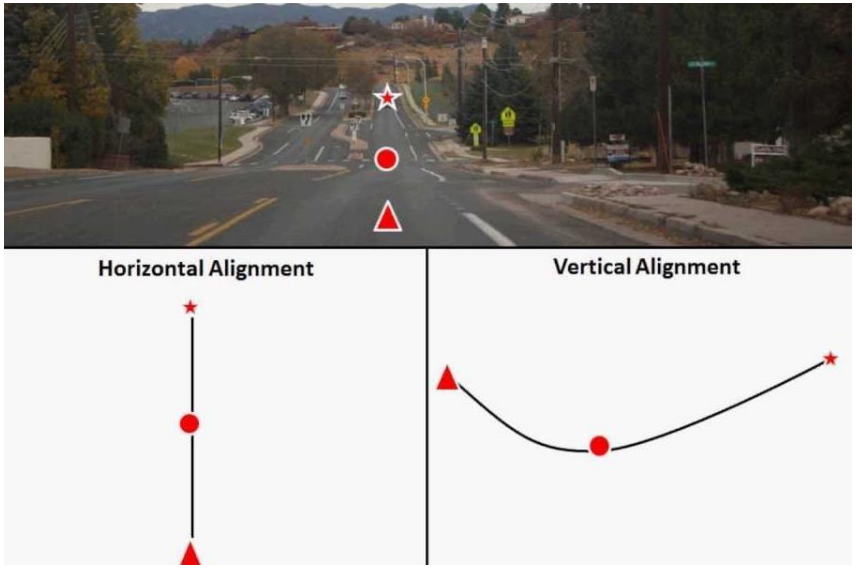
5) Changes in gradient & vertical curves are covered under **vertical alignment** of roads.

6) Improper alignment leads to more **construction cost, maintenance cost, vehicle operation cost & accident rate.**

7) Basic requirements of alignment are **short, easy, safe & economical.**

Highway Alignment







Factors Controlling Highway Alignment:

1) **Obligatory Points:** obligatory points through which the road **alignment has to pass** are generally due to the topographic & other site condition including natural obstruction. **Mountain pass, Bridge to cross a river**, etc. There are obligatory points through which the road should **not pass** like **religious places, costly structure, unsuitable land,** etc.

2) **Traffic:** Road alignment should be decided based on the **requirements of road traffic. Traffic flow, Traffic volume, Future trends, etc.**

3) **Geometric Design: Gradient, radius of curve, & sight distance** also would govern the final alignment of road..

4) **Economics:** While trying to decreasing the **initial construction cost**, either the future road **maintenance cost** or **vehicle operation cost** or both of these may increase considerably. Therefore, economic analysis should be done properly based on life cycle cost.

5) **Other Consideration:** others factors such as **horizontal, vertical alignment, drainage consideration, hydrological factors, political consideration & monotony.**

Survey to be conducted for finalizing the Alignment:

1) **Topographic Map Study:** The process of making several possible alternative alignments of a road on the topographical map of the area is known as map study.

2) **Reconnaissance Survey:** A general examination of a fairly broad stretch of land between the terminal stations in the field, along the proposed alternative alignment marked on the map is known as reconnaissance survey. Engineer visit the site & examine the general characteristics of area. Rapid & rough survey between terminal station.

3) **Preliminary Survey:** The art of finding the details of alternative alignment found suitable during the reconnaissance survey is known as preliminary survey. Field to office work.

4) **Detailed Survey or Location Survey:** The detailed examination of the field along the alignment finally recommend during the preliminary survey is called as location survey.

Curves:

There are two types curves: Horizontal Curve & Vertical Curve.

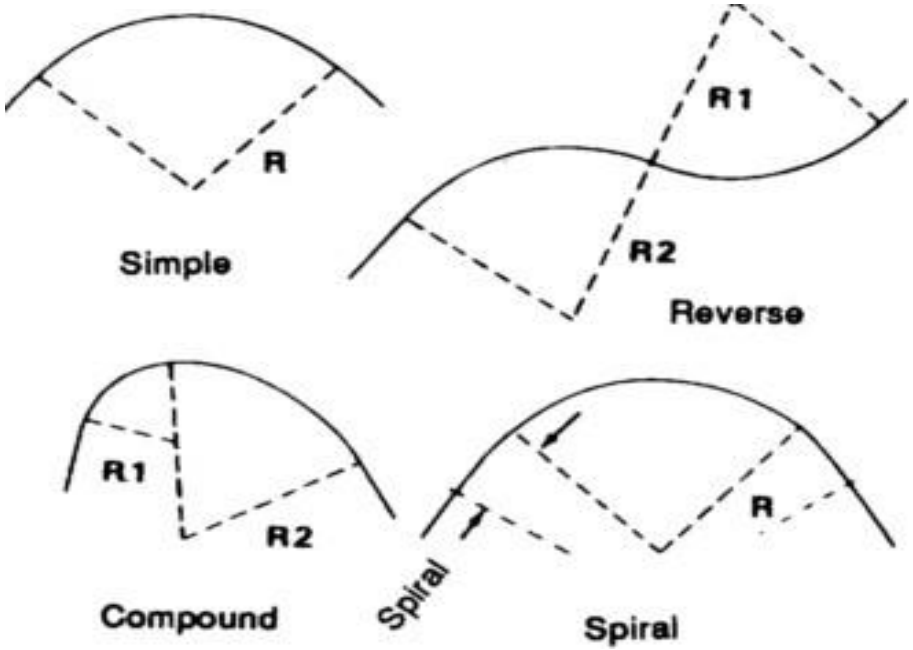
1) **Horizontal Curve:** The curve provided at the turning points in the alignments (in the horizontal plane) of a road are known as horizontal curve. Curve in plan to provide change in direction to the center line of a road.

2) **Vertical Curve:** The curve provided at the change of gradient of a road are called vertical curve.



- Highway curves
  - Horizontal curves
    - Circular curves
      - Simple
      - Reverse
      - Compound
    - Transition Curve
      - Spiral or clothoid
      - Lemniscate
      - Cubic parabolic
  - Vertical curves
    - Summit curve
    - Valley curve





1) **Horizontal Curve:** To achieve gradual change in the direction of alignment.

2) **Simple Curve :** Slow moving traffic & for large radius.

3) **Compound Curve:** Used when compelled by topography of area in order to avoid cutting through hard rock, heavy cutting or filling, etc.

4) **Reverse Curve:** Suitable for alignment of hill road.

5) **Transition Curve :** To provide ease & gradually change in direction of a road alignment**.**

6) **Summit Curve :** Rising gradient intersects a falling gradient or rising gradient meets another rising gradient or rising gradient meets a horizontal curve.

7) **Valley Curve:** Falling gradient intersects a rising gradient or falling gradient meets another falling gradient or a falling gradient meets a horizontal path.

Transition Curve:

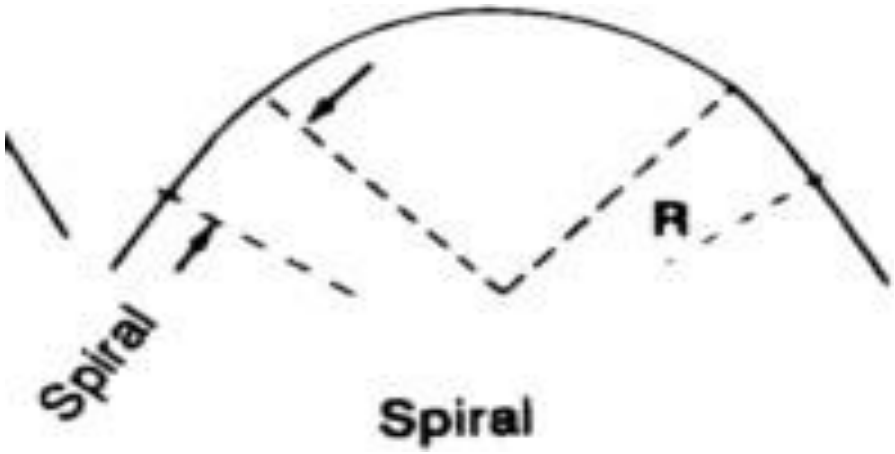
The curve having its radius varying gradually from an infinite to a finite value equal to that of the circular curve to be connected or vice versa is known as transition curve or Spiral Curve or Casement Curve.

1) To provide smooth entry of vehicle from a straight portion to a curved portion of the road, so as to avoid discomfort to the passengers.

2) To permit gradually application of the super elevation and reducing shocks on the vehicle.

3) To minimize wear on pavements.

4) To provide safety to the vehicular traffic.



Super-elevation:

1) In order to counteract the effect of centrifugal force & to reduce the tendency of the vehicle to overturn or kid, **the outer edge of the pavement is raised with respect to the inner edge**, thus providing a **transvers slope** throughout the length of the horizontal curve.

2) This **transverse inclination to the pavement** surface is known as superelevation or cant or banking.

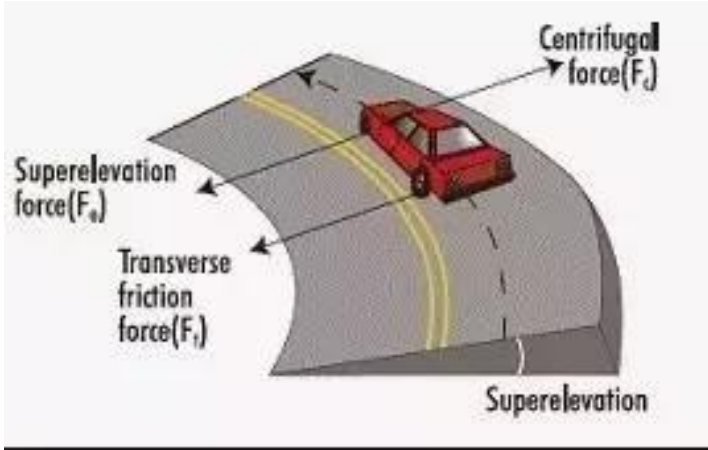
3) The rate of superelevation **'e'** is expressed as the **ratio of the height of outer edge with respect to the horizontal width.**

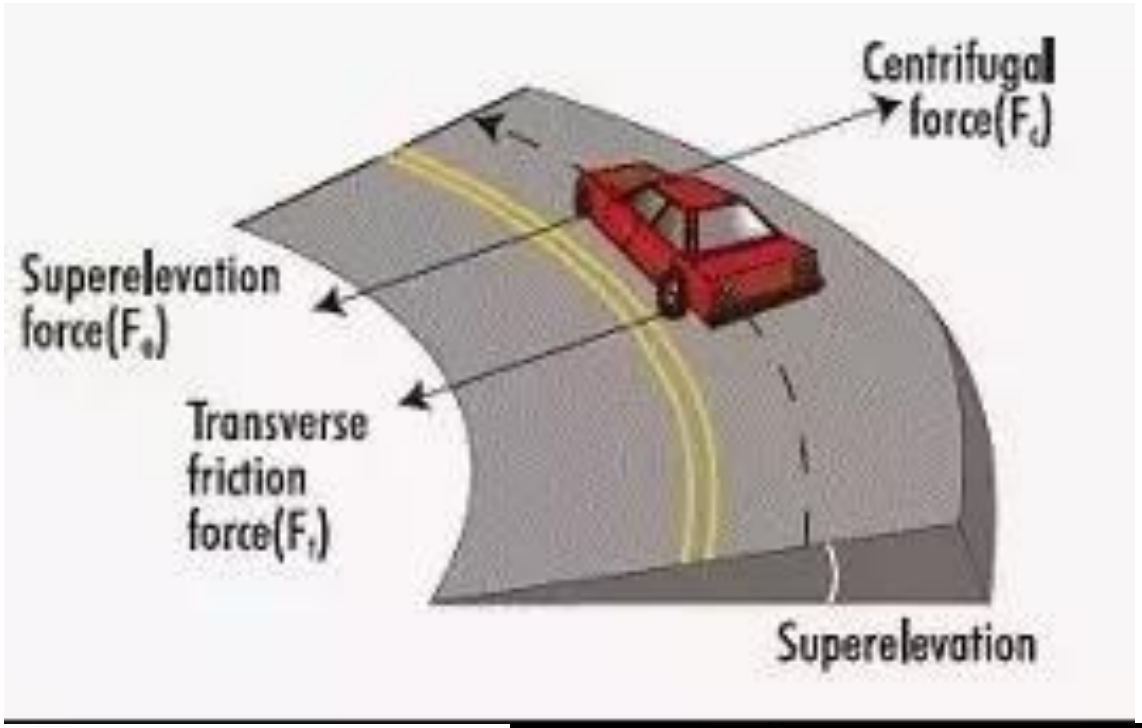
$$e + f = (V^2 / (127 * R)$$

V = Design speed, kmph

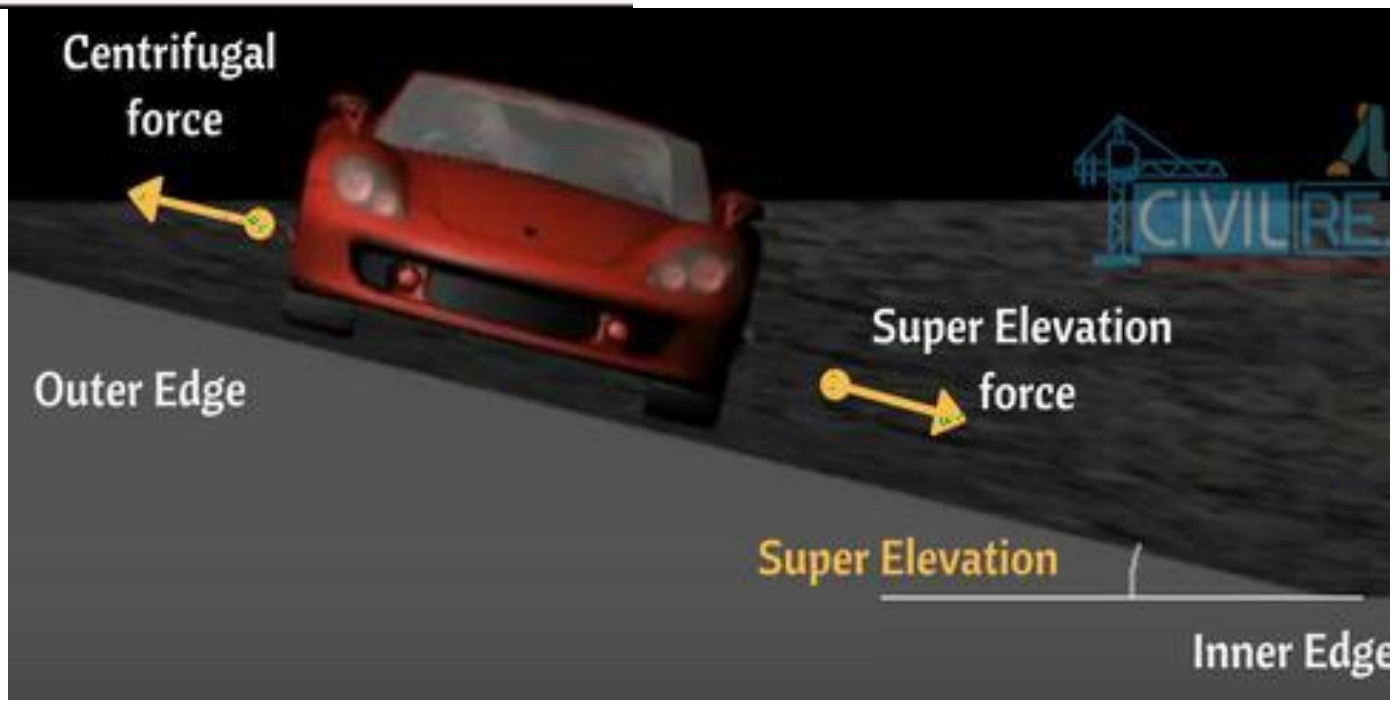
R = Radius of horizontal curve, m

f = lateral friction factor = 0.15




Centrifugal force(F_c)
Superelevation force(F_e)
Transverse friction force(F_t)
Superelevation

$e = \tan \theta$


Centrifugal force
Outer Edge
Super Elevation force
Super Elevation
Inner Edge

Equilibrium Super-elevation:

1) The superelevation determined from fundamental equation assuming coeff. of friction ‘f’ equals to zero & lateral friction insignificant is known as equilibrium superlelevation.

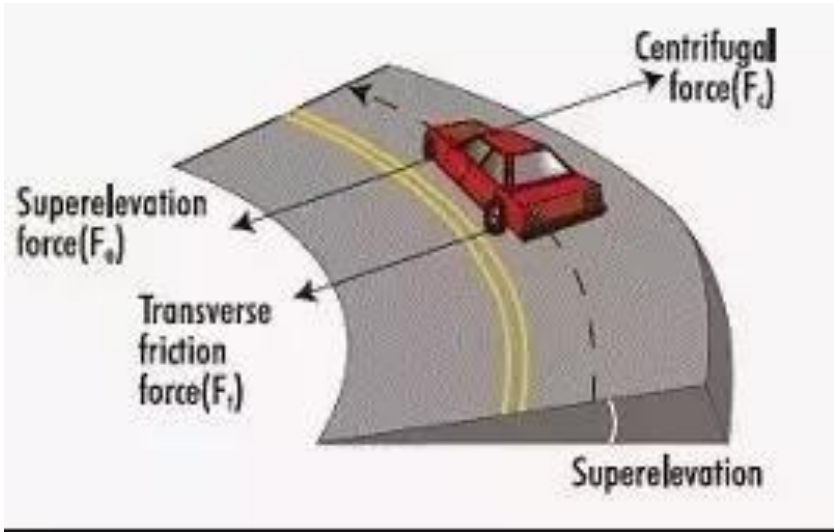
2) It can also be defined as superelevation required to counteract the effect of centrifugal force fully neglecting lateral friction.

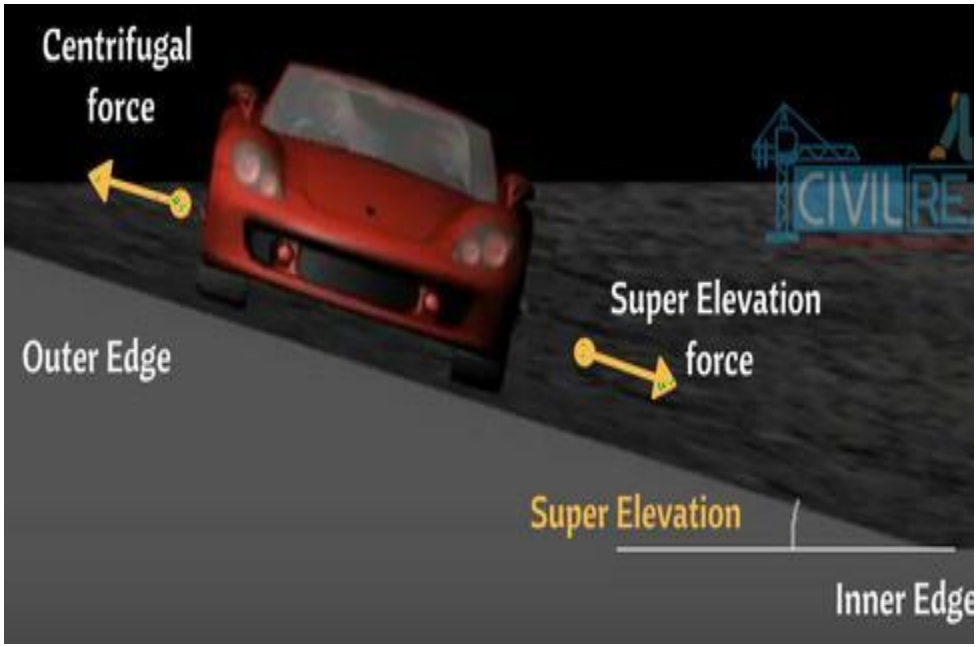
$$e = (V^2 / (127 * R)$$

The design of superlevation is India from practical consideration & is calculated for 75% of design speed neglecting friction . (as per IRC guideline)

$$e = (0.75 * V^2 / (127 * R)$$

$$e = (V^2 / (225 * R)$$





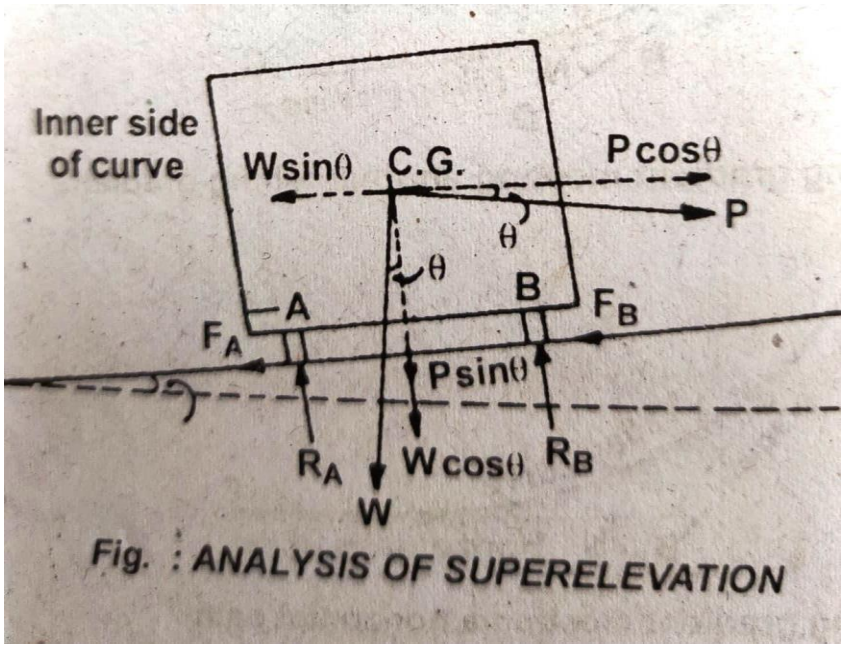
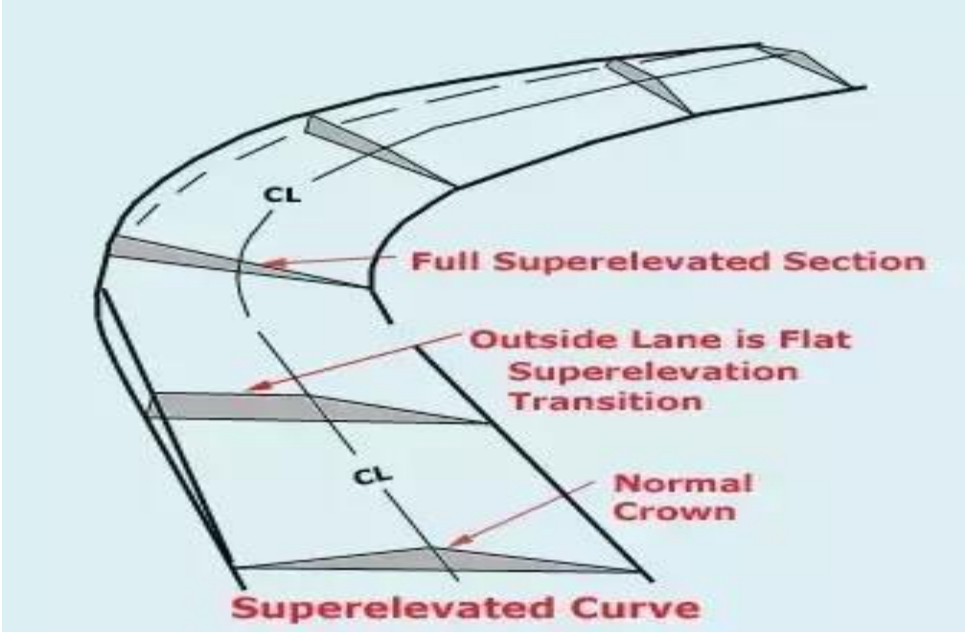


Fig. : ANALYSIS OF SUPERELEVATION





Superelevation :— [Transverse : inclination to the pavement]

The outer edge of the pavement is raised with respect to the inner edge to reduce the tendency of vehicle to overturn or skid, due to the effect of centrifugal force is called superelevation.

The rate of superelevation is expressed as 'e'.

$$e = \tan\theta$$

'e' is the ratio of height of outer edge with respect to the horizontal width.

$$e + f = \frac{V^2}{127 \times R}$$

Where, e = rate of superelevation = $\tan\theta$

f = lateral friction coeff. = 0.15

V = speed of vehicle, kmph.

R = Radius of horizontal curve, m.

Equilibrium Superelevation :-

The superelevation determined from fundamental equation assuming coeff. of friction 'f' equals to zero and lateral friction insignificant, is known as Equilibrium Superelevation.

It can also be defined as superelevation required to counteract the effect of centrifugal force fully, neglecting lateral friction.

$$e = \frac{V^2}{127 \times R}$$

The design of superelevation in India from practical consideration as per IRC guideline are given below:

$$e = \frac{(0.75 \times V)^2}{127 \times R}$$

Superelevation is calculated for 75% of design speed neglecting friction.

$$\therefore \quad e = \frac{V^2}{225 \times R}$$

Que. ⊛ Derive the expression of superelevation?

The forces acting on the vehicle while moving on a circular curve of radius 'R' metres, at speed of 'V' m/sec. are:

(I) The centrifugal force $P = \frac{W \cdot v^2}{g \cdot R}$ acting horizontally outwards through the centre of gravity.

(II) The weight W of the vehicle acting vertically downward through the C.G.

(III) The frictional force developed between the wheels and the pavement counteractions transversely along the pavement surface towards the centre of the curve.

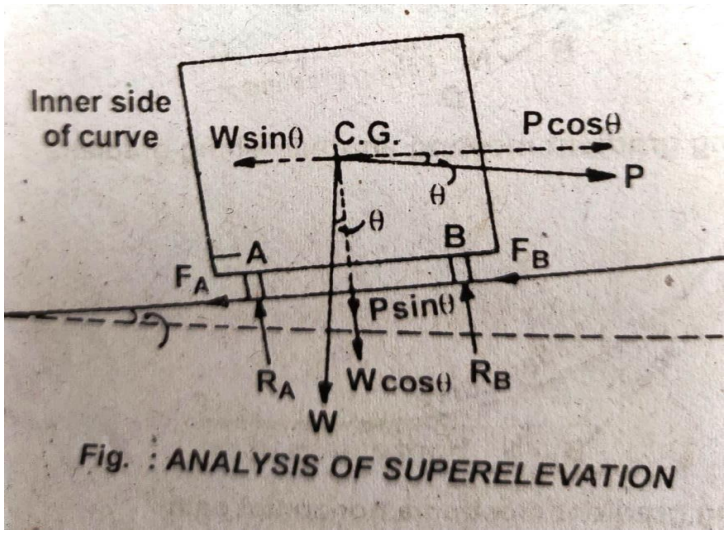


Fig. : ANALYSIS OF SUPERELEVATION

For equilibrium conditions,

$$P\cos\theta = W\sin\theta + F_A + F_B$$

The limiting equilibrium is reached, when the full values of the frictional forces are developed & the values of F_A and F_B reach their maximum value of $(f \times R_A)$ & $(f \times R_B)$ respectively.

Where, f is coeff. of lateral friction.

R_A & R_B are Normal Reactions at wheels A & B.

$\therefore$ $P\cos\theta = W\sin\theta + f(R_A + R_B)$

$P\cos\theta = W\sin\theta + f(W\cos\theta + P\sin\theta)$

$P\cos\theta = W\sin\theta + f \cdot W\cos\theta + f \cdot P\sin\theta$

$P(\cos\theta - f\sin\theta) = W\sin\theta + f \cdot W\cos\theta$

Dividing by $W\cos\theta$

$$\frac{P}{W}(1 - f\tan\theta) = \tan\theta + f$$

$$\frac{P}{W} = \frac{\tan\theta + f}{(1 - f \cdot \tan\theta)}.$$

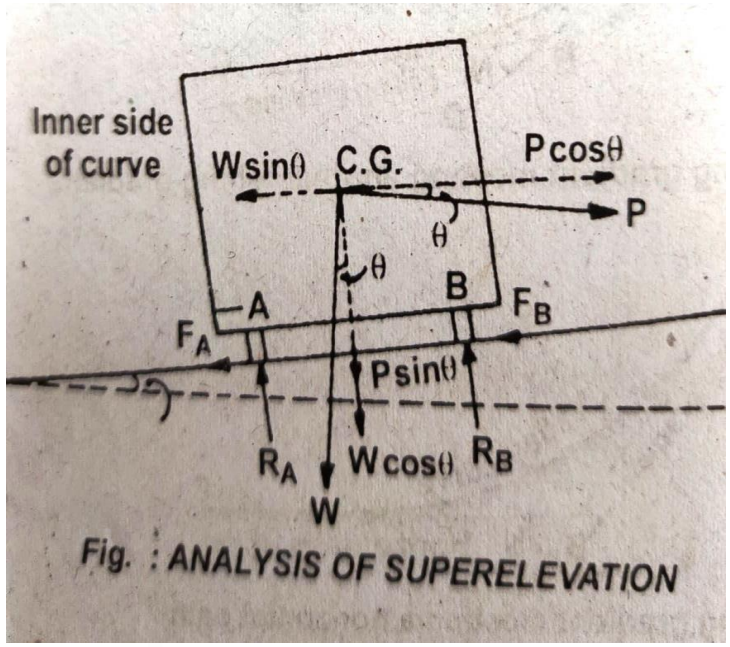


Fig. : ANALYSIS OF SUPERELEVATION

Where, $f = 0.15$ for design purpose

$\tan\theta > 0.07$ or about $1/15$

Hence, $f \cdot \tan\theta = 0.01$

Thus in equation, $(1 - f \cdot \tan\theta) = 0.99 \approx 1.0$

$\therefore$ $\frac{P}{W} = \tan\theta + f = e + f$

$$\frac{P}{W} = \tan\theta + f = e + f$$

But $\frac{P}{W} = \frac{V^2}{g \cdot R}$

$\therefore\ e + f = \frac{V^2}{g \cdot R}$

Here, e = rate of superelevation $= \tan\theta$

$f = 0.15$ $\quad g = 9.8\ m/s^2$ $\quad V = m/s$

If the speed is represented as V in kmph. then equation becomes:

$$e + f = \frac{(0.278V)^2}{9.8\ R} \Rightarrow \frac{V^2}{127 \times R}$$

$$\boxed{e + f = \frac{V^2}{127 \times R}}$$

If $f = 0$ then equilibrium superelevation required to counteract the centrifugal force fully will be given by:

$$\boxed{e = \frac{V^2}{127 \times R}}$$

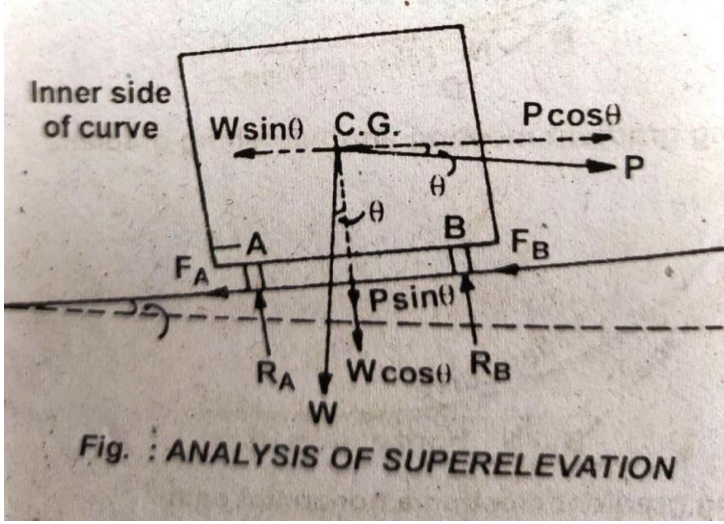


Fig. : ANALYSIS OF SUPERELEVATION

Que. ① The radius of a horizontal circular curve is 100 m. The design speed is 50 kmph and the design coeff. of lateral friction is 0.15.

(a) Calculate the superelevation required if full lateral friction is assumed to develop.

(b) Calculate the coeff. of friction needed if no superelevation is provided.

Sol^n :- $V = 50$ kmph $\quad R = 100$ m $\quad f = 0.15$

(a) Superelevation is given by

$$e + f = \frac{V^2}{127 \times R}$$

$$e + 0.15 = \frac{50^2}{127 \times 100}$$

$$\boxed{e = 0.047}$$

Therefore superelevation rate when full lateral friction is developed 0.047 or 1 in 21.2

(b) If no superelevation is provided, $e = 0$

$$f = \frac{V^2}{127 \times R} \Rightarrow \frac{50^2}{127 \times 100}$$

$$\boxed{f = 0.19}$$

Que.② A two lane road with design speed 80 kmph has horizontal curve of radius 480 m. Design the rate of superelevation for mixed traffic. By how much should the outer edges of the pavement be raised with respect to the inner edge, if the width of pavement at the horizontal curve is 7.5 m.

Sol^n :- $V = 80$ kmph $\quad R = 480$ m $\quad B = 7.5$ m

For mixed traffic condition the superelevation should fully counteract the centrifugal force 75% of design speed.

$$\therefore e = \frac{V^2}{225 \times R} \Rightarrow \frac{80^2}{225 \times 480} \Rightarrow 0.059$$

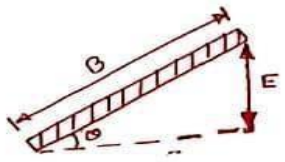
Since this value is less than 0.07, the superelevation of 0.059 may be adopted.

Total width of pavement, $B = 7.5$ m

∴ Raising of outer edge with respect to the inner edge of pavement, $E = B \times e$

$$E = 7.5 \times 0.059$$

$$\boxed{E = 0.44 \text{ m}}$$



$e = \tan\theta = \sin\theta$

Note:- The inclination θ with horizontal is very small and the value of $\tan\theta$ seldom exceeds 0.07. Therefore the value of $\tan\theta$ is practically equal to $\sin\theta$

Que. ③ Design the rate of superelevation for horizontal curve of radius 500m and speed 100 kmph.

Sol^n :- $V = 100$ kmph $\qquad R = 500$ m.

For mixed traffic condition, superelevation is designed for 75% of design speed.

$$e = \frac{V^2}{225R} \Rightarrow \frac{100^2}{225 \times 500} \Rightarrow 0.089$$

As the value is greater than maximum permissible superelevation of 0.07, the actual superelevation to be provided is restricted to **0.07**

Check for lateral coeff of friction developed for full speed, $V = 100$ kmph,

$$f = \frac{V^2}{127 \times R} - 0.07$$

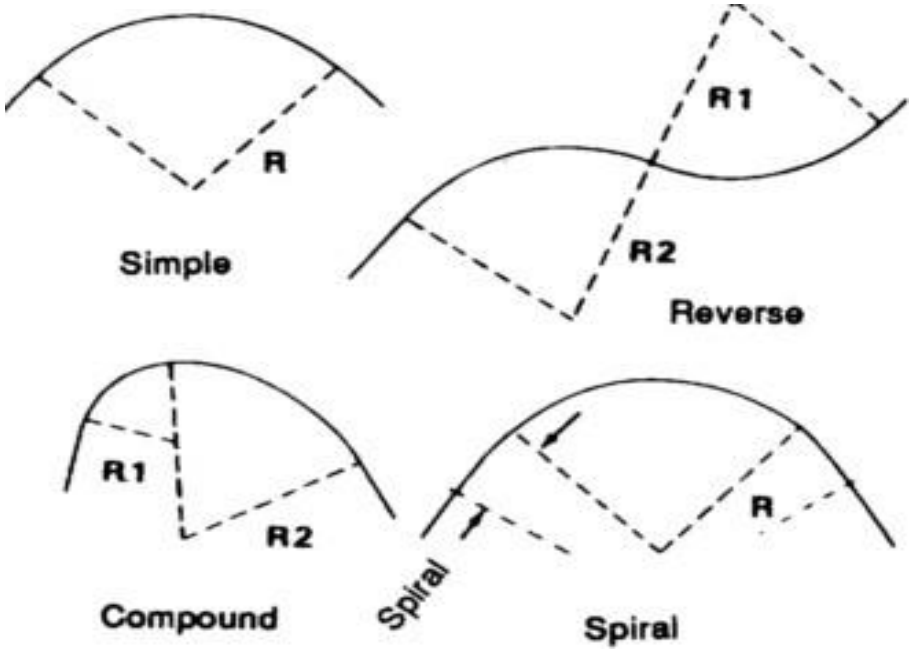
$$f = \frac{100^2}{127 \times 500} - 0.07$$

$$\boxed{f = 0.087}$$

As the value is less than 0.15, design is safe with a superelevation of 0.07. Hence provide a superelevation of 0.07.

- Highway curves
  - Horizontal curves
    - Circular curves
      - Simple
      - Reverse
      - Compound
    - Transition Curve
      - Spiral or clothoid
      - Lemniscate
      - Cubic parabolic
  - Vertical curves
    - Summit curve
    - Valley curve





1) **Horizontal Curve:** To achieve gradual change in the direction of alignment.

2) **Simple Curve :** Slow moving traffic & for large radius.

3) **Compound Curve:** Used when compelled by topography of area in order to avoid cutting through hard rock, heavy cutting or filling, etc.

4) **Reverse Curve:** Suitable for alignment of hill road.

5) **Transition Curve :** To provide ease & gradually change in direction of a road alignment**.**

6) **Summit Curve :** Rising gradient intersects a falling gradient or rising gradient meets another rising gradient or rising gradient meets a horizontal curve.

7) **Valley Curve:** Falling gradient intersects a rising gradient or falling gradient meets another falling gradient or a falling gradient meets a horizontal path.

Transition Curve:

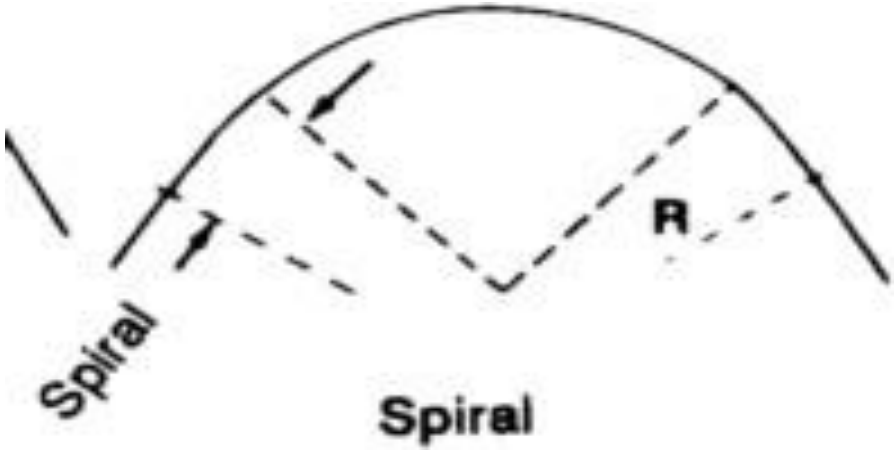
A transition curve is introduced **between a straight road and a circular curve to allow for a gradual change in curvature, improving vehicle stability and comfort**. As per IRC: 38-1988 and IRC: SP: 23-1993, the preferred type of transition curve for highways is the clothoid (spiral curve) due to its property of having a linearly varying radius

1) To provide smooth entry of vehicle from a straight portion to a curved portion of the road, so as to avoid discomfort to the passengers.

2) To permit gradually application of the super elevation and reducing shocks on the vehicle.

3) To minimize wear on pavements.

4) To provide safety to the vehicular traffic.

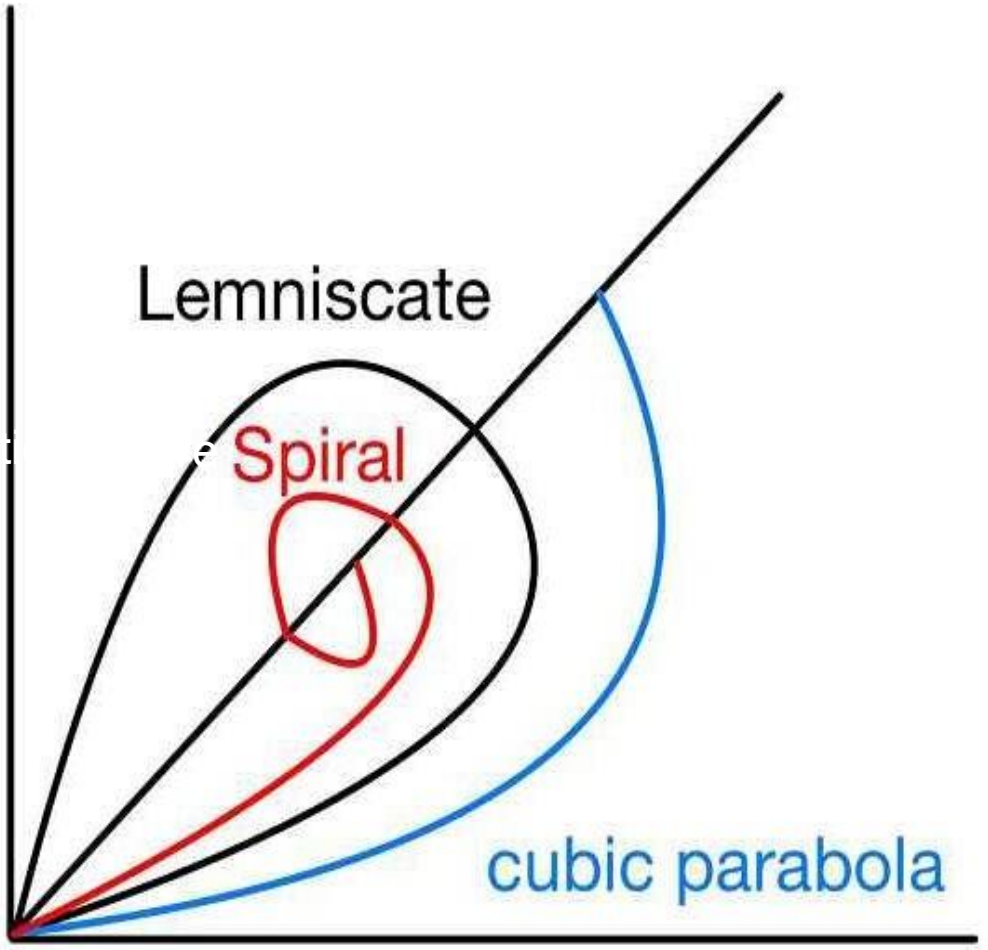


Different Types of Transition Curves

The types of transition curves commonly adopted in horizontal alignment of highways are:

1. Spiral
2. Bernoulli's Lemniscate
3. Cubic Parabola

- In all these curve the radius decreases at length increases.
- But the rate of change of radius or rate of change of centrifugal force is not constant in case of Lemniscate and cubic parabola.
- Hence a spiral curve is used as transition curve as it fulfills the requirement of ideal transition curve.
- **According to IRC an ideal transition curve is spiral because rate of change of radial acceleration (C) remains constant.**



Types of Transition curve

Formulas for Transition Curve Calculation

(A) Based on Rate of Change of Centrifugal Acceleration (Comfort Criteria)

$Ls = V^3 / (C \times R)$

where: - Ls = Length of transition curve (m)

- V = Design speed (m/s) = (Speed in km/h × 1000) / 3600
- C = Rate of change of acceleration (IRC :0.5 to 0.8 m/s^3)

  = 80/(75+V) { V-Kmph }

- R = Radius of the circular curve (m)

(B) Based on Rate of Change of Super-Elevation & Extra Widening

Ls = e * B * N

where:

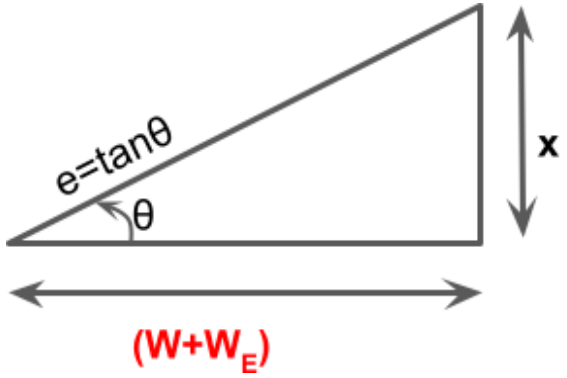
- e = Super-elevation (dimensionless, e.g., 0.07 for 7%)
- B = Total pavement width = $W + W_e$
- W = Pavement Width
- W_e = Extra Widening
- N = Allowable rate of change of super-elevation (Recommended: 150 for plains, 60–100 for hilly areas)

(i) When pavement is rotated about inner edge

Ls= e * (W + W_e) * N

W=Width of pavement

W_E= Extra widening

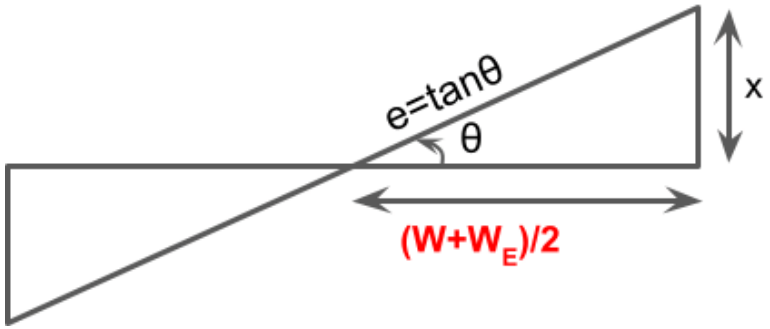


$$e = tan\theta = \frac{x}{W+W_E}$$

$$x = e(W+W_E)$$

ii) When pavement is rotated about centre

Ls=e * N *(W + W_e) / 2



$$e = tan\theta = \frac{x}{\frac{W+W_E}{2}}$$

$$x = \frac{1}{2}e(W+W_E)$$

Widening of Pavement : On horizontal curves, especially when they are not of very large radii, it is a common practice to widen the pavement slightly more than the normal width.

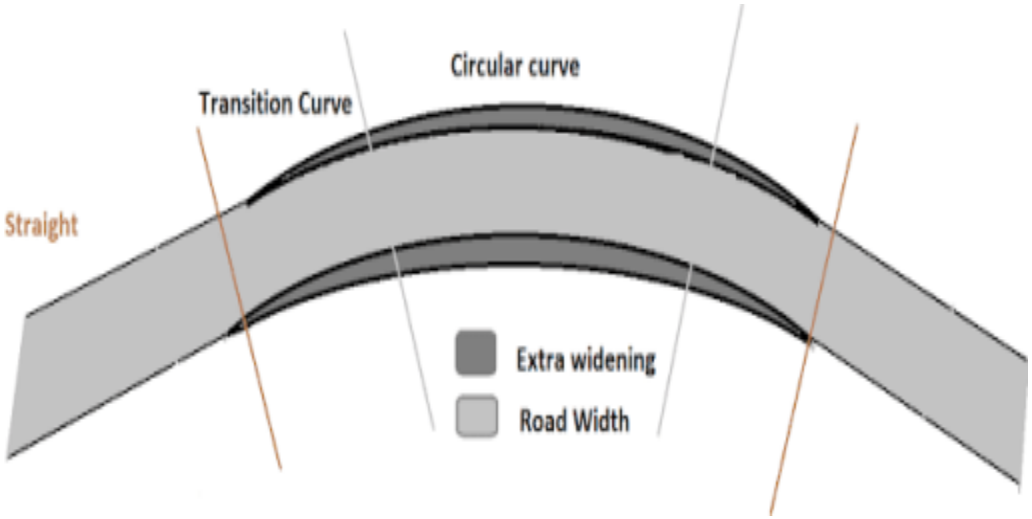
The extra widening of pavements on horizontal curves are due to the following reasons:

1) When the vehicle takes a turn to negotiates a horizontal curve, **the rear wheels do not follow the same path as that of the front wheels**. Inner front wheel takes a path on the inner edge of a pavement at a horizontal curve, inner rear wheel will be off the pavement on the inner shoulder. This phenomenon is called **off-tracking.**

2) When super elevation & lateral friction developed are not fully able to counteract the outwards thrust due to the centrifugal force, **some transvers skidding may occur** & **rear wheels may take paths on the outside of curve**. It can occurs only at excessive high speed.

3) To have **greater visibility**, the driver have tendency not to follow the central path of the lane but to use the outer side at the beginning of a curve.

4) While two vehicles cross or overtake at horizontal curve there is a psychological tendency to maintain a **greater clearance** between the vehicles than on straight for increase safety.



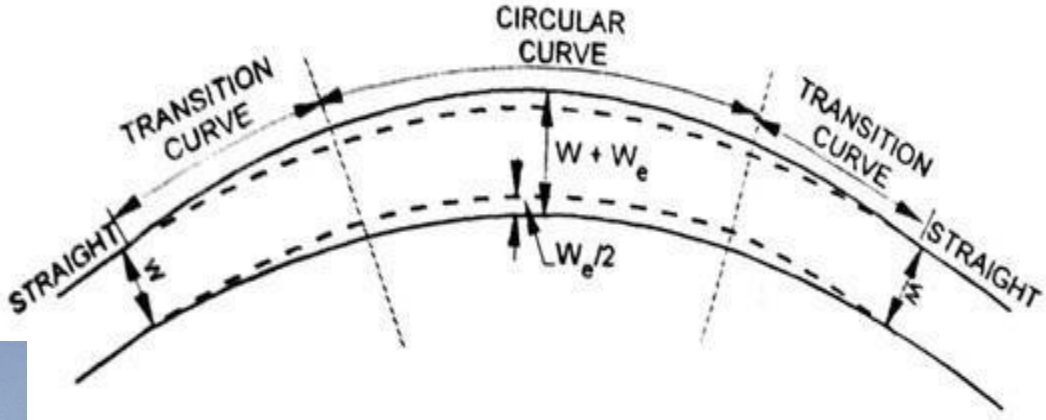


Fig. 4.26 Extra widening of pavement on horizontal curve



Extra Widening Calculation

Extra widening (We) is required on curves to accommodate larger turning radii for vehicles and to provide driver comfort at higher speeds. It consists of two components: Mechanical Widening and Psychological Widening.

Formula for Extra Widening (W)

We = Wm + Wp

where:

- Wm = Mechanical Widening , - Wp = Psychological Widening

(A) Mechanical Widening (Wm)

$Wm = (nL^2) / (2R)$

where:

- n = Number of traffic lanes

- L = Wheelbase of the vehicle (typically 6.0 m for heavy vehicles)

- R = Radius of the circular curve (m)

(B) Psychological Widening (Wp)

$Wp = V / 9.5\sqrt{R}$

where: - V = Design speed (km/h)

Total Extra Widening for Different Terrains

$We = (nL^2) / (2R) + V / 9.5\sqrt{R}$

where:

- We = Total extra widening in meters

(C) Empirical Formula for Transition Curve Length

- For Plain & Rolling Terrain: $Ls = 2.7 \times V^2 / R$
- For Mountainous & Steep Terrain: $Ls = V^2 / R$

MAXIMUM OF ABOVE 3 CRITERIA IS CONSIDER FOR DESIGN OF TRANSITION CURVE

Shift of Curve (S) : The shift is the lateral displacement of the circular curve due to the introduction of the transition curve. It ensures that the alignment remains smooth.

$S = Ls^2 / (24 \times R)$

where:

- S = Shift of the curve (m)

- Ls = Transition curve length (m)

- R = Radius of the circular curve (m)

Vertical Curve:

In India, the **design of vertical curves** in road geometry follows the guidelines specified in the **Indian Roads Congress (IRC), primarily IRC 73:1980 (Geometric Design Standards for Rural (Non-Urban) Highways) and IRC 86:2018 (Geometric Design Standards for Urban Roads).**

Types of Vertical Curves

Vertical curves are used to provide a smooth transition between two different gradients (slopes) in the road profile. As per IRC, two types of vertical curves are considered:

1. Summit Curve (Convex Curve)

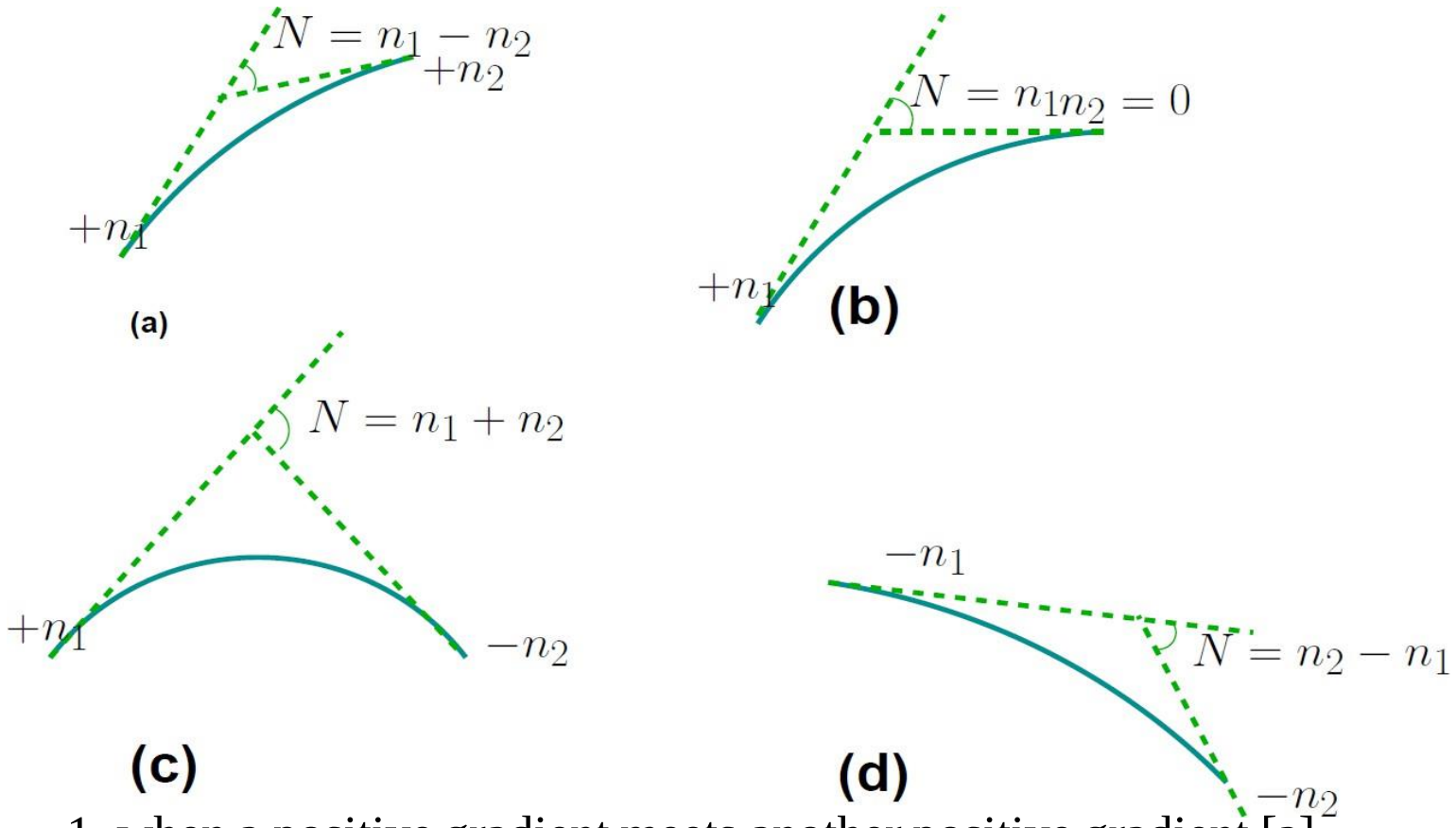
1. Formed when the road transitions from an **upward gradient to a downward gradient** (crest curve).
2. Designed to ensure adequate **sight distance**, especially **stopping sight distance (SSD)**.

2. Valley Curve (Sag Curve)

1. Formed when the road transitions from a **downward gradient to an upward gradient**.
2. Designed for **comfort (rider's perception) and headlight sight distance at night**.

Type of Summit Curve

Summit curves are vertical curves with gradient upwards. They are formed when two gradients meet as illustrated in figure in any of the following four ways:



1. when a positive gradient meets another positive gradient [a].
2. when positive gradient meets a at gradient [b].
3. when an ascending gradient meets a descending gradient [c].
4. when a descending gradient meets another descending gradient [d].

Design Criteria for Summit Curves as per IRC

The design of summit curves depends on:

1. **Stopping Sight Distance (SSD) – Mandatory for all roads**
2. **Overtaking Sight Distance (OSD) – For highways where overtaking is allowed**
3. **Intermediate Sight Distance (ISD) – When OSD is not feasible**
4. **Comfort Criteria – Smooth transition for rider comfort (less critical for summit curves)**

Summit curves are designed as **simple parabolic curves** to provide a smooth transition in vertical alignment.

Length of summit curve for stopping sight distance (SSD)

Two cases are to be considered in deciding the length:

(a) When the length of the curve is greater than the sight distance (L > SSD)

(b) When the length of the curve is less than the sight distance (L < SSD)

(a) When L > SSD

The general equation for length L of the parabolic curve is given by:

$$L = \frac{NS^2}{\left(\sqrt{2H}+\sqrt{2h}\right)^2} \qquad \text{(Eq. 4.34)}$$

Here

L = length of summit curve, m

S = stopping sight distance SSD, m

N = deviation angle, equal to algebraic difference in grades, radians or tangent of the deviation angle

H = height of eye level of driver above roadway surface, m

h = height of subject above the pavement surface, m

The value of H, the height of driver's eye above roadway surface is taken as 1.2 m and the height of object h above the pavement surface for the purpose of safe stopping distance is taken as 0.15 m as per the IRC standard (see Art 4.3.2). Substituting these values in Eq 4.34 the length of summit curve is obtained as:

$$\text{i.e.,} \qquad L = \frac{NS^2}{4.4} \qquad \text{(Eq. 4.35)}$$

(b) When L < SSD

The general equation for the length of the parabolic summit curve, when it is less than the sight distance is given by:

$$L = 2S - \frac{\left(\sqrt{2H} + \sqrt{2h}\right)^2}{N} \qquad \text{(Eq. 4.36)}$$

Here the description for L, S, N, H and h are the same as in equation 4.33. By substituting the values of H = 1.2 and h = 0.15 m the length of the curve is obtained as:

$$L = 2S - \frac{4.4}{N} \qquad \text{(Eq. 4.37)}$$

Thus for the design of vertical summit curve length, to provide for safe stopping, the stopping sight distance is determined, by calculation as given in Art. 4.3.2 or from the recommended values given in Table 4.7. By using any of the two appropriate formulae by trial (Eq 4.35 and Eq 4.37) the required length of summit curve is calculated. The minimum radius of the parabolic summit curve may be calculated from the relation R = L/N.

Length of summit curve for overtaking sight distance (OSD) or intermediate sight distance (ISD)

Two cases to be considered in deciding the length are:

(a) When the length of curve, L is greater than the OSD or ISD (L > S)

(b) When the length of the curve is less than the OSD or ISD (L < S)

(a) When L > S

The same general Eq. 4.34 is applicable in this case also. But in this case, the values of H and h, both are taken equal to 1.2 m. Substituting h = H in the Eq. 4.35 and simplifying.

$$L = \frac{NS^2}{8H}$$

As discussed in Art. 4.3.3 the height of the eye level of the driver as well as the height of the approaching object are taken as 1.2 m. Substituting the value of H, the height of eye level of driver above the pavement surface equal to 1.2 m,

$$L = \frac{NS^2}{9.6} \qquad \text{(Eq. 4.38)}$$

Here,

L = the length of parabolic summit curve, m

N = deviation angle, radians or tangent of the deviation angle,

S = overtaking or intermediate sight distance, m

(b) When L < S

The same general Eq 4.36 may be used. By substituting H = h and simplifying, when L is less than OSD/ISD:

$$L = 2S - \frac{8H}{N}$$

Here again substituting the value of H as 1.2 m, the equation reduces to:

$$L = 2S - \frac{9.6}{N} \qquad \text{(Eq. 4.39)}$$

The minimum lengths of vertical curve for different speeds and for the maximum grade change values (in percent) which do not require vertical curve, as per the recommendations of the Indian Roads Congress are given in Table 4.17.

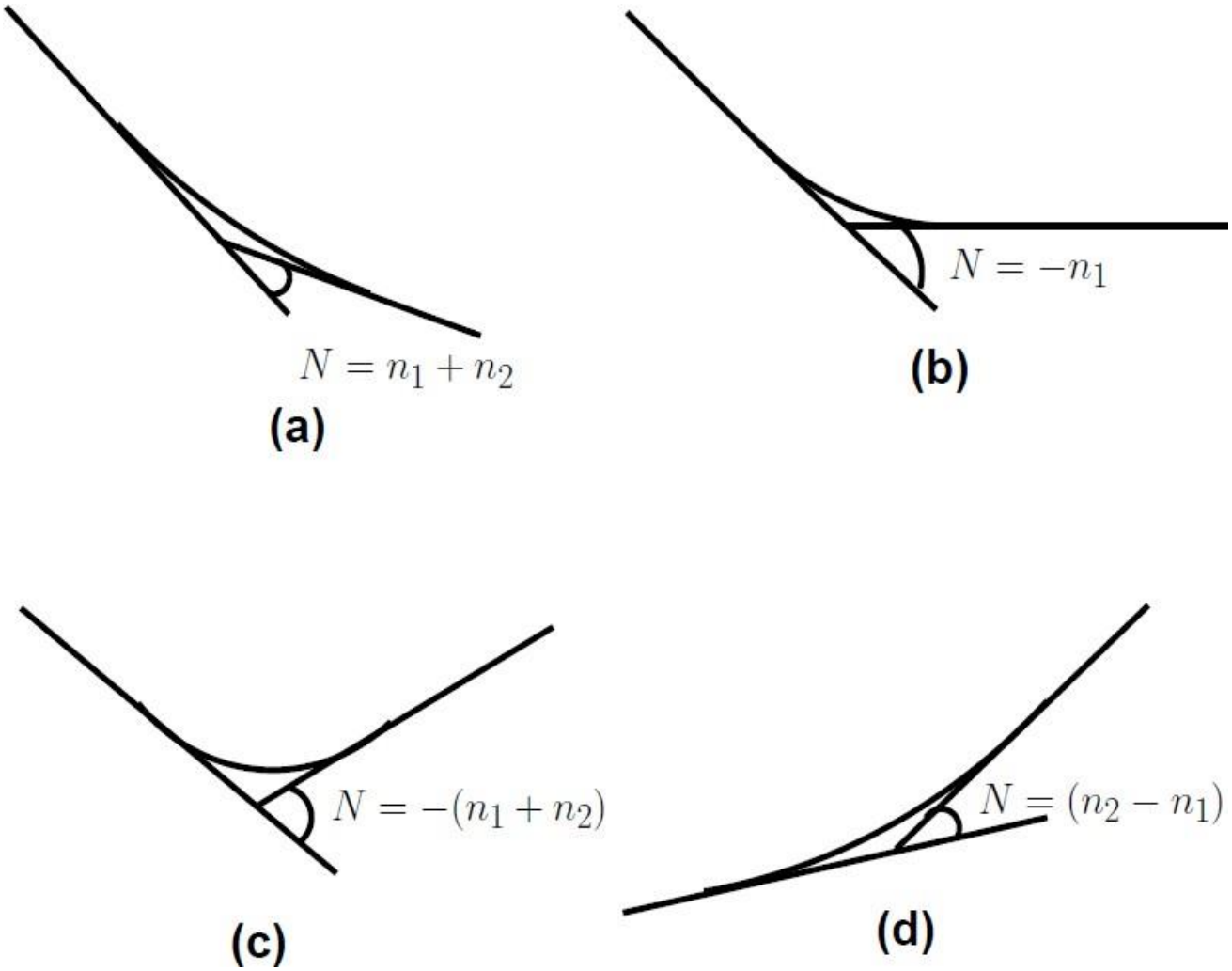
Table 4.17 Minimum length of vertical curves

Design speed, kmph	Maximum grade change (percent) not requiring a vertical curve	Minimum length of vertical curve, m (for higher grade change values)
35	1.5	15
40	1.2	20
50	1.0	30
65	0.8	40
80	0.6	50
100	0.5	60

Valley Curves (Sag Curves)

Valley curve or sag curves are vertical curves with convexity downwards.

They are formed when two gradients meet as illustrated in figure in any of the following four ways



1. when a descending gradient meets another descending gradient [a].
2. when a descending gradient meets a at gradient [b].
3. when a descending gradient meets an ascending gradient [c].
4. when an ascending gradient meets another ascending gradient [d].

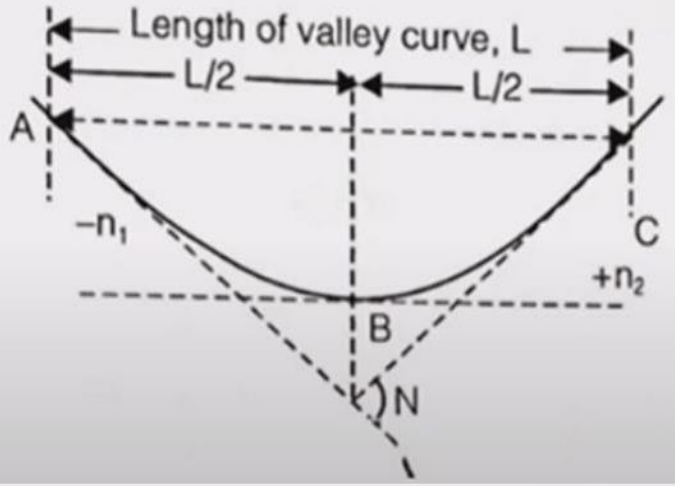
Design Criteria for valley curve as per IRC

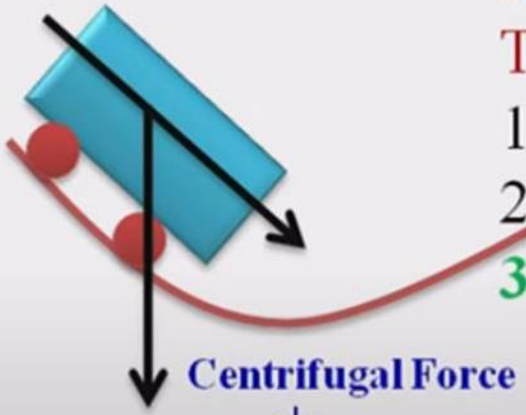
The length of the valley transition curve
is designed based on two criteria:

1. comfort criteria; that is allowable rate of change of centrifugal acceleration (C) is limited to a comfortable level of about 0:6m/sec3.

2. safety criteria; that is the driver should have adequate headlight sight distance at any part of the country.

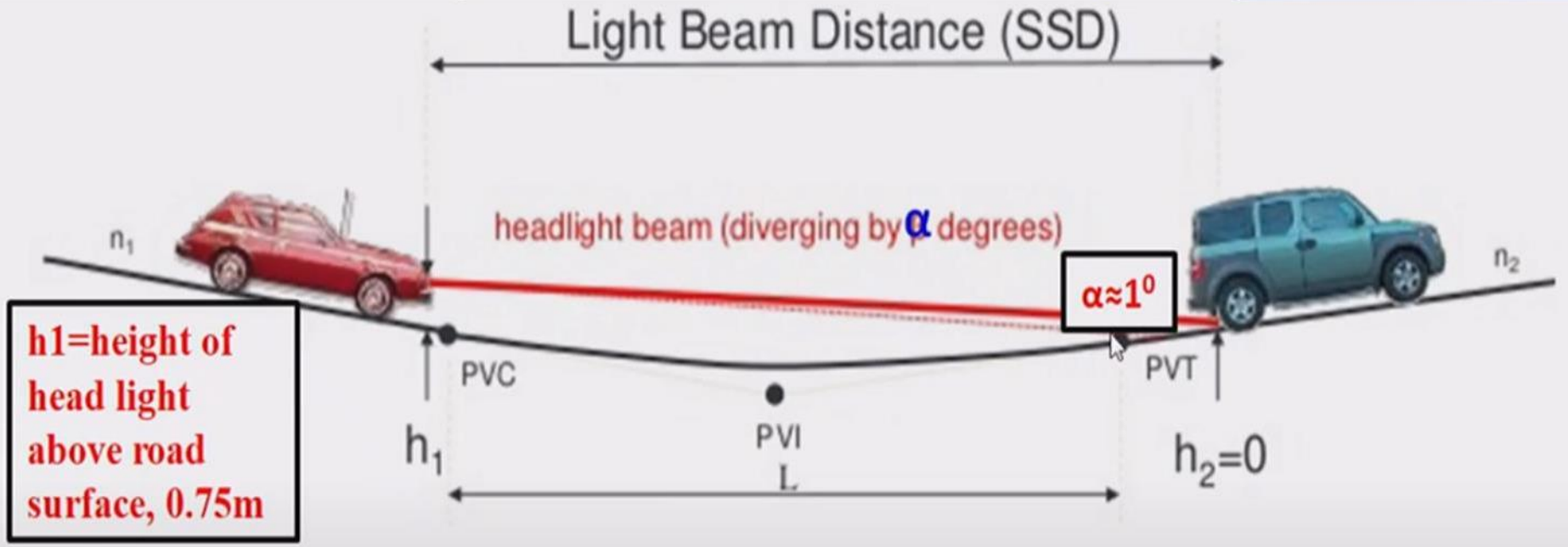
Is there necessary to provide Transition curve in Valley Curve?





Ideal Shape of Transition

1) Lemniscate
2) Spiral
3) Cubic Parabola



1) Comfort criteria

The length of the valley curve based on the rate of change of centrifugal acceleration that will ensure comfort:

$$L = 2\sqrt[2]{\frac{Nv^3}{c}}$$

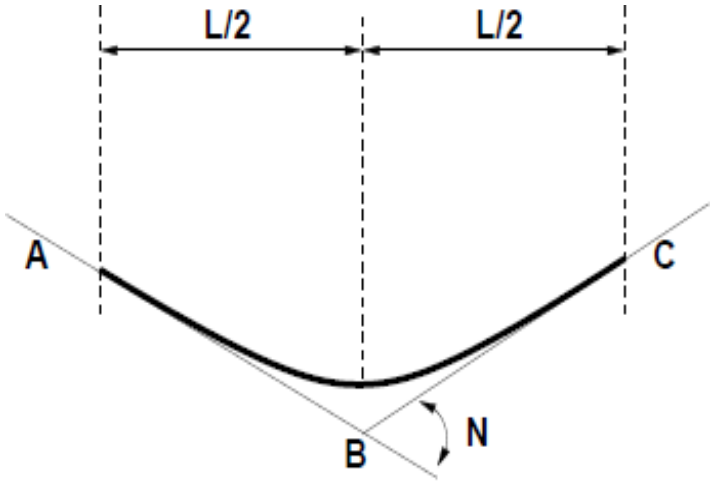


Figure 18:2: Valley curve details

wnere

L = the total length of valley curve

N = the deviation angle in radians or tangent of the deviation angle

v = the design speed- m/s

c = the allowable rate of change of centrifugal acceleration which may be taken as 0.61 m/s^3

2) Safety criteria

Length of the valley curve for headlight distance may be determined for two conditions:

Case 1 : Length of valley curve greater than stopping sight distance (L > S)

L ≥ S = HSD

$$L = \frac{NS^2}{2h_1 + 2S\tan\alpha}$$

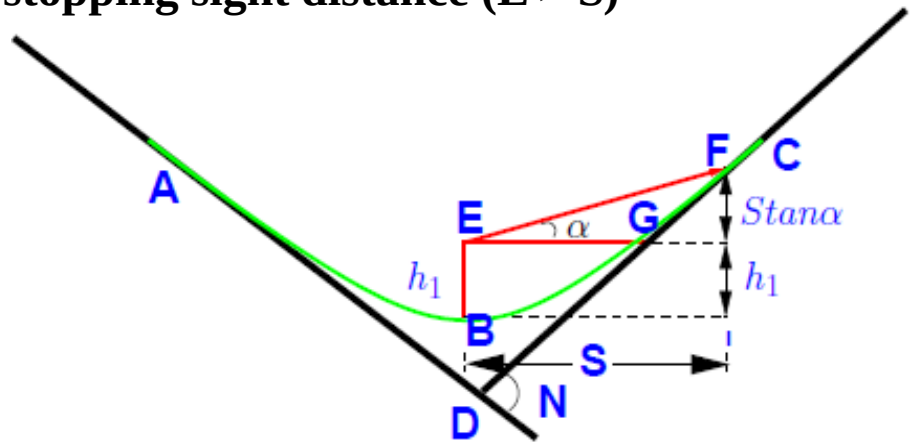


Figure 18:3: Valley curve, case 1, $L > S$

Case 2 Length of valley curve less than stopping sight distance (L < S)

L < S = HSD

$$L = 2S - \frac{2h_1 + 2S\tan\alpha}{N}$$

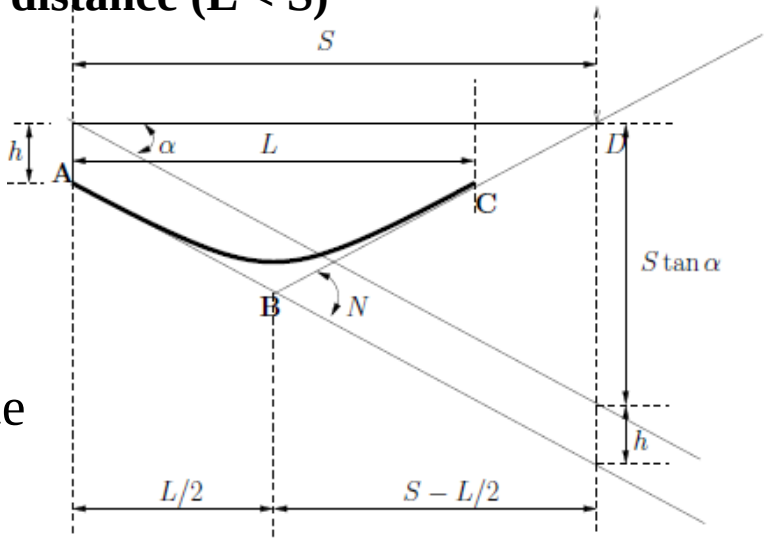


Figure 18:4: Valley curve, case 2, $S > L$

where ,

S= stopping sight distance=head light sight distance

N = Deviation Angle

α = Beam angle of Head light

h_1 = Height of head light above road surface = 0.75 m

Type of Curve	Form	Design Criteria as per IRC
Horizontal Circular Curve	Simple circular arc	- Minimum radius as per IRC 38 and IRC 73 - Superelevation as per IRC 73 - Extra widening required for higher speeds - Transition curves preferred at high speeds
Horizontal Transition Curve	Spiral (clothoid)	- Provided for smooth entry/exit into circular curves - Designed based on rate of change of centrifugal acceleration - Length as per IRC 73 and IRC 38
Summit Curve	Parabolic curve	- Occurs when two gradients meet in convex form - Designed for sight distance criteria (SSD, OSD, ISD) as per IRC 66 - Length based on comfort and visibility
Valley Curve	Parabolic curve	- Occurs when two gradients meet in concave form - Designed for rider comfort (rate of change of acceleration) as per IRC 66 - Length depends on design speed and allowable acceleration

Construction of Pavements

Road Pavement: A layered structure supported by the subgrade soil to form the carriage way of a road is called a road pavement.

Flexible Pavement: The road pavements which can change their shape to some extent without rupture are known as rigid pavement.

Rigid Pavement: The road pavements which can not change their shape without rupture are known as rigid pavement.

Cement Concrete Road: The roads having their wearing surface consisting of cement concrete slab are called cement concrete roads.

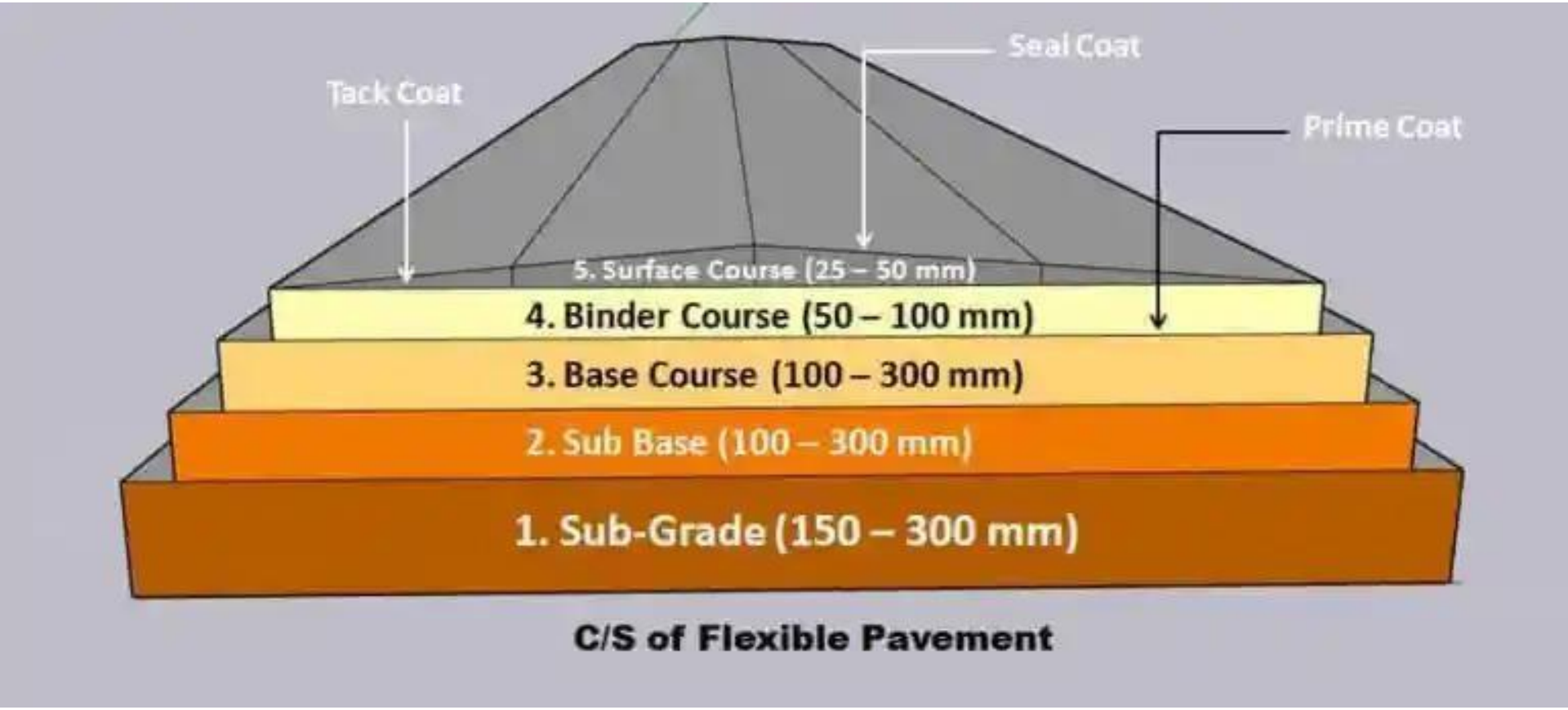
Optimum Moisture Content: The moisture content at which the soil will compact to its greatest density is called OMC.

Earth Road: The road having its foundation & wearing surface consisting of one or two compacted layers of an ordinary or stabilized soil is known as earth road or kutcha road.

Gravel Road: The road having its wearing surface consisting of one or two compacted layers of gravel mixed with sand and clay is known as gravel road.

Water Bound Macadam Road: The road having its wearing surface consists of clean, crushed aggregates, mechanically interlocked by rolling & bound together with filler material & water laid on a prepared base course is called WBM road.

Bituminous Road: The roads having their surface consists of bituminous materials are known as bituminous roads or black top roads.



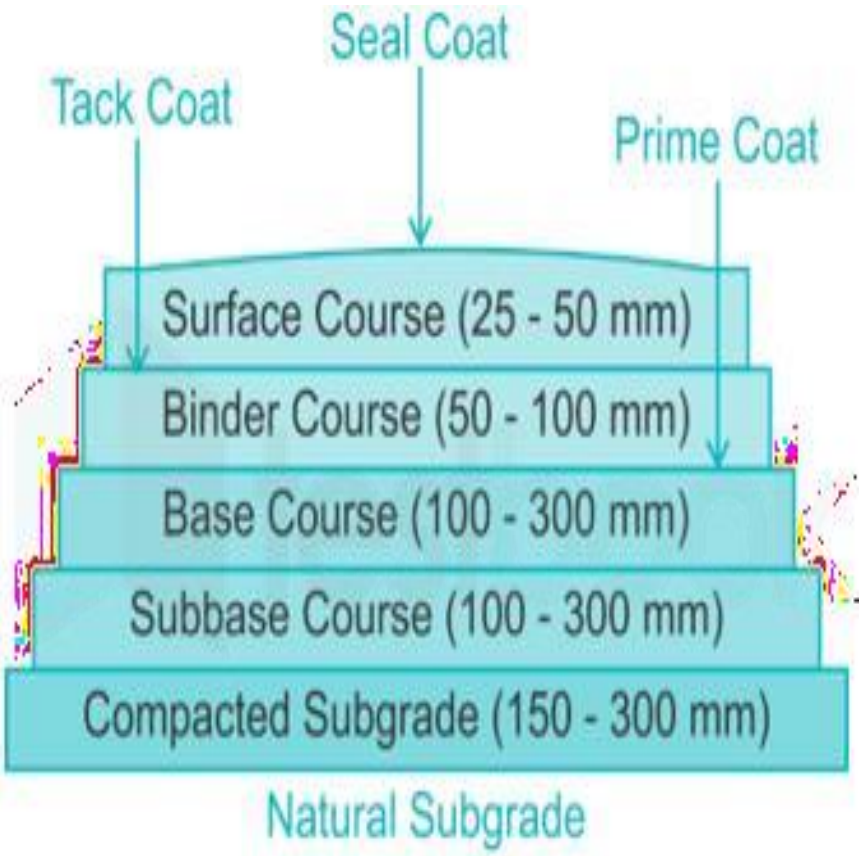
C/S of Flexible Pavement

Subgrade: The finished & compacted surface of earthwork on which a road pavement rest is called subgrade or formation.

Sub-base: A layer of granular material provided in between the subgrade & the base course in a road pavement is called sub-base.

Base Course: Soiling & foundation course.
Wearing Course: The **top most layer of the road pavement** directly exposed to traffic is called wearing course or surfacing.

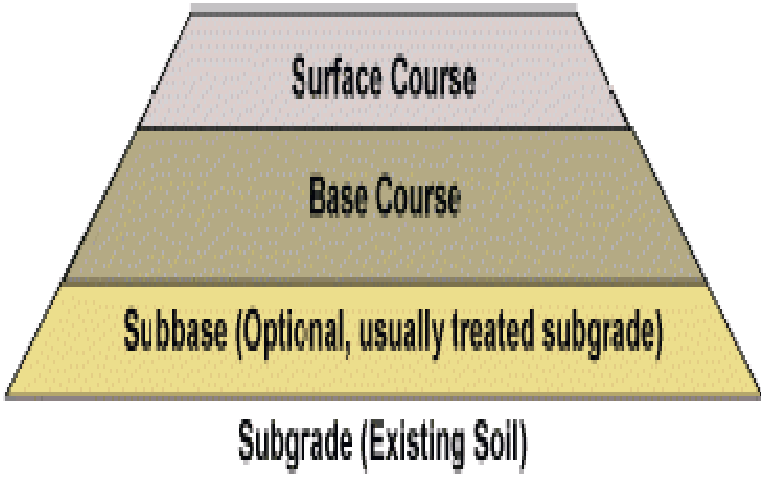
Seal Coat: A final coat of bituminous material provided on the top of surface course for sealing the voids against entry of moisture as known as seal coat.
Prime Coat: A single coat of low viscosity bituminous binder applied to an existing untreated base of previous nature like WBM in order to promote adhesion between the base & the bituminous surface is called prime coat.
Base Coat: The **layer of hard stones** provided in between the **base course & the wearing course** in a road pavement is called base coat or bearing course.

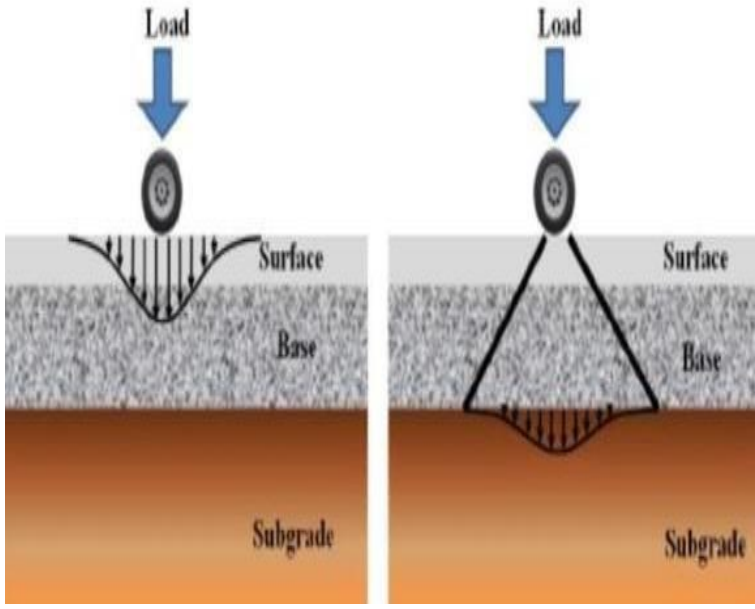


Typical cross section of flexible pavement

1. Flexible Pavements

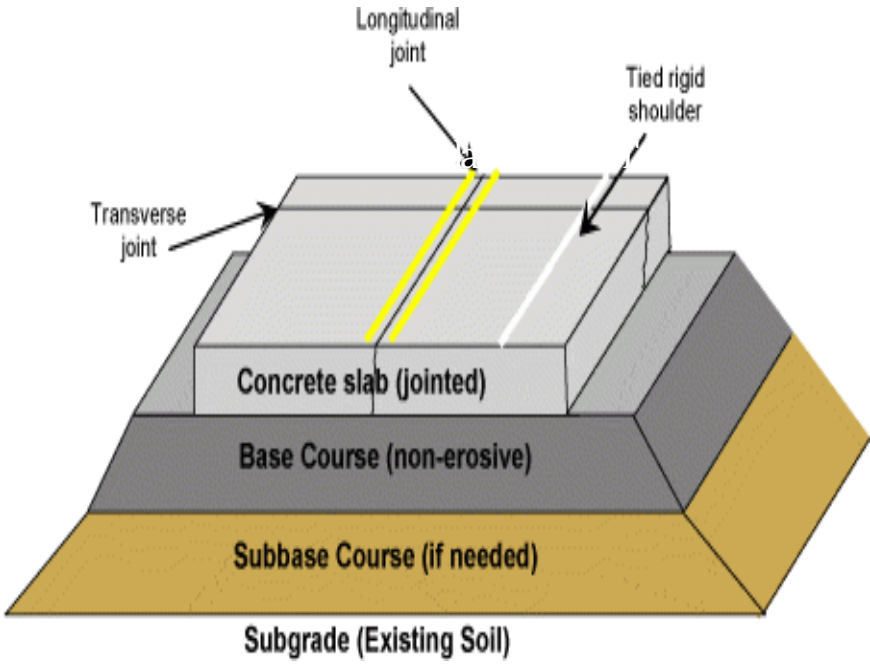
- Flexible pavement consists of a mixture of asphaltic or bituminous material and aggregates placed on a bed of compacted granular material of appropriate quality in layers over the subgrade.
- Water bound macadam roads and stabilized soil roads with or without asphaltic toppings are examples of flexible pavements.
- The **design of flexible pavement** is based on the principle that for a load of any magnitude, the intensity of a load diminishes as the **load is transmitted downwards from the surface by virtue of spreading over an increasingly larger area,** by carrying it deep enough into the ground through successive layers of granular material.



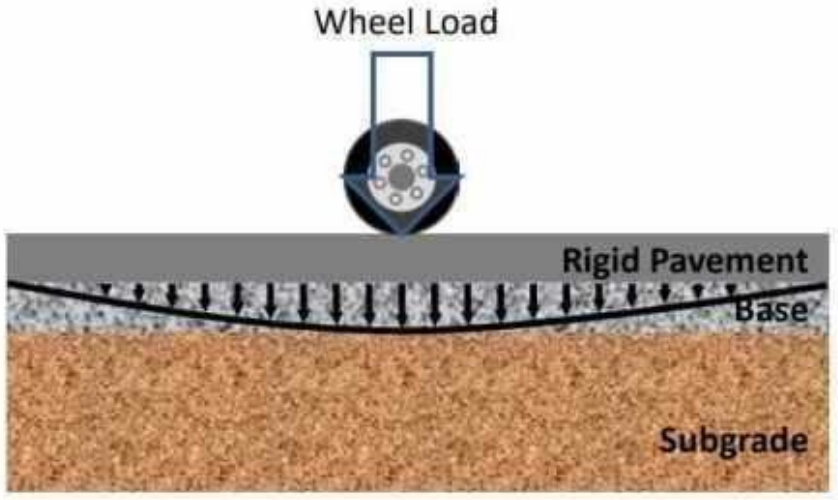


Rigid Pavements

- A rigid pavement is constructed from cement concrete or reinforced concrete slabs.
- Grouted concrete roads are in the category of semi-rigid pavements.
- The design of rigid pavement is based on providing a **structural cement concrete slab of sufficient strength** to resists the loads from traffic.



- The rigid pavement has **rigidity and high modulus of elasticity to distribute the load** over a relatively wide area of soil.



Point	Flexible Pavements	Rigid Pavements
Initial cost	Initial cost is low.	Initial cost is high.
Life span	Short life span.	Long life span.
Thickness	Thickness is more.	Thickness is less.
Joints	Joints are not required.	Joints are essentially required.
Subgrade	A reasonably good subgrade is required.	A good subgrade is not necessary.
Durability	Less durable	More durable.
Temperature effect	Stresses are not induced	Heavy stresses are induced
Repair work	Easy repair work.	Difficult to repair.
Skill and supervision	Moderate skill and less supervision are required.	More skill and supervision is required.
Feasibility of stage development	Stage development is feasible.	Stage development is not feasible.

Point	Flexible Pavements	Rigid Pavements
Openings to traffic after construction	Can be opened to traffic shortly after construction.	Require curing after construction and thus cause delay in opening to traffic.
Behavior with subgrade settlement.	They adjust according to any deformation of subgrade without rupture.	They do not adjust according to any deformation of subgrade without rupture.
Tractive resistance	More tractive resistance.	Less tractive resistance.
Traffic suitability	Suitable for all types of traffic.	Noisy under iron wheeled traffic.
Night visibility	Blacktop pavements provide poor visibility at night.	They offer good visibility at night.
Effect Of loading	They adjust themselves to normal loading by undergoing elastic deformation.	They tend to act as a beam or cantilever and resist deformation.
Maintenance cost	Maintenance cost is high.	Maintenance cost is low.
Glaring effect	They do not usually cause glare due to reflected sunlight.	They often cause glare due to reflected sunlight.

Selection of Flexible & Rigid Road Pavement :

Flexible:

1) When funds are not sufficiently available.
2) Large number of roads are to be constructed.
3) Good subgrade is available.
4) Local material are to be advantageously used.
5) Traffic load is not very heavy.
6) Pavement is to be constructed in stages.

Rigid:

1) When funds are sufficiently available.
2) Long life of the road pavement is desired.
3) Subgrade is poor & varying in nature.
4) Route is very important.
5) Traffic load is very heavy.

Earth Road :-

1) It is cheapest type of road pavement.
2) Camber is very steep and ranges between **1:20 to 1:33.**
3) The maximum cross slope of 1:20 is recommended to avoid erosion.
4) Specification of materials:-

	BASE COURSE	WEARING COURSE
CLAY CONTENT	< 5 %	10 TO 18 %
SILT CONTENT	9 TO 32 %	5 TO 15 %
SAND CONTENT	60 TO 80 %	65 TO 80 %
LIQUID LIMIT	< 35 %	< 35 %
PLASITICITY INDEX	< 6 %	4 TO 10 %

The construction steps of earth roads are as :-

1) Survey is carried out and suitable borrow pits are located within economical haulage distance.

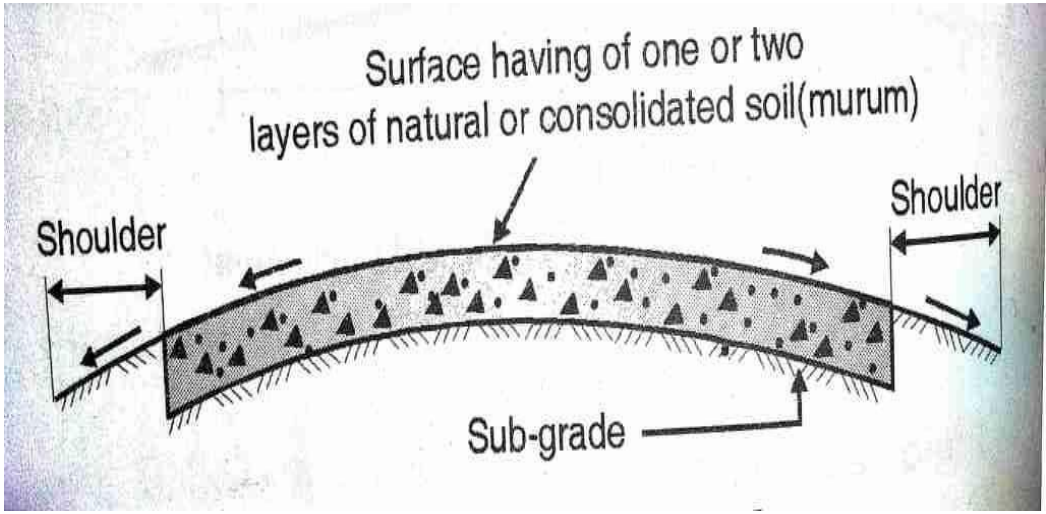
2) Centre line and road edges are marked on the ground along the alignment by driving wooden pegs.

3) Preparation of subgrade:- Clearing site, Excavating and construction of fills, Shaping of subgrade.

4) Pavement construction:- Filled moisture content is checked and additional water is added if necessary to bring it upto **OMC**. The compaction is carried out.

5) Opening to traffic:- The compacted earth road is allowed to dry out for a few days before opening to traffic.





Gravel Road :-



Constructed using soil-aggregates. These roads falls under category of un-surfaced roads. The construction steps are as :-

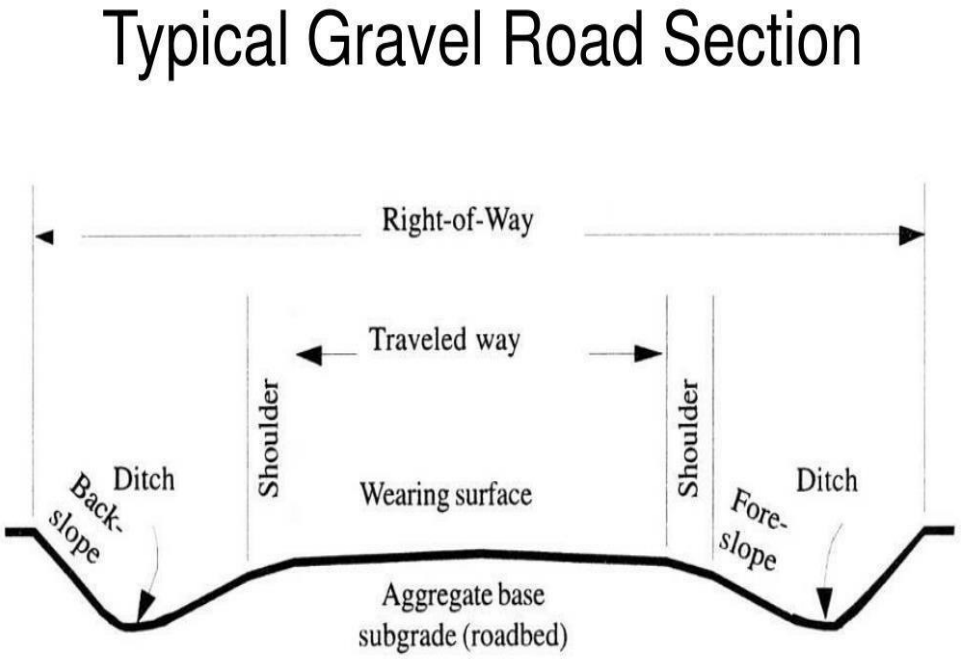
1) Soil subgrade of total thickness 300 mm is constructed by compacting the soil in suitable layers at its OMC.

2) After sub-garde gets sufficiently dry, the gravel base course of specified thickness (100-200mm) is constructed using the selected gravel in suitable layers.

3) The surface course layers is laid using the selected gravel & the material is compacted.

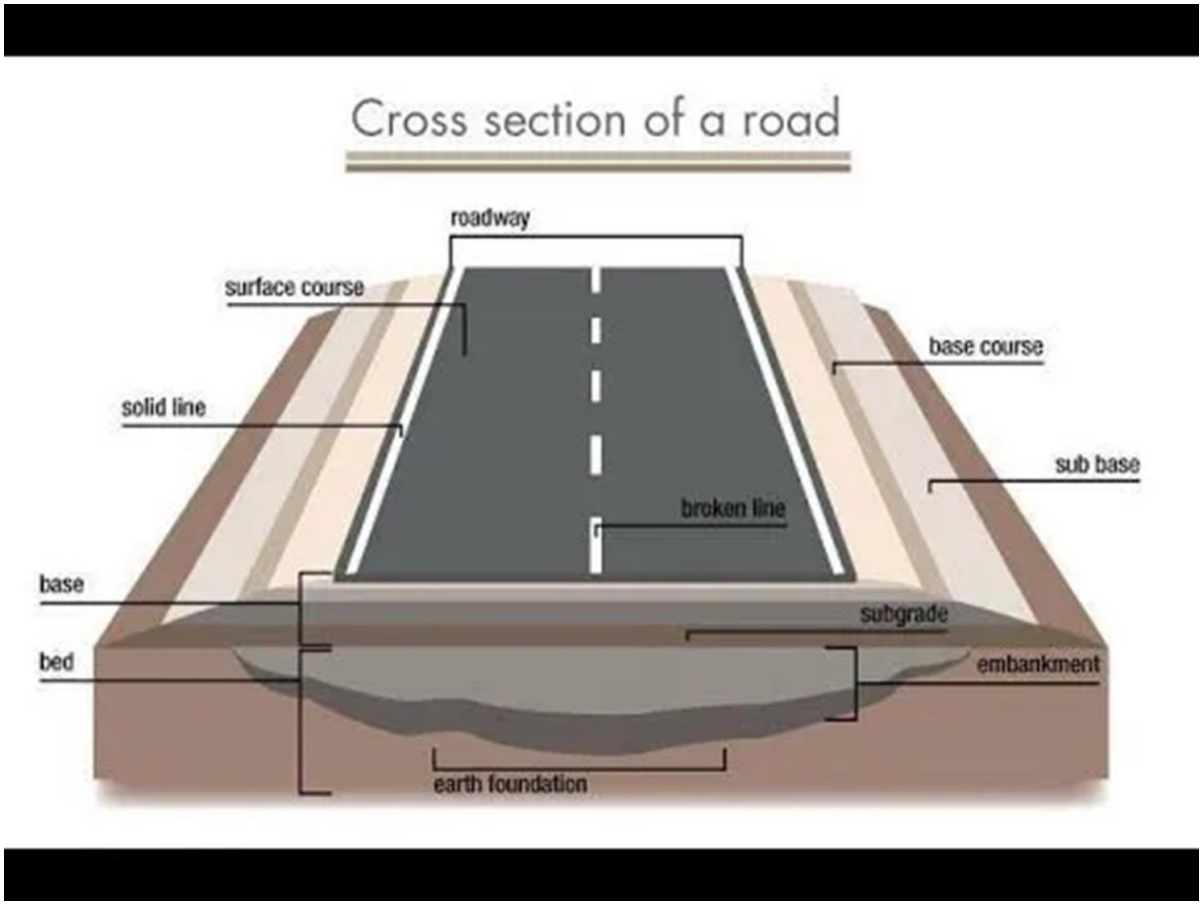
4) Each above layer is compacted at its OMC to obtain 100% dry density by proctor compaction & light compaction.

5) The cross slope of 4 to 5 % is provided.



Water Bound Macadam Roads: -

The term macadam means, the pavement base course made of **crushed or broken aggregate mechanically interlocked** by rolling and the void filled with screening and binding material with the assistance of water. The thickness ranges from 10 cm to 7.5 cm depending on the size and gradation of the aggregate used.



WHAT IS WBM ROAD?

- The water bound macadam, commonly abbreviated as WBM, is a flexible type of road, named after John Macadam.
- It consists of raw materials like clean and crushed aggregates, screening and binding materials.
- All the materials are mixed with water.
- The aggregates are mechanically interlocked by rolling and the screenings and binding materials, are used to fill the voids in the aggregates.
- The thickness of each layer in WBM road ranges between 7.5 to 10 cm.

- According to IRC – 19-1997, a camber of 1 in 48 is provided at the formation level of the subgrade as well as the finished surface if the rainfall in that area is not heavy, and if the rainfall is heavy, a camber of 1 in 36 is suggested.

MATERIALS USED:

The materials used for the preparation of WBM road are as follows:

1. Coarse aggregates,
2. Screenings,
3. Binding materials, and
4. Water.

1. Coarse aggregates:

The coarse aggregates used for WBM road construction are as follows:

- Crushed aggregates
- Over burnt bricks
- Crushed slag
- Kankar
- Broken stones
- Laterite

The table given below shows the maximum permissible values of tests for aggregates, required for various layers of pavement.

Sr. No.	Test	Sub base	Base course	Surfacing course
1	Los Angeles abrasion test	60	50	40
2	Aggregate impact value test	50	40	30
3	Flakiness index test	-	15	15

2. Screenings:

- The screenings are basically used for filling up the voids in the layer of coarse aggregates. Screenings include aggregates of smaller sizes. The screenings are generally the same material as coarse aggregates used in WBM road construction.

3. Binding Materials:

- To avoid raveling of aggregates, fine grained materials such as kankar nodules or lime stone dust are used as binding materials for the construction of water bound macadam road construction.
- When WBM is used as a surface course, the binding materials used in it should have a plasticity index between 4 to 9.
- When and where crushable materials such as moorum or soft gravel are used as screenings, binding materials are not required.

The construction steps of water bound macadam road are as follows:

1) **Preparation** of foundation for receiving the WBM course:- it consist of subgrade or subbase or base course. The foundation layer is prepared to the required grade and camber and also dust and loose material are clean.
2) **Provision of lateral confinement:-** This is done by constructing the shoulder to the advance ,to a thickness equal to that of compacted WBM layer.
3) **Spreading of coarse aggregates:-** It is spread uniformly to proper profile. The WBM course is normally constructed to compacted thickness of 7.5 cm.
4) **Rolling:-** it is done by Three Wheeled Power roller of capacity 6 to 10 tonnes.
5) **Application of Screenings**:- Dry screenings are applied gradually over the surface to fill the interstices in the three or more applications.
6) **Sprinkling and grouting:-** After screening, the surface the sprinkled with water, swept and rolled.
7) **Application of binding material**: it is applied at a uniform and slow rate in two or more successive thin layers. **Murrum or gravel** are used for binding.
8) **Setting and drying**:-After final compaction the WBM course is allow to set over night.

WBM ROAD CONSTRUCTION PROCEDURE:

The following steps are adopted for the construction of WBM roads:

1. Preparing the foundation for receiving the WBM course:

- The foundation supporting the layer of WBM is prepared to the required grade and camber.
- The depressions on the surface of the road are properly filled up and compacted.
- It is cleaned of all the dust particles and loose materials present in it.
- The foundation supporting the WBM road should be dried.



2. Lateral confinement of aggregates:

- The aggregates in WBM should be confined because if they are placed on the subgrade, there are possibilities that the aggregates may come out due to the load of traffic.
- It is done by construction of shoulders of thickness similar to the thickness of the compacted layer of WBM (7.5 cm).

3. Spreading coarse aggregates:

- The aggregates should be evenly and uniformly spread on the prepared base in the required amount.
- They may be spread manually or mechanically. The thickness of the layers (compacted) of WBM road constructed should be 75 mm.



4. Rolling:

- Once the aggregates are spread evenly, the rollers are deployed to carry out the process of compaction. The rollers used shall be of 6 to 10 tonnes capacity.
- The compaction of aggregates should start from the edge of the surface. The aggregates are compacted partially then the compaction is paused to permit the application of screenings.



5. Application of screenings:

- After partial compaction of aggregates, screenings are applied so that the voids in the aggregates are filled completely.
- They are applied gradually over the surface in three or more applications.
- Again after applying screenings, dry rolling is resumed so that they are completely filled in the voids.

6. Sprinkling and grouting:

The surface is sprinkled with water after applying the screenings. The surface is then swept and rolled so that the voids are completely filled. If any voids are left empty, additional screenings may be applied.



7. Application of binding materials:

After applying the screenings, binding materials are applied in thin layers. Two or more thin layers of binders are applied.

After the application of each layer of binding materials, water is sprinkled, followed by compaction of layers by rollers of 6 to 10 tonnes capacity.



8. Setting and drying:

- After all the above-mentioned processes, the layer of WBM is allowed to set overnight.
- Next day the road is inspected and if any voids are empty, they are filled and compacted.

Advantages of WBM Road:

The advantages of WBM roads are as follows:

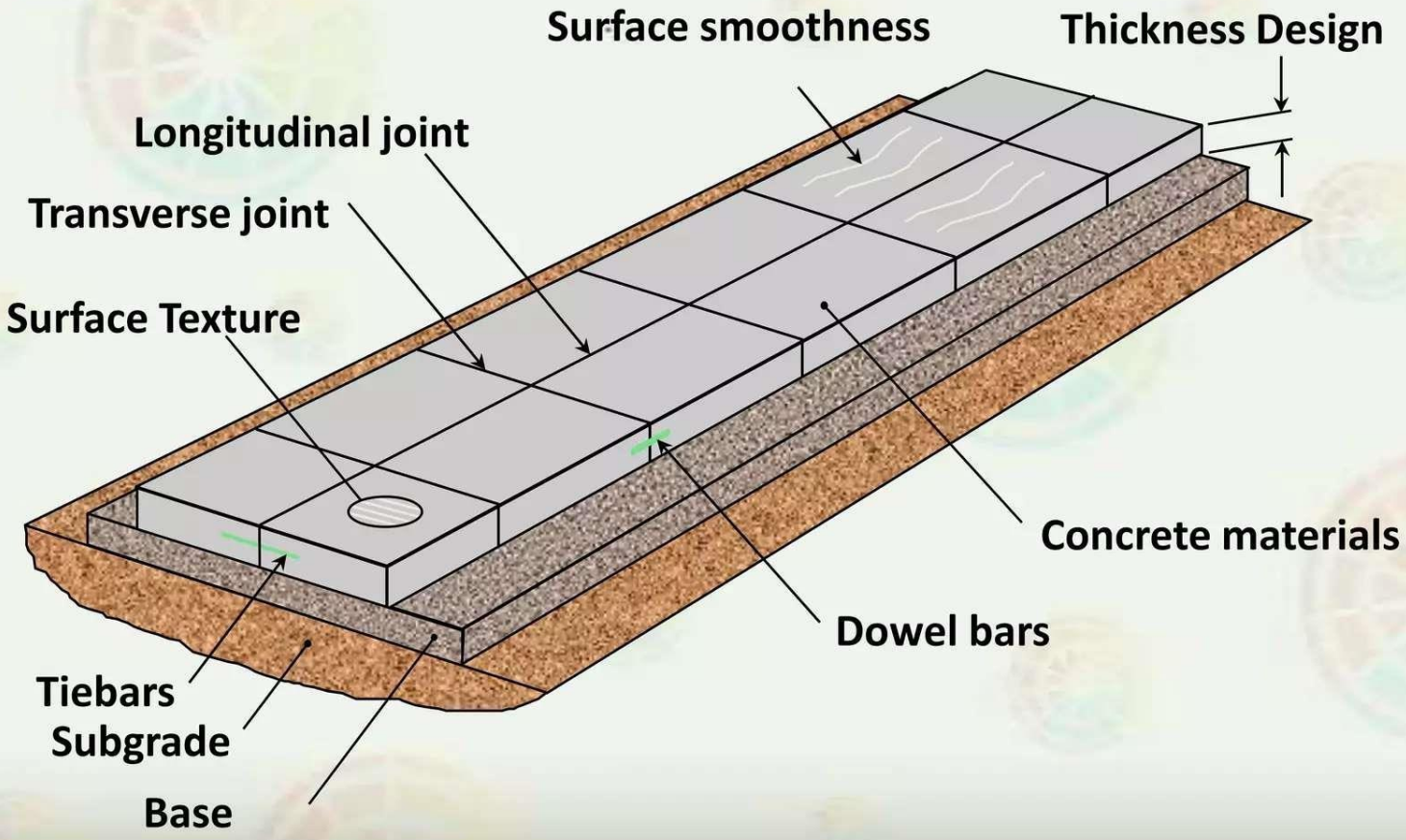
- Initially, the construction of WBM road is cheaper.
- Locally available materials are used.
- No specially trained labours required.
- If it is maintained properly, it can take up to 900 tonnes of traffic/day.

Disadvantages Of WBM Road:

The disadvantages of WBM roads are as follows:

- Higher maintenance cost.
- It is permeable. So it causes softening of subsoil and may get damaged during the rainy season.
- The life span of WBM roads is less.

Basic Components of a Concrete Pavement



Types of Rigid Pavements

- **Jointed Plain Concrete Pavement (JPCP)**
  – does not use any reinforcing steel

- **Jointed Reinforced Concrete Pavement (JRCP)**
  – Reinforcing steel placed at mid height and discontinued at the joints.

- **Continuously Reinforced Concrete Pavement (CRCP)**
  – This method is very costly and generally not used in India.

- **Pre-stressed Concrete Pavement (PCP)**
  – Comprises new and innovative construction methods

– Precast pavement components are fabricated and installed on a prepared foundation (existing pavement or re-graded foundation).

MINERALS USED

Cement

- **Ordinary Portland cement (OPC)**

Ordinary Portland cement and High Strength Ordinary Portland Cement are most widely used for concrete pavements.

- **Rapid hardening Portland cement**

In general, this cement would be used only where time is a critical factor and the road is required to be opened to traffic at an earlier date than would be possible if Ordinary Portland Cement or High Strength Ordinary Portland Cement is used.

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Mineral used(cont.)

Aggregates

- There are two types of Aggregates are used during construction of road i.e. Fine and Coarse Aggregate depending on requirements.

- Maximum size of aggregate should not exceed 1/4th of the pavement slab thickness.

Water

- Water used in mixing or curing of concrete shall be clean and free from injurious amounts of oil, salt, acid, vegetable matter or other substances harmful to the finished concrete.

- Potable waters are generally considered satisfactory for mixing or curing.

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Equipments used

A list of tools, equipment and appliances required for the different phases of concrete road construction is given below

- Three wheeled or vibratory roller for compaction purpose
- Shovels, spades and Sieving screens
- Concrete mixer for mixing of concrete
- Formwork and iron stakes
- Watering devices - Water Lorries, water carriers or watering cans
- Wooden hand tampers for concrete compaction
- Cycle pump/pneumatic air blower for cleaning of joint
- Mild steel sections and blocks for making joint grooves for finishing purpose

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Site Preparation

Before construction begins, the construction site must be carefully prepared, This includes preparing the grade or road base, sub grade and sub base-

- First the site is graded to cut high points and fill low areas to the desired roadway profile elevations.
- Generally, cut material can be used as embankment fill.
- A course of material is placed on the sub grade to provide drainage and stability.
- A course of fairly rigid material, sometimes cement- or asphalt-treated, that is placed on the sub base to provide a stable platform for the concrete pavement slab.





Construction Process

After site preparation, Construction process consists of various steps, which are described below-

FORM WORK

Wooden sheets, battens, plywood, fibre hard board, steel plates, angles, rope etc are generally use for form work.

- before using form work, it should free from all type material like as dust ,cement.
- To placing the concrete in appropriate depth used 2.5-5cm thick and 3mtr long wooden sheeting
- The depth of wooden block must be same as level of slab thick.
- After 24hrs, form work displaced next length of road.



Watering of base

If the base of the road is dry or construction is taken place during summer seasons Then watering is done using the sprinkling process, after this concrete should be Placed.

Construction of Joints

- There are three basic joint types used in concrete pavement: contraction, construction and isolation.

- Except for some isolation joints, all joints provide a means to mechanically connect slabs. The connection helps to spread a load applied on one slab onto slabs along its perimeter.

- Where it is necessary to provide transverse and Longitudinal joints; there wedge of woods, metals fix on level of concrete.

- After setting of concrete it should be pull out.

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EFFECT OF DOWEL BARS ON CONCRETE PAVEMENT

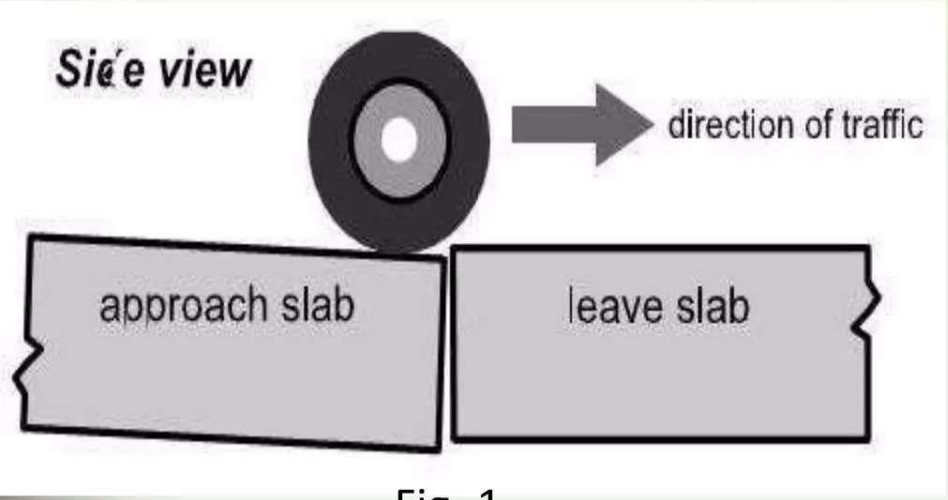


Fig.-1

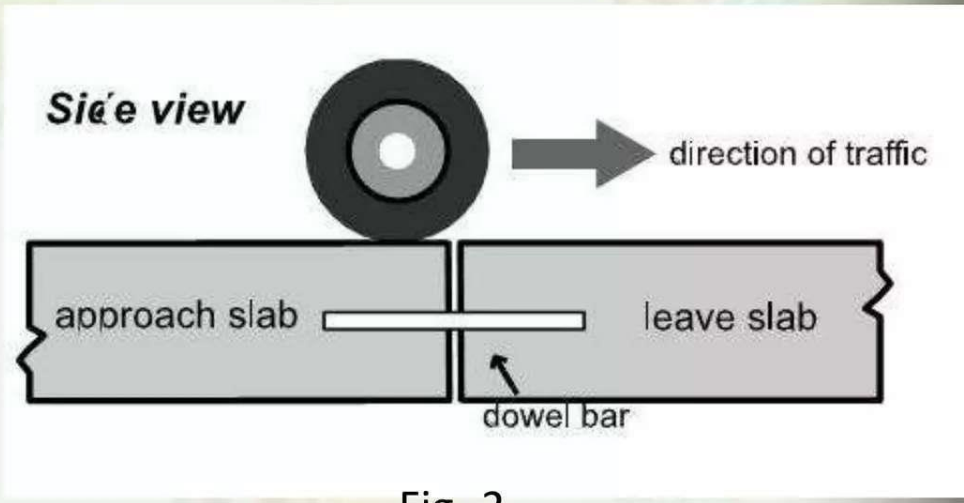


Fig.-2

MATERIAL MIXING AND PLACING OF CONCRETE

Mixer is equipment that mix the concrete using distinct amount of cement , concrete, sand and water. There are two type of mixer generally used during construction,

Batch mixer- used for small road construction
Continuous mixer- used for large construction .

- If distance is more from site , mix concrete transported at site within setting time.

PLACING OF CONCRETE

Two methods generally used in placing of concrete-

1- Alternate bay method-

Placed the concrete on both side of road alternatively like as1,3,5… part at one side and 2,4,6… part other side, This method have slow process due to road traffic problems.

1st side	1	2	3	4	5	6	7
2nd side	1	2	3	4	5	6	7

2- Continuous bay method-

Construct one side of road regularly, and after construction of some part of first side than construct other side, this method have fast process without no obstruction of traffic

COMPACTION OF CONCRETE

Purpose of compaction is that to pull out air from void and make concrete harden.
Compaction done by-
1.Mechanically surface vibrator

2. Manually hand tempers

➢ After this, **Floating, Belting, Screeding or strike off, Bull floating and Brooming** is done as per requirement for the finishing purpose of concrete pavement to smoothen the top surface.

CURING

After completion of the finishing operations, the surface of the pavement shall be entirely covered with wet hessian cloth, burlap or jute mats
Curing is the process of increasing the hydration of cement, after setting the concrete curing process done till 14-28days.

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OPENING OF THE TRAFFIC

Generally after a month, road should be open to traffic, but If we used rapid hardening cement it take 7 days to open traffic.

Advantages of Concrete Road

- Durability and maintenance free life
- Vehicles consume less fuel
- Resistant to automobile fuel spillage and extreme weather
- Greener process
- Saving of natural resources
- Eco friendly process

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Procedure for Rigid Pavement Construction by Alternate Bay Method :

1) The subgrade is prepared & checked at least two days in advance of correcting.
2) The sub—grade or sub-base is kept in moist condition at the time when the cement concrete is placed.
3) The steel or wooden forms are placed. These have minimum base width of 10 cm for slab thickness of 20 cm & minimum base width of 10cm for slab thickness of 20cm & minimum base width of 15 cm for slab over 20 cm & minimum base width of 15cm for slabs over 20 cm thickness.
4) Material is batched & mixed as per design, then it is transported to the site & placing of concrete is carried out.
5) Spreading is done uniformly. Needle vibrator is employed in lieu of rodding & splicing of the concrete.
6) The surface of pavement is compacted either by means of power driven finishing machine or by a vibrating hand screed. At small width or corner hand consolidation & finishing may be adopted.

Procedure for Rigid Pavement Construction by Alternate Bay Method :

1) The surface of pavement is compacted either by means of power driven finishing machine or by a vibrating hand screed. At small width or corner hand consolidation & finishing may be adopted.
2) The concrete is further compacted by means of longitudinal float. Just before the concrete becomes hard, the surface is belted with two ply canvas belt. After belting pavement is given broom finish with fibre broom brush.
3) Finally curing of newly laid pavement surface is done. For initial curing burlap or cotton mats saturated with water are placed in connect with the surface. In initial curing edges of the slab are banked with soil berm. The soil is thoroughly kept saturated with water for 14 days.

Construction and Maintenance of Drainage

- The road drainage system is used for the removal and disposal of water from the surface as well as subsurface water.
- It maintains the stability and durability of the road, by keeping it dry as possible.
- The longer water sits on a surface, the more susceptible it is to contamination.
- The installation of suitable surface and sub-surface drainage systems is an essential part of highway design and construction.

- Sources of water ingress on roads;
  - ✓Capillary action of water
  - ✓Rainwater falling on the road surface
  - ✓Water accumulated from the surrounding area
  - ✓Sub-soil water collection

Why do we need road drainage?

- Road drainage systems are installed to capture surface water run-off to alleviate flooding and protect the surface and sub-surface of the road. It is required for the following reasons:

    - Prevent flooding, ponding and seepage
    - Keep the carriageway, cycleway and footway free of water
    - Keep the underlying road structure as dry as possible.
    - Prevent damage caused by hazardous surface water
    - Stopping surface water flooding adjacent properties

Defects caused due to improper road drainage

- The improper highway drainage leads to the deterioration of the road in the form of the following defects:
- The formation of erosion, causing disembarkment along the roadsides
- Dislocation of pavement edge due to the presence of excess water.
- Stripping of bitumen from aggregates like loosening or detachment of some of the bituminous pavement layers
- Formation of potholes
- Failure of soil slopes because excess moisture causes an increase in weight and
- Formation of waves and corrugations in flexible pavements.
- Softening of subgrade soil bearing capacity
- Reduction in soil mass
- Shoulders are pavement edge distress

Requirements of Road Drainage System

Following are the essential requirements of a good highway drainage system:

- ✓ The surface water from the **carriageway and shoulder** should effectively be drained off
- ✓ The surface water from the **adjoining land** should be prevented from entering the roadway.
- ✓ The side drain should have **sufficient capacity and longitudinal slope** to carry away all the surface water collected from the roadway.
- ✓ The flow of surface water across the road and shoulders and along slopes **should not cause erosion** or form cross ruts
- ✓ **Seepage and other sources** of underground water should be effectively intercepted and drained off
- ✓ In water-logged areas, **special precautions** should be taken
- ✓ The road surface should be provided with **suitable camber** to drain off quickly

Types of Road Drainage System;

There are many different types of drainage systems with different design features and attributes that can be used to manage flows and treat water quality.



1. Surface drainage system

- Surface drainagesystem refers to the drainage wherein surface water is collected and disposed of.
- It is useful to carry **rainwater away** from the carriageway.
- It is applicable in **heavy rainfall** areas.
- The water in this drainage system is removed by **providing a camber and cross slope** to the pavement
- The camber and slope depend upon the type of pavement and the intensity of rainfall.

1- Closed drains
- Drains provided with gratings
- Jelly drains

2- Ditches or open drains
- Shallow surface drains
- Medium-sized drains

3 Side drains
4 Catch basins
5 Inlets

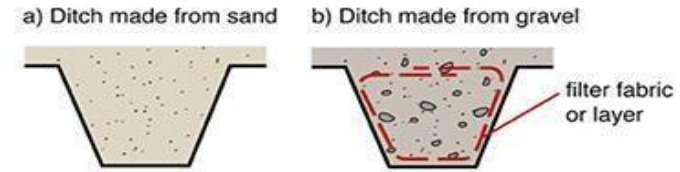


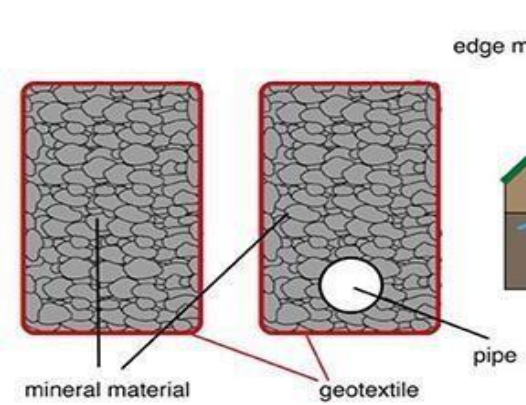
2. Subsurface drainage system

- Subsurface drainage is used to remove water that has **infiltrated into the soil** in excess of the amount.
- It is useful to **carry ground water away** from the subsoil.
- It is feasible in **water-logging** areas.

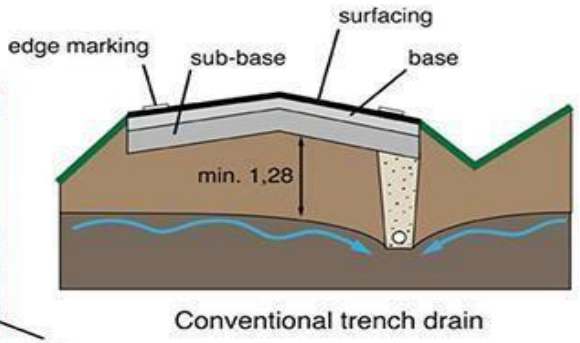
Different types of drainage used for surface drainage systems are;

- Subgrade Drain
- Longitudinal Drain Trenches
- Subsurface pipes
- Interceptor drains





Trench drain with and without pipe



Conventional trench drain



3. Cross drainage system

- Cross drainage is provided whenever streams have to **cross the roadway** facility.
- The water from the side drains is also often taken across these structures to **divert the water away** from the road to a water course or a valley.
- The need for constructing cross-drainage structures arises to ensure that the **water flows beneath the road without causing any inconvenience or instability** to the highway structure.

Types of Cross-drainage Structures:

- Box Culverts
- Aqueduct
- Syphon Aqueduct
- Super passage
- Canal Syphon
- Level Crossing
- Canal Inlets and Outlets
- Minor and major bridges
- Causeways

Road Arboriculture

Definition of Road Arboriculture:

Road arboriculture refers to the systematic planning, planting, and maintenance of trees and vegetation along roads, highways, and streets to enhance aesthetics, provide environmental benefits, and improve safety.

It involves selecting suitable plant species, ensuring proper spacing, and maintaining greenery to support ecological balance and road user comfort.

As per IRC:SP:21-2009 ("Manual on Landscaping of Roads"), road arboriculture is an essential part of highway landscaping, aimed at improving environmental quality, reducing pollution, and enhancing road safety.

Benefits of Road Arboriculture as per IRC Codes

Category	Benefits	Key IRC References	Key Words
Environmental	Reduces air pollution (absorbs CO_2, dust, NOx, SOx)	IRC:SP:21-2009, IRC:SP:103-2014	**Air purification, pollution control**
	Acts as a noise and dust barrier	IRC:SP:21-2009	**Noise reduction, dust control**
	Enhances biodiversity (supports birds & insects)	IRC:SP:103-2014	**Habitat, ecological balance**
Aesthetic & Psychological	Improves roadside aesthetics	IRC:SP:21-2009, IRC:SP:50-2013	**Greenery, scenic beauty**
	Reduces driver fatigue and stress	IRC:SP:50-2013	**Psychological relief, comfort**
Road Safety	Prevents vehicle glare, improves visibility	IRC:35-2015	**Anti-glare, visibility**
	Provides shade, reduces road surface temperature	IRC:SP:103-2014	**Heat reduction, thermal comfort**
	Defines road edges, helps control vehicle speed	IRC:SP:21-2009	**Speed calming, edge marking**
Soil & Water Conservation	Prevents soil erosion on slopes & embankments	IRC:SP:21-2009, IRC:SP:103-2014	**Slope protection, erosion control**
	Enhances groundwater recharge, reduces runoff	IRC:SP:103-2014	**Water retention, groundwater recharge**
Economic & Social	Reduces road maintenance costs	IRC:SP:103-2014	**Cost savings, infrastructure upkeep**
	Provides employment (nurseries, maintenance)	IRC:SP:50-2013	**Jobs, community participation**
	Encourages public involvement & social benefits	IRC:SP:103-2014	**Community engagement, social value**

Objects of Roadside Development and Arboriculture as per Indian Roads Congress (IRC) specifications:

Objective	Details	Relevant IRC Specification
Safety & Traffic Management	- Improves visibility and reduces glare. - Prevents soil erosion and stabilizes slopes. - Acts as windbreaks to reduce dust and storm impact.	IRC:SP:21-2009 (Guidelines on Landscaping and Tree Plantation) IRC:35-2015 (Code of Practice for Road Markings)
Environmental Protection	- Reduces air and noise pollution. - Provides shade and lowers temperature. - Enhances groundwater recharge and prevents flooding.	IRC:SP:21-2009 (Landscape Planning) IRC:SP:50-2013 (Guidelines for Urban Roads)
Aesthetic & Psychological Benefits	- Enhances visual appeal of roads. - Reduces driver fatigue and stress. - Provides recreational and aesthetic value.	IRC:SP:21-2009 (Guidelines on Roadside Landscaping)
Functional Aspects	- Demarcates road boundaries and traffic lanes. - Serves as buffer zones between highways and adjacent land. - Supports ecological balance with biodiversity.	IRC:SP:21-2009 (Tree Plantation Techniques) IRC:36-2010 (Highway Landscaping)
Economic & Social Benefits	- Supports agroforestry and employment generation. - Promotes tourism by enhancing scenic beauty. - Provides timber, fuel, and medicinal plants.	IRC:SP:21-2009 (Agroforestry and Roadside Tree Management)